RTL Design Sherpa

RAPIDS Beats Micro-Architecture Specification 0.6

July 2, 2026

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2 Document Information

This document describes the RAPIDS “Beats” architecture, a Phase 1 implementation of the Rapid AXI Programmable In-band Descriptor System. The beats architecture provides network-to-memory and memory-to-network data transfer capabilities using descriptor-based DMA with 8-channel support. It emphasizes “beat-level” tracking for precise flow control and latency management.

2.1 References

2.1.1 Related Documents

Table 0.1: Related Documents and Specifications

Source	Title	Version
RTL Design Sherpa	RAPIDS Product Requirements Document	1.0
RTL Design Sherpa	RAPIDS Original Specification	0.25
RTL Design Sherpa	STREAM MAS (Reference Architecture)	0.90
ARM	AMBA AXI and ACE Protocol Specification	IHI0022H
ARM	AMBA AXI-Stream Protocol	IHI0051A

Source	Title	Version
	Specification	

2.2 Terminology

AXIS

AXI-Stream. ARM's streaming protocol for high-bandwidth, low-latency data transfer without addressing.

AXI4

Version 4 of the AXI protocol, supporting burst transfers up to 256 beats.

Beat

A single data transfer within an AXI burst. For RAPIDS with 512-bit data width, one beat = 64 bytes.

Beats Architecture

The Phase 1 RAPIDS implementation that tracks transfers at beat granularity for precise flow control.

Burst

A group of consecutive data transfers (beats) initiated by a single address phase.

Channel

One of 8 independent DMA channels in RAPIDS. Each channel has its own descriptor chain and SRAM buffer allocation.

Descriptor

A 256-bit data structure containing transfer parameters: address, length, channel ID, and control flags.

Descriptor Chain

A linked list of descriptors where each descriptor points to the next, enabling scatter-gather operations.

DMA

Direct Memory Access. Hardware-controlled data movement between memory locations without CPU intervention.

FUB

Functional Unit Block. A self-contained RTL module with defined interfaces and testbench.

Latency Bridge

A module that provides buffering to compensate for pipeline latency between alloc/drain controllers.

MAC

Macro. An integration-level block that instantiates and connects multiple FUBs.

MonBus

Monitor Bus. A 64-bit internal bus for performance monitoring and debug event reporting.

RAPIDS

Rapid AXI Programmable In-band Descriptor System. A DMA accelerator for network-to-memory transfers.

Sink Path

The data path from network (AXIS slave) to system memory (AXI write master).

Source Path

The data path from system memory (AXI read master) to network (AXIS master).

SRAM

Static Random Access Memory. Used for internal data buffering between network and memory interfaces.

Virtual FIFO

A tracking mechanism (alloc_ctrl/drain_ctrl) that manages SRAM space without moving actual data.

2.3 “Beats” Architecture Concept

The “beats” architecture name reflects the key design decision to track all transfers at **beat granularity** (data-width units) rather than byte or chunk granularity:

1. **Beat-level Allocation:** Space is allocated in beats (e.g., 64-byte units for 512-bit datapath)
2. **Beat-level Drain:** Data availability is tracked in beats
3. **Beat-level Latency:** The latency bridge compensates for pipeline delays in beat counts

4. **Beat-level Completion:** AXI engines report completion in beats

This simplifies the math and aligns naturally with AXI burst semantics where each beat represents one data-width transfer.

2.4 Revision History

Revisions follow the convention x.y, where x is the major version and y is the minor version.

Table 0.2: RAPIDS Beats MAS Document Revision History

Rev	Date	Author	Notes
0.25	2025-01-10	seang	Initial RAPIDS Beats MAS structure
0.6	2026-07-02	seang	Resynced to RTL: scheduler recoverable write-progress timeout (`cfg_sched_timeout_limit`, strike escalation, `scheduler_idle` excludes CH_ERROR) and B-response commit gating; functional descriptor prefetch (`cfg_prefetch_enable`/`cfg_fifo_threshold`, deferred chaining); AXI write engine COMMIT strobes; commit/timeout-limit plumbing through `rapids_core_beats`; added

Rev	Date	Author	Notes
			`rapids_regs` register block (monitors @ 0x1000), `rapids_config_block`, and `rapids_beats_top` integration with the MonBus AXI-Lite group
0.7	2026-07-13	seang	Documented the control-read / control-write engine subsystem that was in the RTL but undocumented: new fub-block specs for `ctrlrd_engine` (Section 2.8, consumer gate: poll-until-`(rd & mask)==expected`, retry budget) and `ctrlwr_engine` (Section 2.9, producer doorbell: single-beat write); scheduler Control Interface + three-opcode (`DATA`/`CTRL_READ`/`CTRL_WRITE`) operation (Section 2.1); `CTRL_CONFIG` @ 0x240 (`CTRLRD_MAX_TRY`)

Rev	Date	Author	Notes
			added to the register map (Section 3.12); `scheduler_group_beats` control-engine instantiation. Naming cleanup: the per-instance register file is `rapids_engine_regs` (was `rapids_half_regs`).

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3 Architecture Overview

Module: `rapids_core_beats.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

3.1 Overview

The RAPIDS “Beats” architecture is a Phase 1 implementation providing network-to-memory and memory-to-network data transfer capabilities. The name “beats” reflects the design decision to track all transfers at beat granularity (data-width units) for simplified flow control.

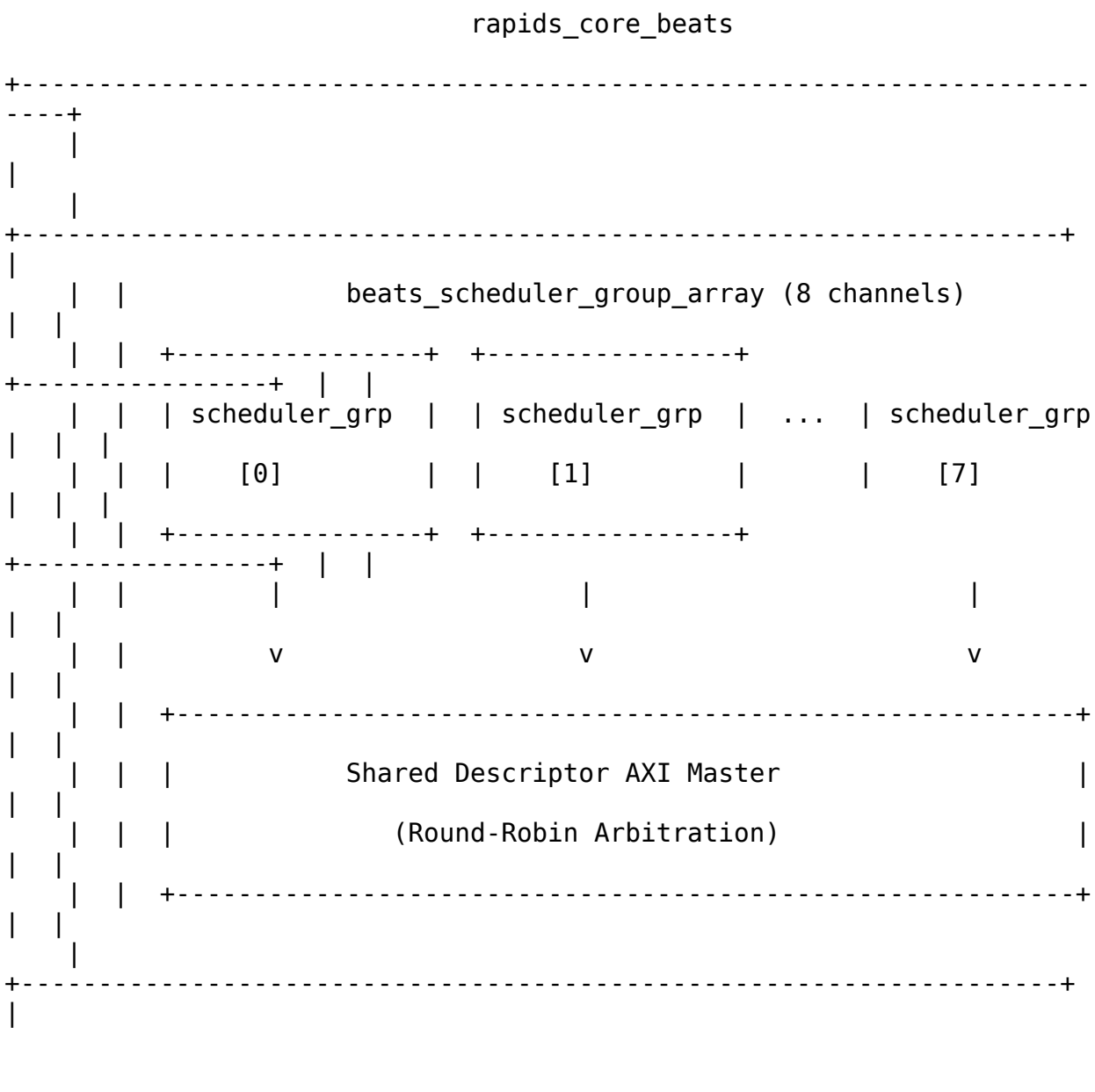
3.1.1 Key Features

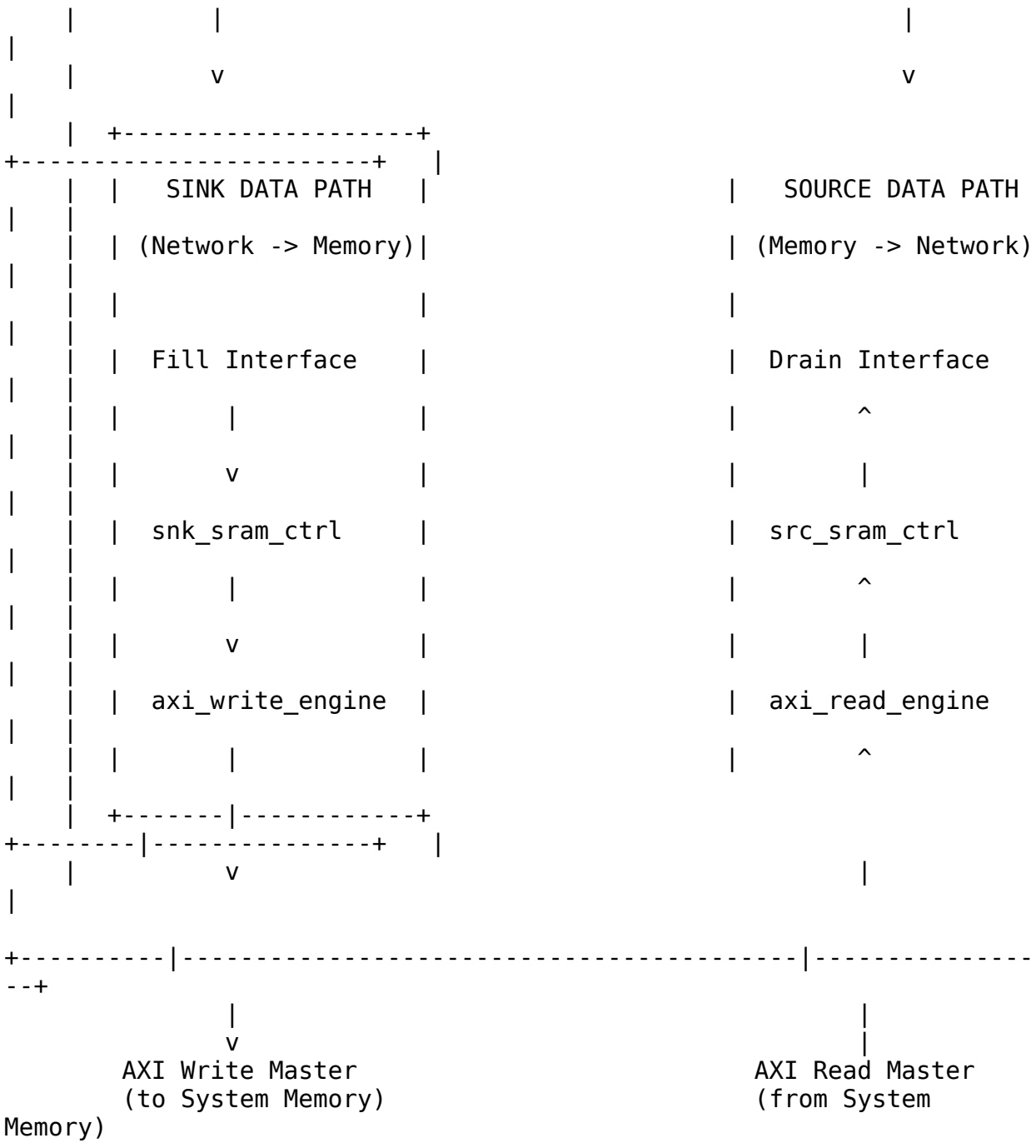
- **8 Independent Channels:** Each channel has its own descriptor chain and SRAM allocation

- **Separate Data Paths:** Sink (network-to-memory) and source (memory-to-network) paths
- **Beat-Level Tracking:** All flow control operates at beat granularity
- **SRAM Buffering:** Separate sink and source SRAM buffers for data staging
- **MonBus Integration:** Comprehensive monitoring for all subsystems
- **Streaming Pipelines:** No FSM in data engines - pure streaming for maximum throughput

3.1.2 Block Diagram

3.1.3 Figure 1.1.1: RAPIDS Beats Architecture Block Diagram





Source: 01_architecture_block.mmd

3.2 Data Flow Overview

3.2.1 Sink Path (Network to Memory)

The sink path receives data from an external source (via Fill interface) and writes it to system memory:

3.2.2 Figure 1.1.2: Sink Path Data Flow

```
1. External Fill Valid
        |
        v
2. beats_alloc_ctrl (Space Allocation)
        |
        v
3. snk_sram_controller_unit (SRAM Write)
        |
        v
4. beats_drain_ctrl (Data Available)
        |
        v
5. axi_write_engine (AXI Burst Write)
        |
        v
6. System Memory
```

Source: 01_sink_path_flow.mmd

3.2.3 Source Path (Memory to Network)

The source path reads data from system memory and sends it to an external destination (via Drain interface):

3.2.4 Figure 1.1.3: Source Path Data Flow

```
1. Scheduler Request
        |
        v
2. beats_alloc_ctrl (Space Allocation)
        |
        v
3. axi_read_engine (AXI Burst Read)
        |
        v
4. src_sram_controller_unit (SRAM Write)
        |
        v
5. beats_drain_ctrl (Data Available)
        |
```

```
        v
6. External Drain Ready
```

Source: 01_source_path_flow.mmd

3.3 Key Architectural Decisions

3.3.1 Beat-Level Tracking

All flow control uses beat granularity:

Table 1.1.1: Beat Granularity Examples

Operation	Granularity	Example (512-bit DW)
Space allocation	Beats	Request 8 beats = 512 bytes
Data availability	Beats	16 beats ready = 1024 bytes
AXI burst length	Beats	ARLEN=15 = 16 beats
Latency compensation	Beats	2-beat pipeline delay

3.3.2 Concurrent Read/Write

To prevent deadlock with large transfers:

```
Example: 100MB transfer with 2KB SRAM buffer

Sequential operation (WRONG):
1. Read 100MB -> DEADLOCK at 2KB (SRAM full, can't complete read)

Concurrent operation (CORRECT):
1. Read starts filling SRAM -> SRAM becomes full (2KB)
2. Read pauses (natural backpressure)
3. Write drains SRAM -> SRAM has free space
4. Read resumes -> Both continue until 100MB complete
```

3.3.3 Virtual FIFOs (Alloc/Drain)

The alloc_ctrl and drain_ctrl modules are “virtual FIFOs” that track space/data without storing actual data:

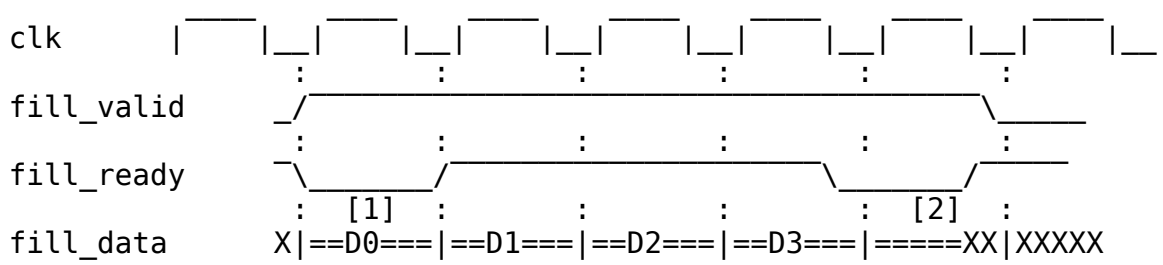
- **beats_alloc_ctrl:** Tracks allocated space (write pointer advances on allocation, read pointer on actual write)
- **beats_drain_ctrl:** Tracks available data (write pointer advances on data arrival, read pointer on drain)

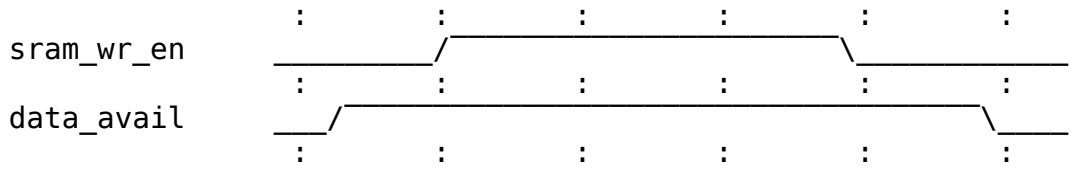
3.4 Module Hierarchy

```
rapids_core_beats
├── beats_scheduler_group_array
│   ├── beats_scheduler_group [0..7]
│   │   ├── descriptor_engine
│   │   └── scheduler
│   └── Shared Descriptor AXI Master
│
├── sink_data_path
│   ├── snk_sram_controller
│   │   └── snk_sram_controller_unit [0..7]
│   │       ├── beats_alloc_ctrl
│   │       ├── simple_sram
│   │       └── beats_drain_ctrl
│   └── axi_write_engine
│       └── beats_latency_bridge
│
└── source_data_path
    ├── axi_read_engine
    │   └── beats_latency_bridge
    └── src_sram_controller
        └── src_sram_controller_unit [0..7]
            ├── beats_alloc_ctrl
            ├── simple_sram
            └── beats_drain_ctrl
```

3.5 Timing Diagram: Basic Transfer

3.5.1 Figure 1.1.4: Basic Sink Path Transfer Timing






```
[1] = fill_ready deasserted (SRAM full or allocation pending)
[2] = fill_ready deasserted (end of burst)
```

TODO: Replace with simulation-generated waveform showing actual signals

---

3.6 Related Documentation

- **Top-Level Port List** - Complete port specification
- **Clocks and Reset** - Timing requirements
- **Beats Scheduler Group** - Scheduler integration
- **Sink Data Path** - Sink path details

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4 Top-Level Port List

Module: `rapids_core_beats.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

4.1 Parameters

4.1.1 Primary Parameters

```
parameter int NUM_CHANNELS = 8;                // Number of DMA
channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS);
parameter int ADDR_WIDTH = 64;                 // Address bus width
```

```
parameter int DATA_WIDTH = 512;                // Data bus width
parameter int AXI_ID_WIDTH = 8;                // AXI ID width
parameter int SRAM_DEPTH = 512;                // SRAM depth per
channel
parameter int SEG_COUNT_WIDTH = $clog2(SRAM_DEPTH) + 1;
parameter int PIPELINE = 0;                    // Pipeline stages
parameter int AR_MAX_OUTSTANDING = 8;          // Max outstanding AR
transactions
parameter int AW_MAX_OUTSTANDING = 8;          // Max outstanding AW
transactions
parameter int W_PHASE_FIFO_DEPTH = 64;         // Write data FIFO depth
parameter int B_PHASE_FIFO_DEPTH = 16;         // Write response FIFO
depth
```

: Table 1.2.1: Primary Parameters

4.1.2 Monitor Bus Parameters

```
parameter int DESC_MON_BASE_AGENT_ID = 16;     // 0x10 - Descriptor
Engines (16-23)
parameter int SCHED_MON_BASE_AGENT_ID = 48;    // 0x30 - Schedulers
(48-55)
parameter int DESC_AXI_MON_AGENT_ID = 8;       // 0x08 - Descriptor AXI
Master Monitor
parameter int MON_UNIT_ID = 1;                 // 0x1
parameter int MON_MAX_TRANSACTIONS = 16;
```

: Table 1.2.2: Monitor Bus Parameters

4.2 Clock and Reset

Table 1.2.3: Clock and Reset Signals

Signal	Direction	Width	Description
`clk`	input	1	System clock (100-500 MHz)
`rst_n`	input	1	Active-low asynchronous reset

4.3 APB Programming Interface

Table 1.2.4: APB Programming Interface

Signal	Direction	Width	Description
`apb_valid`	input	NC	Per-channel kick-off valid
`apb_ready`	output	NC	Per-channel ready
`apb_addr`	input	NC x AW	Per-channel descriptor start address

4.4 Configuration Interface

4.4.1 Per-Channel Configuration

Table 1.2.5: Per-Channel Configuration

Signal	Direction	Width	Description
`cfg_channel_enable`	input	NC	Enable each channel
`cfg_channel_reset`	input	NC	Per-channel soft reset

4.4.2 Scheduler Configuration

Table 1.2.6: Scheduler Configuration

Signal	Direction	Width	Description
`cfg_sched_enable`	input	1	Global scheduler enable
`cfg_sched_timeout_cycles`	input	16	Timeout threshold in cycles
`cfg_sched_timeout_enable`	input	1	Enable timeout detection
`cfg_sched_err_enable`	input	1	Enable error reporting
`cfg_sched_compl_enable`	input	1	Enable completion reporting

Signal	Direction	Width	Description
cfg_sched_perf_enable	input	1	Enable performance monitoring

4.4.3 Descriptor Engine Configuration

Table 1.2.7: Descriptor Engine Configuration

Signal	Direction	Width	Description
cfg_desceng_enable	input	1	Global descriptor engine enable
cfg_desceng_prefetch	input	1	Enable descriptor prefetching
cfg_desceng_fifo_thresh	input	4	Prefetch FIFO threshold
cfg_desceng_addr0_base	input	AW	Address range 0 base
cfg_desceng_addr0_limit	input	AW	Address range 0 limit
cfg_desceng_addr1_base	input	AW	Address range 1 base
cfg_desceng_addr1_limit	input	AW	Address range 1 limit

4.4.4 AXI Transfer Configuration

Table 1.2.8: AXI Transfer Configuration

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	AXI read burst length (beats)
cfg_axi_wr_xfer_beats	input	8	AXI write burst length (beats)

4.5 Status Interface

Table 1.2.9: Status Interface

Signal	Direction	Width	Description
system_idle	output	1	All components idle
descriptor_engine_idle	output	NC	Per-channel descriptor engine idle
scheduler_idle	output	NC	Per-channel scheduler idle
scheduler_state	output	NC x 7	Per-channel FSM state (one-hot)
sched_error	output	NC	Per-channel error flag

4.6 Sink Path Interface (Network to Memory)

4.6.1 Fill Interface (External to SRAM)

Table 1.2.10: Sink Fill Interface

Signal	Direction	Width	Description
snk_fill_alloc_req	input	1	Allocation request
snk_fill_alloc_size	input	8	Size in beats to allocate
snk_fill_alloc_id	input	CIW	Channel ID for allocation
snk_fill_space_free	output	NC x SCW	Available space per channel
snk_fill_valid	input	1	Fill data valid
snk_fill_ready	output	1	Ready to accept fill data
snk_fill_data	input	DW	Fill data
snk_fill_last	input	1	Last beat of fill burst
snk_fill_id	input	CIW	Fill channel ID

4.6.2 Sink AXI Write Master

Table 1.2.11: Sink AXI Write Master Interface

Signal	Direction	Width	Description
m_snk_axi_awid	output	IW	Write address ID
m_snk_axi_awaddr	output	AW	Write address
m_snk_axi_awlen	output	8	Burst length (beats - 1)
m_snk_axi_awsize	output	3	Burst size
m_snk_axi_awburst	output	2	Burst type
m_snk_axi_awvalid	output	1	Write address valid
m_snk_axi_awready	input	1	Write address ready
m_snk_axi_wdata	output	DW	Write data
m_snk_axi_wstrb	output	DW/8	Write strobes
m_snk_axi_wlast	output	1	Last write beat
m_snk_axi_wvalid	output	1	Write data valid
m_snk_axi_wready	input	1	Write data ready
m_snk_axi_bid	input	IW	Response ID
m_snk_axi_bresp	input	2	Write response
m_snk_axi_bvalid	input	1	Response valid
m_snk_axi_bready	output	1	Response ready

4.7 Source Path Interface (Memory to Network)

4.7.1 Source AXI Read Master

Table 1.2.12: Source AXI Read Master Interface

Signal	Direction	Width	Description
m_src_axi_arid	output	IW	Read address ID
m_src_axi_araddr	output	AW	Read address
m_src_axi_arlen	output	8	Burst length (beats - 1)
m_src_axi_arsize	output	3	Burst size
m_src_axi_arburst	output	2	Burst type
m_src_axi_arvalid	output	1	Read address valid
m_src_axi_arready	input	1	Read address ready
m_src_axi_rid	input	IW	Read data ID
m_src_axi_rdata	input	DW	Read data
m_src_axi_rresp	input	2	Read response
m_src_axi_rlast	input	1	Last read beat
m_src_axi_rvalid	input	1	Read data valid
m_src_axi_rready	output	1	Read data ready

4.7.2 Drain Interface (SRAM to External)

Table 1.2.13: Source Drain Interface

Signal	Direction	Width	Description
src_drain_valid	output	1	Drain data valid

Signal	Direction	Width	Description
src_drain_ready	input	1	External ready for drain
src_drain_data	output	DW	Drain data
src_drain_last	output	1	Last beat of drain burst
src_drain_id	output	CIW	Drain channel ID

4.8 Descriptor AXI Master

Table 1.2.14: Descriptor AXI Master Interface

Signal	Direction	Width	Description
m_desc_axi_arid	output	IW	Read address ID
m_desc_axi_araddr	output	AW	Read address
m_desc_axi_arlen	output	8	Burst length
m_desc_axi_arsize	output	3	Burst size
m_desc_axi_arburst	output	2	Burst type
m_desc_axi_arvalid	output	1	Read address valid
m_desc_axi_arready	input	1	Read address ready
m_desc_axi_rid	input	IW	Read data ID
m_desc_axi_rdata	input	256	Descriptor data (256-bit)
m_desc_axi_rresp	input	2	Read response
m_desc_axi_rlast	input	1	Last read beat
m_desc_axi_rv	input	1	Read data

Signal	Direction	Width	Description
`alid`			valid
`m_desc_axi_rready`	output	1	Read data ready

4.9 MonBus Interface

Table 1.2.15: MonBus Interface

Signal	Direction	Width	Description
`monbus_valid`	output	1	Monitor packet valid
`monbus_ready`	input	1	Monitor consumer ready
`monbus_data`	output	64	Monitor packet data

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5 Clocks and Reset

Module: `rapids_core_beats.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

5.1 Clock Domain

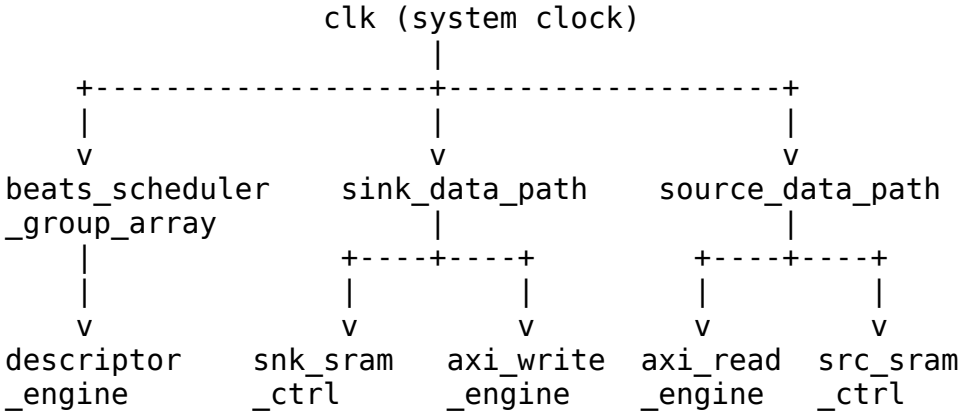
RAPIDS Beats operates in a single clock domain for simplified design and maximum throughput.

5.1.1 Clock Specifications

Table 1.3.1: Clock Specifications

Parameter	Specification
Clock Signal	`clk` (also `axi_aclk` in some submodules)
Frequency Range	100 - 500 MHz
Duty Cycle	45% - 55%
Clock Jitter	< 100 ps peak-to-peak

5.1.2 Clock Distribution



5.2 Reset Specification

RAPIDS uses asynchronous assert, synchronous deassert reset methodology.

5.2.1 Reset Signal

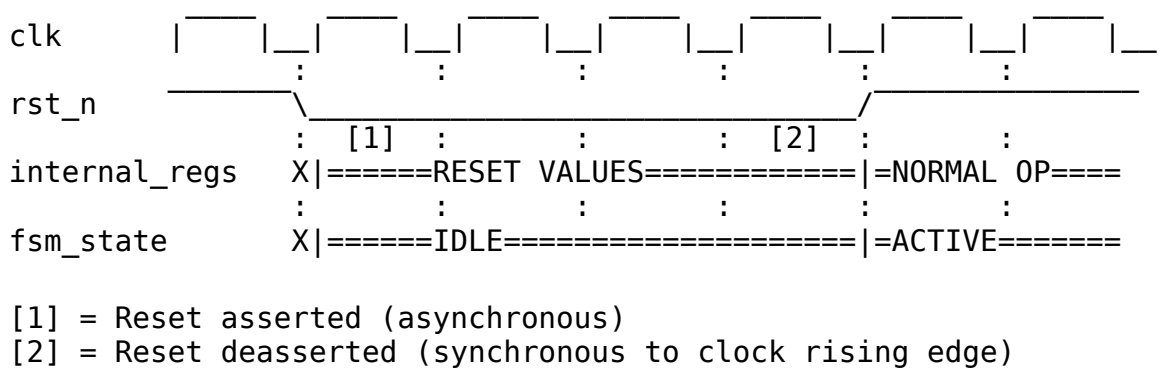
Table 1.3.2: Reset Specifications

Parameter	Specification
Reset Signal	`rst_n` (also `axi_aresetn` in some submodules)
Polarity	Active-low
Assert	Asynchronous (immediate)
Deassert	Synchronous (on rising clock edge)
Minimum Duration	10 clock cycles

Parameter	Specification

5.2.2 Reset Timing Diagram

5.2.3 Figure 1.3.1: Reset Timing



[1] = Reset asserted (asynchronous)
[2] = Reset deasserted (synchronous to clock rising edge)

TODO: Replace with simulation-generated waveform

5.3 Reset Behavior

5.3.1 On Reset Assert

All modules initialize to known states:

Table 1.3.3: Reset States

Component	Reset State
Scheduler FSM	IDLE
Descriptor Engine	IDLE, FIFOs empty
AXI Read Engine	Idle, no outstanding transactions
AXI Write Engine	Idle, no outstanding transactions
SRAM Pointers	All zeros
Alloc/Drain Controllers	Empty, full space available
MonBus	No packets pending

5.3.2 Reset Sequence

```
1. External reset asserted (rst_n = 0)
   - All flip-flops async reset
   - All outputs driven to safe states
   - No AXI transactions

2. System stabilization (minimum 10 cycles)
   - Clock running and stable
   - Configuration registers programmed (optional)

3. Reset deasserted (rst_n = 1, sync to clk rising edge)
   - FSMs begin operation
   - Ready to accept descriptors/data
```

5.4 Configuration During Reset

Some configuration signals MUST be stable before reset deassertion:

Table 1.3.4: Configuration Timing Requirements

Signal	Requirement
`cfg_channel_enable`	Stable before rst_n=1
`cfg_sched_timeout_cycles`	Stable before rst_n=1
`cfg_axi_rd_xfer_beats`	Stable before rst_n=1
`cfg_axi_wr_xfer_beats`	Stable before rst_n=1

5.5 Reset Macros

RAPIDS uses standardized reset macros from `reset_defs.svh`:

```
`include "reset_defs.svh"

// Standard reset pattern
`ALWAYS_FF_RST(clk, rst_n,
    if (`RST_ASSERTED(rst_n)) begin
        r_state <= IDLE;
        r_counter <= '0;
    end else begin
        r_state <= w_next_state;
        r_counter <= w_next_counter;
```

```
    end
)
```

5.6 Power-On Reset Considerations

- External reset should be held for minimum 10 clock cycles after clock is stable
- Configuration registers should be programmed before or during reset
- No AXI transactions until reset is deasserted and channels are enabled

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6 Scheduler Specification

Module: `scheduler.sv` **Location:** `projects/components/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

6.1 Overview

The Scheduler coordinates descriptor-based data transfers for a single RAPIDS channel. It receives descriptors from the descriptor engine, parses transfer parameters, and coordinates read/write operations through the data paths.

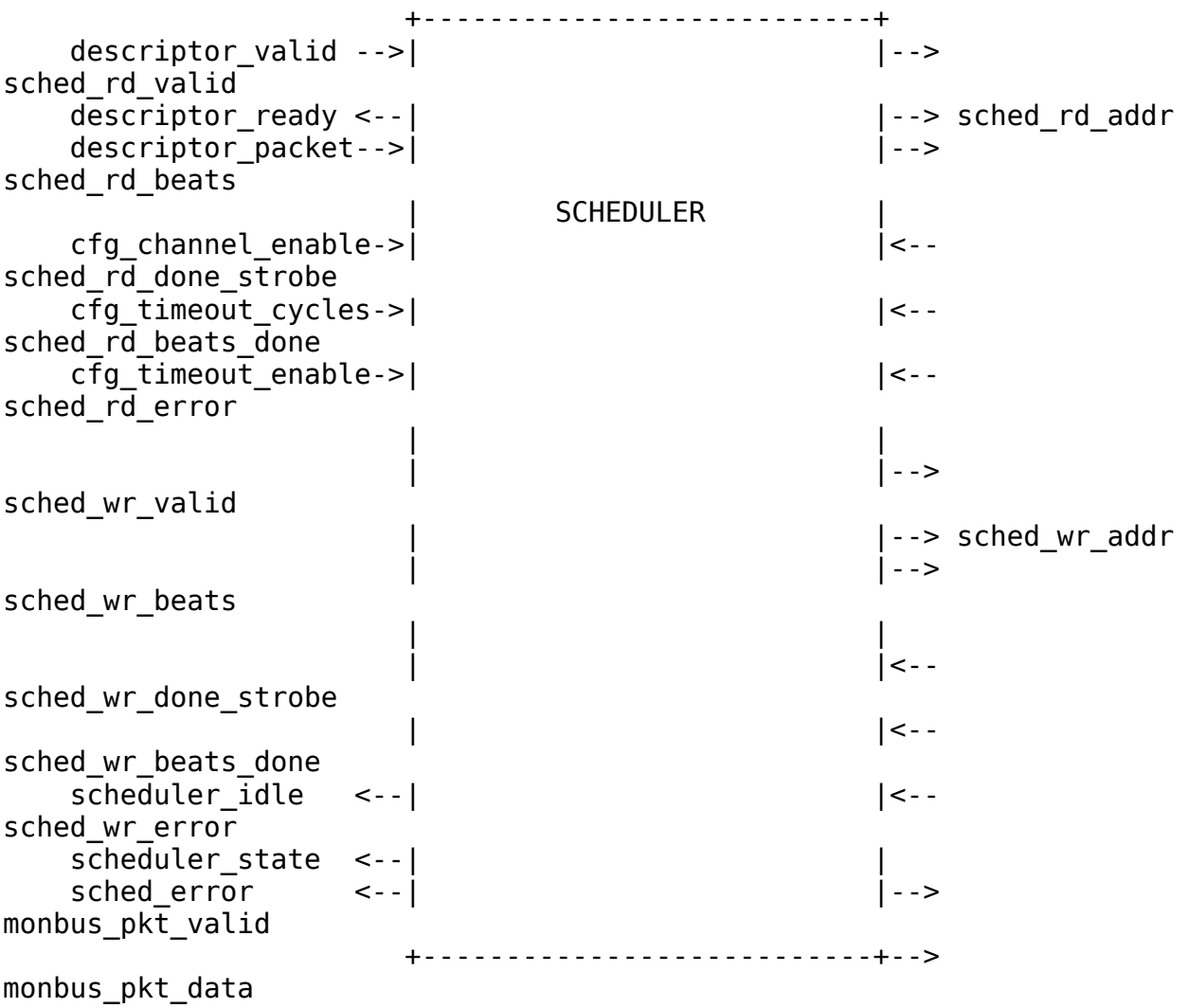
6.1.1 Key Features

- **Descriptor Processing:** Parses 256-bit descriptors for transfer parameters
- **Beat-Based Tracking:** All lengths tracked in beats (data-width units)
- **Concurrent Read/Write:** Simultaneous read and write to prevent deadlock
- **Three descriptor opcodes:** `DATA` (concurrent read/write), `CTRL_READ` (consumer gate via the control-read engine), `CTRL_WRITE` (producer doorbell via the control-write engine) – see Control Interface below

- **Timeout Detection:** Configurable watchdog for stalled transfers
- **MonBus Integration:** State transition and error event reporting
- **Error Aggregation:** Combines errors from descriptor engine and data paths

6.1.2 Block Diagram

6.1.3 Figure 2.1.1: Scheduler Block Diagram



Source: 02_scheduler_block.mmd

6.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 8;                  // Total channels in
system
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Channel ID width
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int DATA_WIDTH = 512;                  // Data bus width
(beats)

// Monitor Bus Parameters
parameter logic [7:0] MON_AGENT_ID = 8'h30;      // RAPIDS Scheduler
Agent ID
parameter logic [3:0] MON_UNIT_ID = 4'h1;        // Unit identifier
parameter logic [5:0] MON_CHANNEL_ID = 6'h0;     // Base channel ID

// Descriptor Width
parameter int DESC_WIDTH = 256;                  // RAPIDS descriptor
width
```

: Table 2.1.1: Scheduler Parameters

6.3 Port List

6.3.1 Clock and Reset

Table 2.1.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

6.3.2 Configuration Interface

Table 2.1.3: Configuration Interface

Signal	Direction	Width	Description
`cfg_channel_enable`	input	1	Enable this channel
`cfg_channel_reset`	input	1	Channel soft reset
`cfg_sched_ti`	input	32	Write-progress timeout

Signal	Direction	Width	Description
meout_cycles			window (cycles)
cfg_sched_timeout_limit	input	8	Consecutive-timeout windows before fatal escalation (0 = never)
cfg_sched_timeout_enable	input	1	Enable timeout detection

6.3.3 Descriptor Engine Interface

Table 2.1.4: Descriptor Engine Interface

Signal	Direction	Width	Description
descriptor_valid	input	1	Descriptor valid
descriptor_ready	output	1	Ready to accept descriptor
descriptor_packet	input	256	256-bit descriptor
descriptor_error	input	1	Error from descriptor engine

6.3.4 Data Read Interface

Table 2.1.5: Data Read Interface

Signal	Direction	Width	Description
sched_rd_valid	output	1	Read request valid
sched_rd_addr	output	AW	Source address
sched_rd_beats	output	32	Beats to read
sched_rd_done_strobe	input	1	Read complete strobe
sched_rd_beats_done	input	32	Beats completed
sched_rd_error	input	1	Read error

6.3.5 Data Write Interface

Table 2.1.6: Data Write Interface

Signal	Direction	Width	Description
sched_wr_valid	output	1	Write request valid (gated on remaining ISSUE beats)
sched_wr_addr	output	AW	Destination address
sched_wr_beats	output	32	Beats to write
sched_wr_done_strobe	input	1	Write AW-issue strobe (burst accepted on AW)
sched_wr_beats_done	input	32	Beats issued this strobe
sched_wr_commit_strobe	input	1	Write COMMIT strobe (B response received)
sched_wr_commit_beats	input	32	Beats committed this strobe
sched_wr_error	input	1	Write error

6.3.6 Control Interface

For CTRL_READ / CTRL_WRITE descriptors the scheduler drives two per-channel control engines instead of the data path. Both are request/response handshakes whose completion is signalled by the engine's *_idle going high after issue.

Table 2.1.7: Control Interface (CTRL_READ / CTRL_WRITE)

Signal	Direction	Width	Description
ctrlrd_valid	output	1	Control-read request valid (to ctrlrd_engine)
ctrlrd_ready	input	1	Engine accepted the poll request
ctrlrd_addr	output	AW	Gate poll address (descriptor poll_addr)
ctrlrd_data	output	32	Expected value (descriptor expected)

Signal	Direction	Width	Description
ctrlrd_mask	output	32	Compare mask (descriptor mask)
ctrlrd_error	input	1	Engine error (poll retries exhausted / AXI error)
ctrlrd_idle	input	1	ctrlrd_engine_idle – gate satisfied (completion)
ctrlwr_valid	output	1	Control-write request valid (to ctrlwr_engine)
ctrlwr_ready	input	1	Engine accepted the doorbell request
ctrlwr_addr	output	AW	Doorbell address (descriptor wr_addr)
ctrlwr_data	output	32	Doorbell value (descriptor wr_data)
ctrlwr_error	input	1	Engine error (AXI error)
ctrlwr_idle	input	1	ctrlwr_engine_idle – doorbell written (completion)

A CTRL_READ descriptor is a **consumer gate**: the scheduler asserts ctrlrd_valid and holds off the descriptor's completion (and the chain) until the control-read engine reports the masked poll matched (ctrlrd_idle). The poll retry budget is bounded by CTRL_CONFIG.CTRLRD_MAX_TRY (Section 3.12) so a never-matching gate cannot hang the channel – exhaustion raises ctrlrd_error. A CTRL_WRITE descriptor is a **producer doorbell**: a single-beat write of ctrlwr_data to ctrlwr_addr, after which the chain continues. The scheduler decodes the opcode from the descriptor (rapids_pkg DESC_OP_DATA / DESC_OP_CTRL_READ / DESC_OP_CTRL_WRITE); DATA descriptors ignore this interface. The engines are instantiated per channel in scheduler_group_beats (u_ctrlrd_engine / u_ctrlwr_engine); see Sections 2.8 and 2.9.

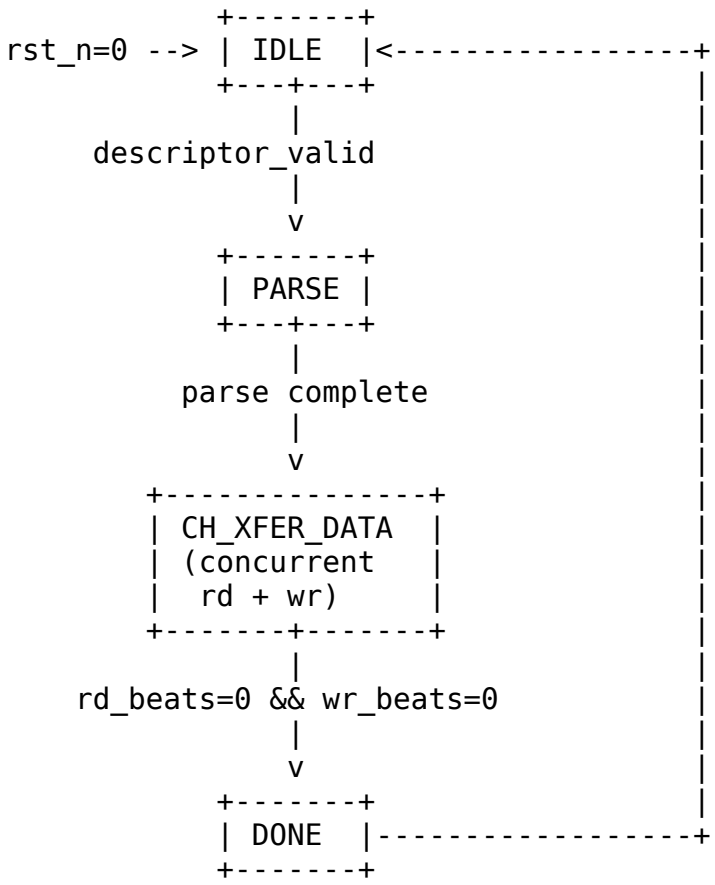
Completion is gated on write-data COMMIT, not AW issue. The scheduler maintains two independent beat counters for the write side:

- r_write_beats_remaining decrements on sched_wr_done_strobe (AW handshake) and drives the destination address and sched_wr_valid (writes are requested only while ISSUE beats remain).

- `r_write_beats_to_commit` decrements on `sched_wr_commit_strobe` (B response). A transfer is considered write-complete only when this counter reaches zero (`w_write_complete`), so a channel is never reported done until every issued burst has been acknowledged on the B channel.

6.4 FSM States

6.4.1 Figure 2.1.2: Scheduler FSM State Diagram



Source: 02_scheduler_fsm.mmd

6.4.2 State Encoding

Table 2.1.7: FSM State Encoding

State	Encoding	Description
IDLE	7'b0000001	Waiting for descriptor

State	Encoding	Description
PARSE	7'b0000010	Parsing descriptor fields
CH_XFER_DATA	7'b0000100	Concurrent read/write transfer
DONE	7'b0001000	Transfer complete, MonBus report
ERROR	7'b0010000	Error handling
TIMEOUT	7'b0100000	Timeout occurred
SOFT_RESET	7'b1000000	Soft reset handling

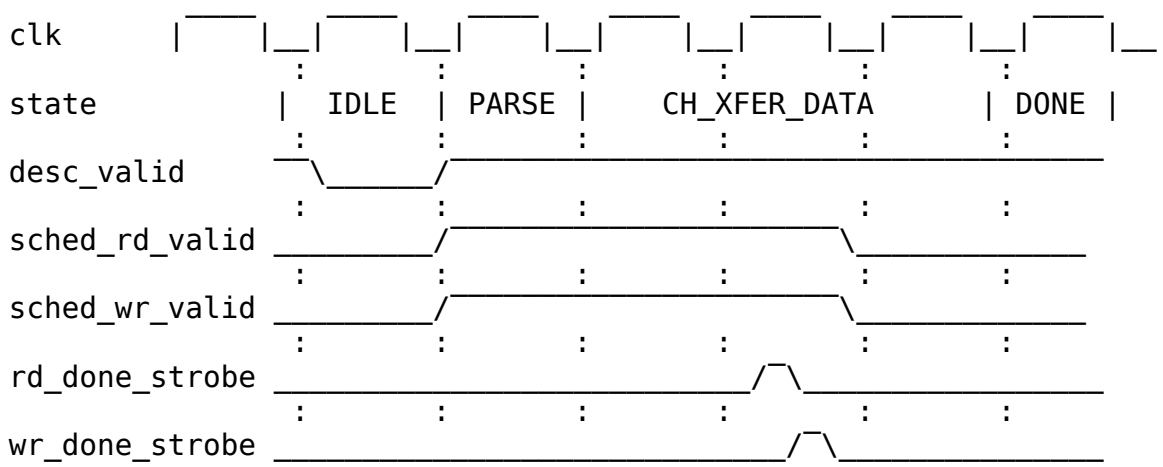
6.5 Operation

6.5.1 Descriptor Format

```
Bits [63:0]    - src_addr: Source memory address
Bits [127:64]  - dst_addr: Destination memory address
Bits [159:128] - length: Transfer length in beats
Bits [191:160] - next_ptr: Next descriptor pointer (0 = last)
Bits [195:192] - channel_id: Channel identifier
Bits [196]     - last: Last descriptor flag
Bits [197]     - gen_irq: Generate interrupt
Bits [255:198] - reserved
```

6.5.2 Transfer Sequence

6.5.3 Figure 2.1.3: Basic Transfer Timing



TODO: Replace with simulation-generated waveform

6.6 Error Handling

Table 2.1.8: Error Handling

Error Source	Detection	Response
Descriptor engine	`descriptor_error`	Transition to CH_ERROR state
Read engine	`sched_rd_error`	Set sticky error flag
Write engine	`sched_wr_error`	Set sticky error flag
Write-progress timeout	Recoverable, escalates after limit	See below

6.6.1 Recoverable Write-Progress Timeout

The write-progress watchdog is recoverable and only escalates to a fatal fault after a configurable number of consecutive stalled windows:

1. **Window counting:** `r_timeout_counter` increments while `sched_wr_valid && !sched_wr_ready` (a write request is stalled). The window expires when it reaches `cfg_sched_timeout_cycles`.
2. **Re-arm:** On expiry the counter re-arms (resets to 0) and the scheduler keeps waiting; a single expired window is not fatal.
3. **Strike counting:** `r_timeout_strikes` (8-bit) increments once per expired window. Any write progress – `sched_wr_done_strobe` (AW issue) or `sched_wr_commit_strobe` (B response) – clears both the window counter and the strike counter.
4. **Escalation:** When `cfg_sched_timeout_limit != 0` and `r_timeout_strikes >= cfg_sched_timeout_limit`, the channel escalates to a sticky `CH_ERROR` (surfaced on `sched_error`). Total time to escalate is approximately `cfg_sched_timeout_limit * cfg_sched_timeout_cycles`.
5. **Never-escalate mode:** `cfg_sched_timeout_limit == 0` is a pure soft timeout – the event may be reported, but the channel never faults.

`scheduler_idle` excludes `CH_ERROR`. Only `CH_IDLE` (with no active channel reset) is reported idle; a faulted channel in `CH_ERROR` is deliberately NOT reported idle so a wedged channel cannot read back as idle. The fault is surfaced via `sched_error` instead.

6.7 MonBus Events

Table 2.1.9: MonBus Event Codes

Event Code	Description
0x01	Descriptor received
0x02	Transfer started
0x03	Transfer complete
0x04	Error occurred
0x05	Timeout

Last Updated: 2026-07-02

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7 Descriptor Engine Specification

Module: `descriptor_engine.sv` **Location:** `projects/components/rapids/rtl/fub_beats/`
Status: Implemented **Last Updated:** 2025-01-10

7.1 Overview

The Descriptor Engine fetches and manages 256-bit descriptors from memory via AXI4. It maintains a prefetch FIFO to hide memory latency and provides parsed descriptors to the scheduler.

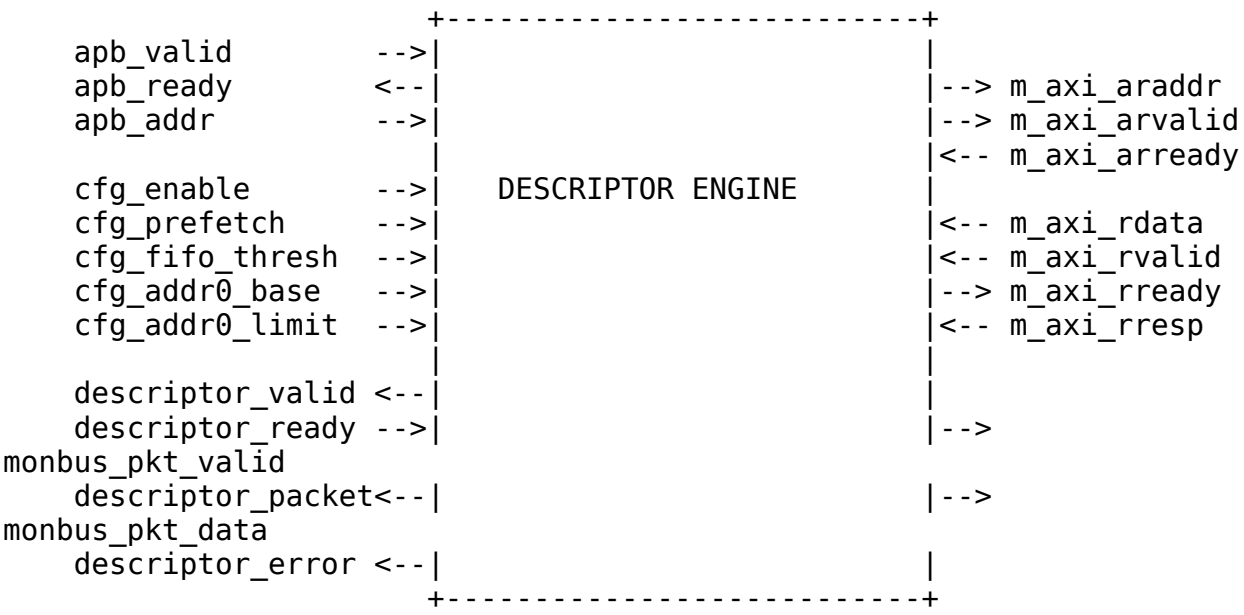
7.1.1 Key Features

- **AXI4 Descriptor Fetch:** Single-beat 256-bit reads
- **Prefetch FIFO:** Configurable depth for latency hiding
- **Address Range Checking:** Validates descriptor addresses
- **Error Detection:** Address violations, AXI errors

- **MonBus Integration:** Fetch events and error reporting

7.1.2 Block Diagram

7.1.3 Figure 2.2.1: Descriptor Engine Block Diagram



Source: 02_descriptor_engine_block.mmd

7.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 8;                  // Total channels
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int FIFO_DEPTH = 4;                    // Prefetch FIFO
depth

// Monitor Bus Parameters
parameter logic [7:0] MON_AGENT_ID = 8'h10;      // Descriptor Engine
Agent ID
parameter logic [3:0] MON_UNIT_ID = 4'h1;        // Unit identifier
```

: Table 2.2.1: Descriptor Engine Parameters

7.3 Port List

7.3.1 Clock and Reset

Table 2.2.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

7.3.2 APB Programming Interface

Table 2.2.3: APB Programming Interface

Signal	Direction	Width	Description
`apb_valid`	input	1	Kick-off valid (start descriptor chain)
`apb_ready`	output	1	Ready to accept kick-off
`apb_addr`	input	AW	First descriptor address

7.3.3 Configuration Interface

Table 2.2.4: Configuration Interface

Signal	Direction	Width	Description
`cfg_prefetch_enable`	input	1	Enable prefetch chaining (0 = on-demand, ~1 ahead)
`cfg_fifo_threshold`	input	4	Prefetch depth (descriptors buffered ahead when enabled)
`cfg_addr0_base`	input	AW	Valid address range 0 base
`cfg_addr0_limit`	input	AW	Valid address range 0 limit

Signal	Direction	Width	Description
cfg_addr1_base	input	AW	Valid address range 1 base
cfg_addr1_limit	input	AW	Valid address range 1 limit

7.3.4 AXI4 Read Master Interface

Table 2.2.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
m_axi_araddr	output	AW	Read address
m_axi_arvalid	output	1	Address valid
m_axi_arready	input	1	Address ready
m_axi_rdata	input	256	Read data (descriptor)
m_axi_rvalid	input	1	Data valid
m_axi_rready	output	1	Data ready
m_axi_rresp	input	2	Read response

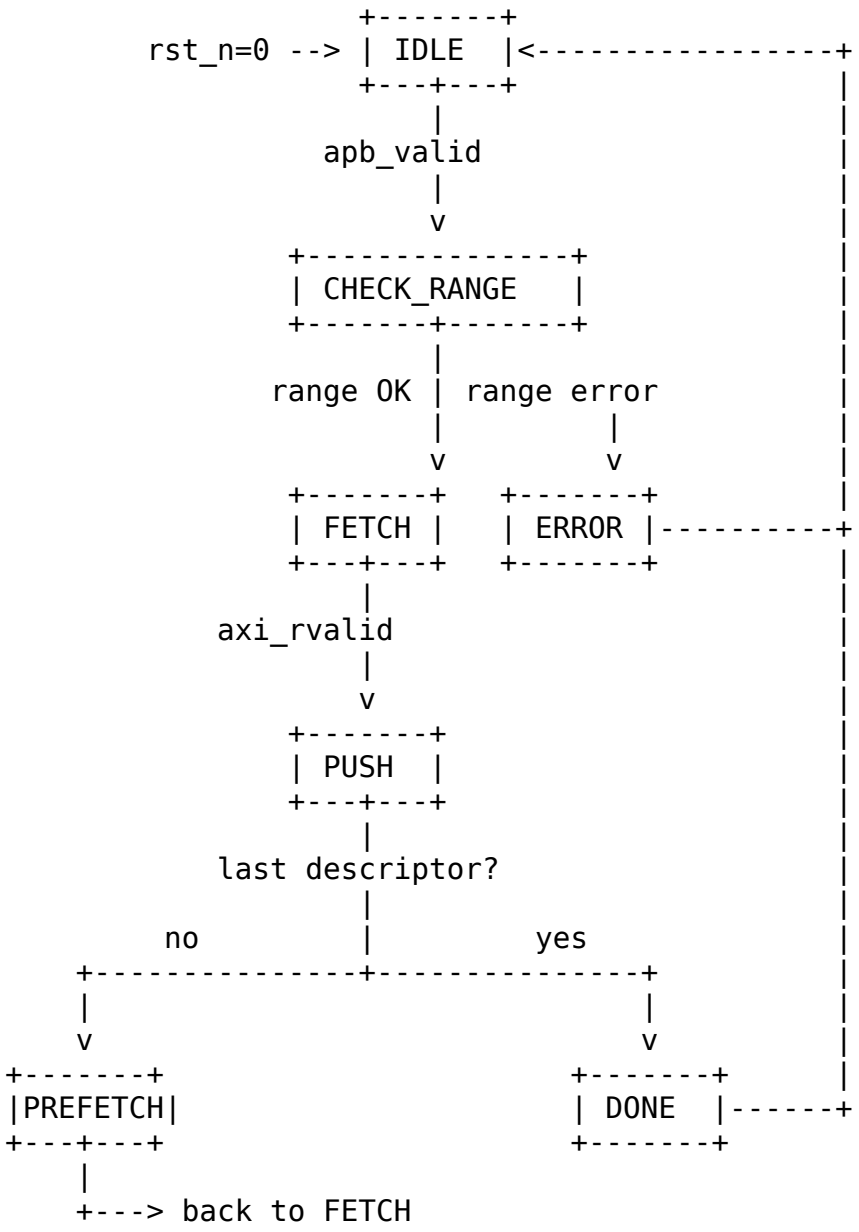
7.3.5 Scheduler Interface

Table 2.2.6: Scheduler Interface

Signal	Direction	Width	Description
descriptor_valid	output	1	Descriptor valid
descriptor_ready	input	1	Scheduler ready
descriptor_packet	output	256	Parsed descriptor
descriptor_error	output	1	Error flag

7.4 FSM States

7.4.1 Figure 2.2.2: Descriptor Engine FSM

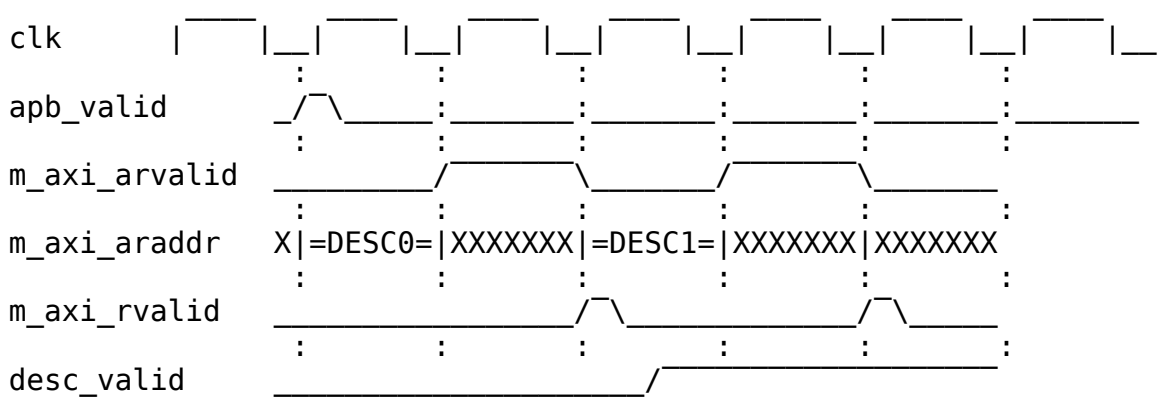


Source: 02_descriptor_engine_fsm.mmd

7.5 Operation

7.5.1 Descriptor Chain Processing

7.5.2 Figure 2.2.3: Descriptor Chain Timing



TODO: Replace with simulation-generated waveform

7.6 Address Range Checking

Descriptors must fall within configured valid address ranges:

```
Valid if: (addr >= cfg_addr0_base && addr <= cfg_addr0_limit)
       || (addr >= cfg_addr1_base && addr <= cfg_addr1_limit)
```

Invalid addresses generate an error and halt the descriptor chain.

7.7 Prefetch and Autonomous Chaining

`cfg_prefetch_enable` and `cfg_fifo_threshold` are functional prefetch controls (previously declared-but-unused, which left chaining permanently greedy). They throttle how far ahead the engine fetches chained descriptors by comparing the output descriptor FIFO occupancy (`w_desc_fifo_count`) against a computed `w_prefetch_limit`:

```
if      (!cfg_prefetch_enable)      w_prefetch_limit = 1;   // on-
demand
else if (cfg_fifo_threshold == 0)   w_prefetch_limit = 1;   // guard
(0 would stall)
else                                w_prefetch_limit =
cfg_fifo_threshold;
```

```
w_prefetch_allows = (w_desc_fifo_count < w_prefetch_limit);
```

- **Prefetch OFF (`cfg_prefetch_enable = 0`):** limit forced to 1, so a chained fetch is only allowed when the output FIFO has drained – the engine stays roughly one descriptor ahead of the scheduler (on-demand).
- **Prefetch ON:** the engine buffers up to `cfg_fifo_threshold` descriptors ahead.

7.7.1 Chaining Decision

A chain fetch issues immediately when all of the following hold:

```
w_should_chain = w_chain_eligible     // valid non-last descriptor, in
range, no error
               && w_desc_committed      // current descriptor accepted
into output FIFO
               && w_prefetch_allows     // occupancy below prefetch
limit
               && w_desc_addr_fifo_wr_ready;  // address FIFO has a
slot
```

7.7.2 Deferred (Throttle-Blocked) Chain Fetch

If a descriptor commits and is chain-eligible but the fetch cannot issue this cycle (throttled by `w_prefetch_allows`, or the address FIFO is full), the owed fetch is recorded rather than dropped:

- `r_chain_pending` is set and the next-descriptor pointer is held in `r_pending_chain_addr`.
- The deferred fetch is reissued (`w_pending_push_fire`) once the output FIFO drains enough that `w_prefetch_allows` is true again and the address FIFO has a slot, clearing `r_chain_pending`.

Last Updated: 2026-07-02

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8 AXI Read Engine Specification

Module: `axi_read_engine.sv` **Location:** `projects/components/rapids/rtl/fub_beats/`
Status: Implemented **Last Updated:** 2025-01-10

8.1 Overview

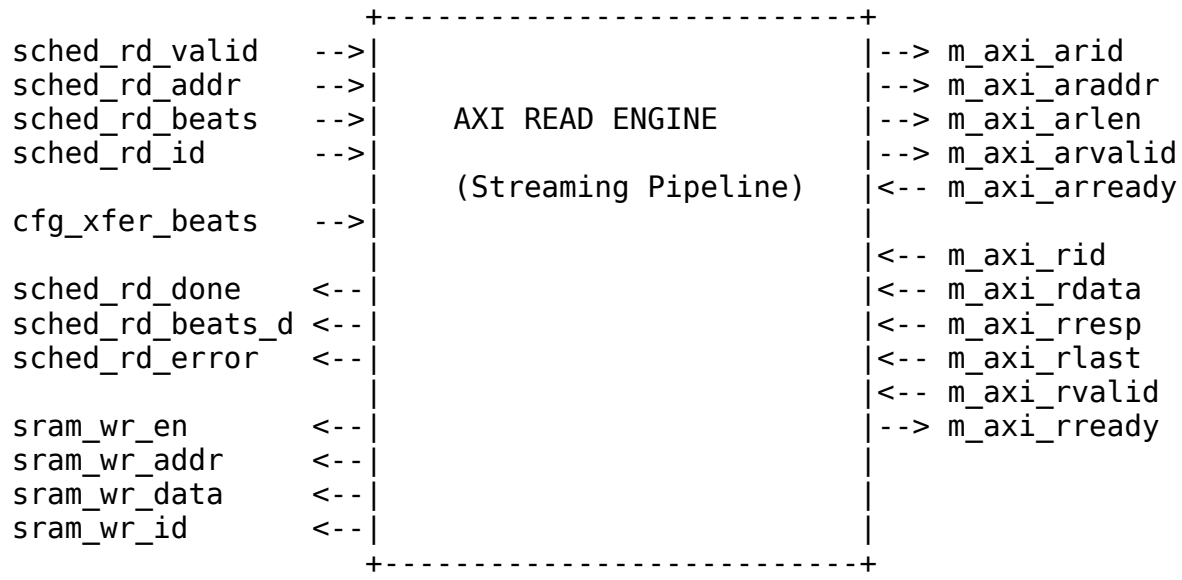
The AXI Read Engine performs burst reads from system memory and writes data to the SRAM buffer. It operates as a streaming pipeline with no FSM, maximizing throughput through continuous data flow.

8.1.1 Key Features

- **Streaming Pipeline:** No FSM - pure streaming architecture
- **Multi-Channel Support:** Channel ID passed through AXI ID field
- **Configurable Burst Length:** Up to 256 beats per burst
- **Latency Compensation:** Integrated beats_latency_bridge
- **Error Handling:** AXI RRESP error detection and reporting

8.1.2 Block Diagram

8.1.3 Figure 2.3.1: AXI Read Engine Block Diagram



Source: 02_axi_read_engine_block.mmd

8.2 Parameters

```
parameter int NUM_CHANNELS = 8;                    // Number of channels
parameter int ADDR_WIDTH = 64;                     // Address bus width
parameter int DATA_WIDTH = 512;                    // Data bus width
parameter int AXI_ID_WIDTH = 8;                    // AXI ID width
parameter int MAX_OUTSTANDING = 8;                 // Max outstanding AR
transactions
parameter int PIPELINE = 0;                        // Pipeline stages

// Derived
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS);
```

: Table 2.3.1: AXI Read Engine Parameters

8.3 Port List

8.3.1 Clock and Reset

Table 2.3.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

8.3.2 Scheduler Interface

Table 2.3.3: Scheduler Interface

Signal	Direction	Width	Description
`sched_rd_valid`	input	1	Read request valid
`sched_rd_addr`	input	AW	Source address
`sched_rd_beats`	input	32	Total beats to read
`sched_rd_id`	input	CW	Channel ID
`sched_rd_done_strobe`	output	1	Burst complete strobe
`sched_rd_beats_done`	output	32	Beats completed

Signal	Direction	Width	Description
sched_rd_error	output	1	Error flag

8.3.3 AXI4 Read Master Interface

Table 2.3.4: AXI4 Read Master Interface

Signal	Direction	Width	Description
m_axi_arid	output	IW	Read address ID
m_axi_araddr	output	AW	Read address
m_axi_arlen	output	8	Burst length (beats - 1)
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type (INCR)
m_axi_arvalid	output	1	Address valid
m_axi_arready	input	1	Address ready
m_axi_rid	input	IW	Read data ID
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Read response
m_axi_rlast	input	1	Last beat
m_axi_rvalid	input	1	Data valid
m_axi_rready	output	1	Data ready

8.3.4 SRAM Write Interface

Table 2.3.5: SRAM Write Interface

Signal	Direction	Width	Description
sram_wr_en	output	1	SRAM write enable
sram_wr_addr	output	AW	SRAM write address
sram_wr_data	output	DW	SRAM write data
sram_wr_id	output	CW	Channel ID for

Signal	Direction	Width	Description
			data

8.4 Operation

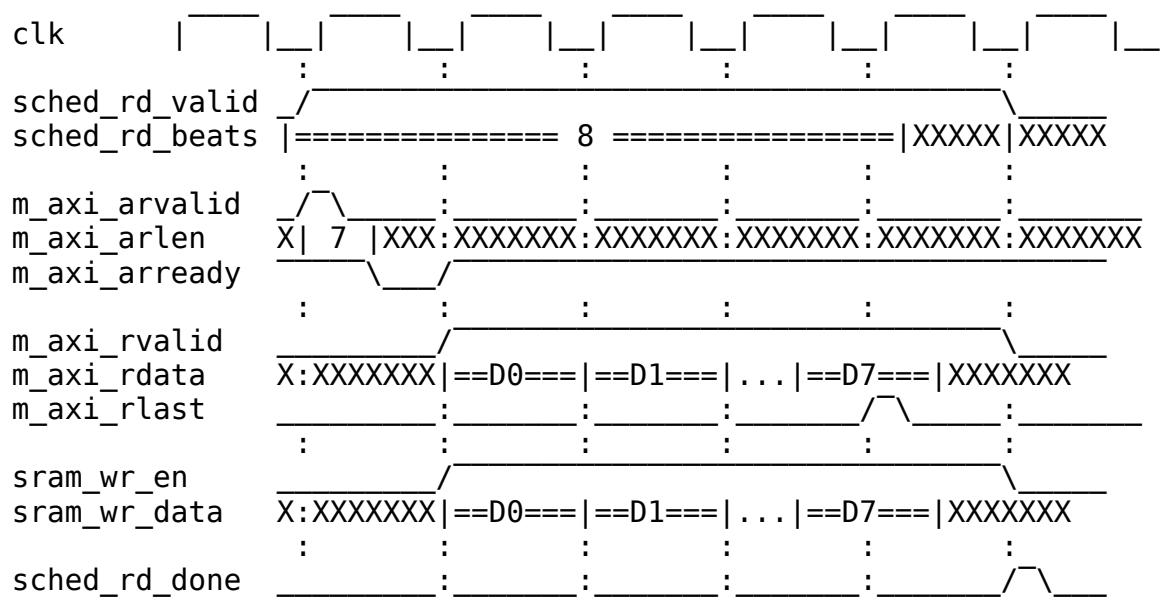
8.4.1 Streaming Pipeline Architecture

The AXI read engine uses a **streaming pipeline** with no FSM:

```
Scheduler Request --> AR Channel --> R Channel --> SRAM Write
      |                   |             |              |
      +--- beat count     +-- addr      +-- data       +-- wr_en
           tracking           issue         receive        generate
```

8.4.2 Timing Diagram

8.4.3 Figure 2.3.2: AXI Read Burst Timing



TODO: Replace with simulation-generated waveform

8.5 Burst Segmentation

Large transfers are segmented into AXI-compliant bursts:

```
Total beats = 256
cfg_xfer_beats = 64
```

```
Bursts issued:
  AR[0]: addr=0x1000, len=63 (64 beats)
  AR[1]: addr=0x2000, len=63 (64 beats)
  AR[2]: addr=0x3000, len=63 (64 beats)
  AR[3]: addr=0x4000, len=63 (64 beats)
```

8.6 Error Handling

Table 2.3.6: Error Handling

Error	Detection	Response
AXI SLVERR	`m_axi_rresp == 2'b10`	Set `sched_rd_error`, continue
AXI DECERR	`m_axi_rresp == 2'b11`	Set `sched_rd_error`, continue

Last Updated: 2025-01-10

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9 AXI Write Engine Specification

Module: `axi_write_engine.sv` **Location:** `projects/components/rapids/rtl/fub_beats/`
Status: Implemented **Last Updated:** 2025-01-10

9.1 Overview

The AXI Write Engine reads data from SRAM and performs burst writes to system memory. It operates as a streaming pipeline and includes write response handling for completion tracking.

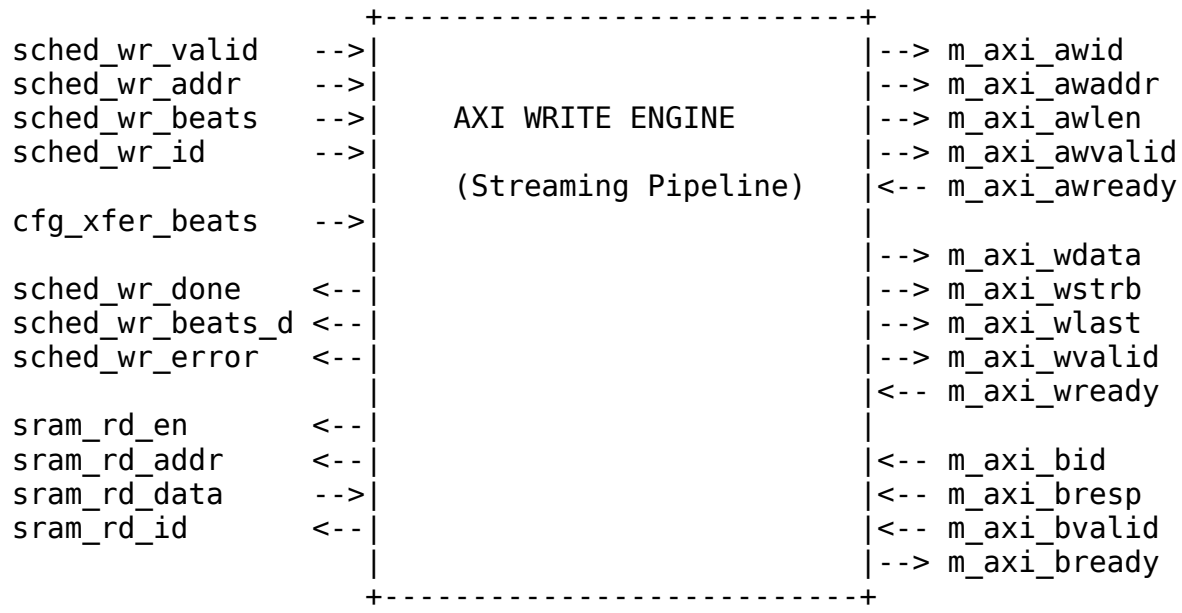
9.1.1 Key Features

- **Streaming Pipeline:** No FSM - pure streaming architecture
- **Multi-Channel Support:** Channel ID in AXI ID field

- **Write Response Tracking:** B channel completion handling
- **Configurable Burst Length:** Up to 256 beats per burst
- **Latency Compensation:** Integrated beats_latency_bridge

9.1.2 Block Diagram

9.1.3 Figure 2.4.1: AXI Write Engine Block Diagram



Source: 02_axi_write_engine_block.mmd

9.2 Parameters

```
parameter int NUM_CHANNELS = 8;                 // Number of channels
parameter int ADDR_WIDTH = 64;                  // Address bus width
parameter int DATA_WIDTH = 512;                 // Data bus width
parameter int AXI_ID_WIDTH = 8;                 // AXI ID width
parameter int MAX_OUTSTANDING = 8;              // Max outstanding AW
transactions
parameter int W_FIFO_DEPTH = 64;                // Write data FIFO
depth
parameter int B_FIFO_DEPTH = 16;                // Write response
FIFO depth
parameter int PIPELINE = 0;                     // Pipeline stages

// Derived
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS);
```

: Table 2.4.1: AXI Write Engine Parameters

9.3 Port List

9.3.1 Clock and Reset

Table 2.4.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

9.3.2 Scheduler Interface

Table 2.4.3: Scheduler Interface

Signal	Direction	Width	Description
`sched_wr_valid`	input	1	Write request valid
`sched_wr_addr`	input	AW	Destination address
`sched_wr_beats`	input	32	Total beats to write
`sched_wr_id`	input	CW	Channel ID
`sched_wr_done_strobe`	output	NC	AW-issue strobe (pulses on AW handshake)
`sched_wr_beats_done`	output	NC*32	Beats issued this strobe (`awlen + 1`)
`sched_wr_commit_strobe`	output	NC	COMMIT strobe (pulses on B response)
`sched_wr_commit_beats`	output	NC*32	Beats committed this strobe
`sched_wr_error`	output	1	Error flag

Two completion strobes per channel. The engine reports write progress to the scheduler at two points, letting the scheduler gate completion on data actually landing in memory:

- `sched_wr_done_strobe` / `sched_wr_beats_done` pulse when the **AW command handshakes** (`m_axi_awvalid && m_axi_awready`); the beat count is taken directly from `m_axi_awlen + 1`. This advances the scheduler's destination address (ISSUE tracking).
- `sched_wr_commit_strobe` / `sched_wr_commit_beats` pulse when the matching **B response** arrives (`m_axi_bvalid && m_axi_bready` for the channel); the beat count is recovered from the per-channel B-phase transaction FIFO. This is the COMMIT the scheduler uses to declare the transfer complete.

9.3.3 AXI4 Write Master Interface

Table 2.4.4: AXI4 Write Master Interface

Signal	Direction	Width	Description
`m_axi_awid`	output	IW	Write address ID
`m_axi_awaddr`	output	AW	Write address
`m_axi_awlen`	output	8	Burst length (beats - 1)
`m_axi_awsize`	output	3	Burst size
`m_axi_awburst`	output	2	Burst type (INCR)
`m_axi_awvalid`	output	1	Address valid
`m_axi_awready`	input	1	Address ready
`m_axi_wdata`	output	DW	Write data
`m_axi_wstrb`	output	DW/8	Write strobes
`m_axi_wlast`	output	1	Last beat
`m_axi_wvalid`	output	1	Data valid
`m_axi_wready`	input	1	Data ready
`m_axi_bid`	input	IW	Response ID
`m_axi_bresp`	input	2	Write response
`m_axi_bvalid`	input	1	Response valid
`m_axi_bready`	output	1	Response ready

9.3.4 SRAM Read Interface

Table 2.4.5: SRAM Read Interface

Signal	Direction	Width	Description
sram_rd_en	output	1	SRAM read enable
sram_rd_addr	output	AW	SRAM read address
sram_rd_data	input	DW	SRAM read data
sram_rd_id	output	CW	Channel ID for read

9.4 Operation

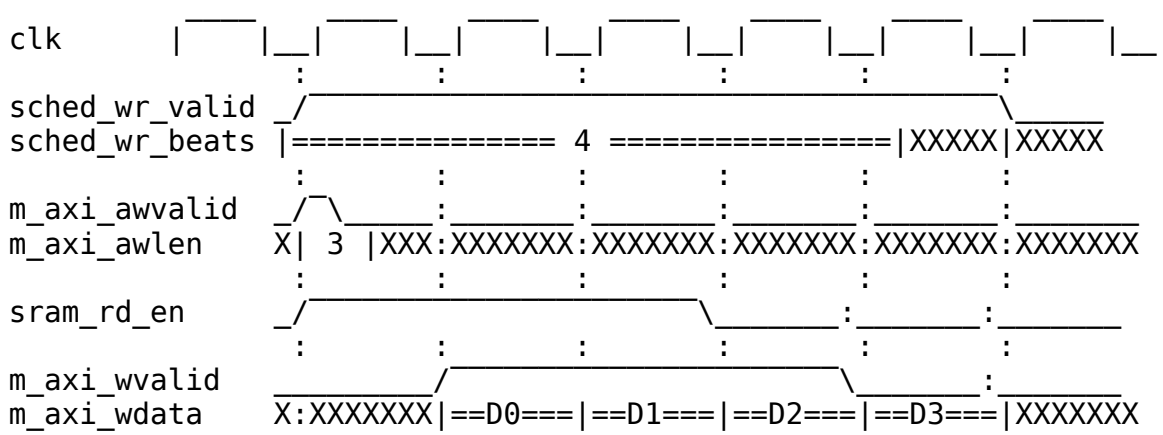
9.4.1 Write Transaction Phases

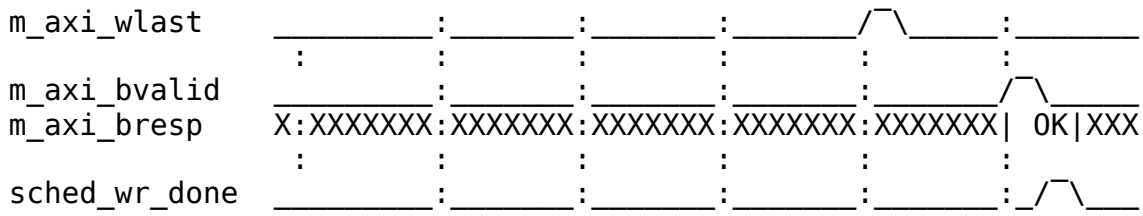
AXI writes have three phases that can overlap:

```
Phase 1: AW (Address)    Phase 2: W (Data)    Phase 3: B (Response)
    |                        |                       |
    +-- Issue address        +-- Stream data         +-- Receive
response
        before data              from SRAM               track
completion
```

9.4.2 Timing Diagram

9.4.3 Figure 2.4.2: AXI Write Burst Timing





TODO: Replace with simulation-generated waveform

9.5 Error Handling

Table 2.4.6: Error Handling

Error	Detection	Response
AXI SLVERR	`m_axi_bresp == 2'b10`	Set `sched_wr_error`
AXI DECERR	`m_axi_bresp == 2'b11`	Set `sched_wr_error`

Last Updated: 2026-07-02

---

10 Beats Alloc Control Specification

Module: `beats_alloc_ctrl.sv` **Location:** `projects/components/rapids/rtl/fub_beats/`
Status: Implemented **Last Updated:** 2025-01-10

10.1 Overview

The Beats Alloc Control is a “virtual FIFO” that tracks space allocation without storing actual data. It uses FIFO pointer logic to manage pre-allocation for AXI engines, preventing race conditions between allocation and actual data arrival.

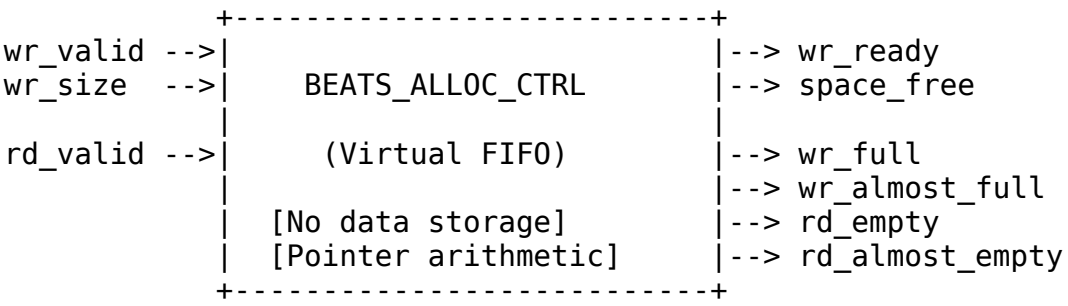
10.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage

- **Variable-Size Allocation:** Supports multi-beat allocation requests
- **Single-Beat Release:** Data writes release one beat at a time
- **Status Flags:** Full, almost full, empty, almost empty
- **Space Tracking:** Reports available space for flow control

10.1.2 Block Diagram

10.1.3 Figure 2.5.1: Beats Alloc Control Block Diagram



Source: 02_beats_alloc_ctrl_block.mmd

10.2 Concept: Virtual FIFO

Unlike a traditional FIFO that stores data, beats_alloc_ctrl only tracks **space reservations**:

```
Traditional FIFO:          Virtual FIFO (alloc_ctrl):

  wr_data --> [storage] -> rd_data    wr_size --> [pointers] -->
space_free

  Data moves through FIFO             Only pointers move, no data
```

10.2.1 Use Case: Pre-Allocation for AXI Reads

```
1. AXI engine requests N beats of space
   - alloc_ctrl.wr_valid = 1
   - alloc_ctrl.wr_size = N
   - wr_ptr advances by N

2. As AXI read data arrives (1 beat at a time)
   - alloc_ctrl.rd_valid = 1 (each beat)
   - rd_ptr advances by 1

3. Space becomes available again
   - space_free reflects available capacity
```

10.3 Parameters

```
parameter int DEPTH = 512;                    // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;           // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;           // Almost empty margin
parameter int REGISTERED = 1;                 // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);             // Address width
```

: Table 2.5.1: Beats Alloc Control Parameters

10.4 Port List

10.4.1 Clock and Reset

Table 2.5.2: Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low reset

10.4.2 Write Interface (Allocation Requests)

Table 2.5.3: Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	Allocate space
wr_size	input	8	Number of beats to allocate
wr_ready	output	1	Space available for allocation

10.4.3 Read Interface (Data Written)

Table 2.5.4: Read Interface

Signal	Direction	Width	Description
`rd_valid`	input	1	One beat of data written
`rd_ready`	output	1	Not full (can accept)

10.4.4 Status Outputs

Table 2.5.5: Status Outputs

Signal	Direction	Width	Description
`space_free`	output	AW+1	Available space in beats
`wr_full`	output	1	No space available
`wr_almost_full`	output	1	Near full
`rd_empty`	output	1	No allocations pending
`rd_almost_empty`	output	1	Near empty

10.5 Operation

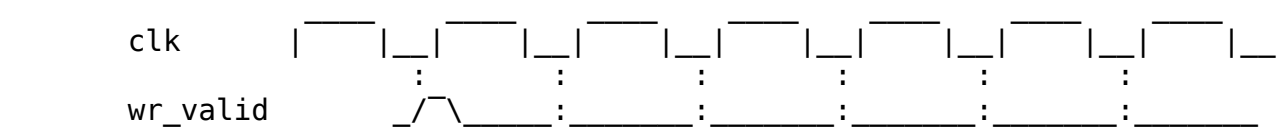
10.5.1 Pointer Arithmetic

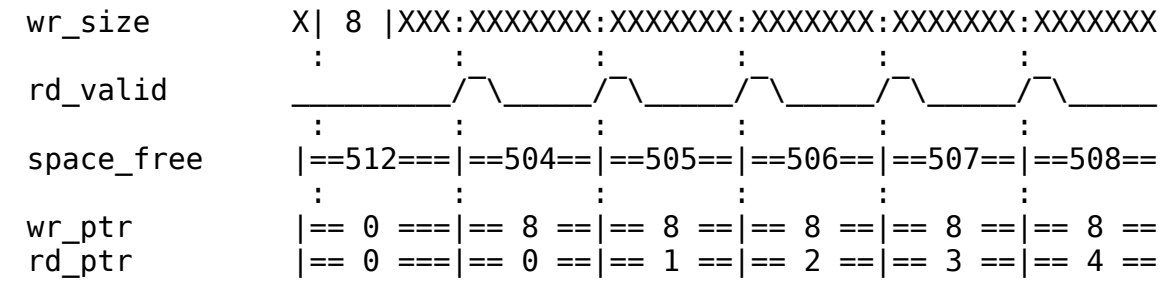
```
space_free = DEPTH - (wr_ptr - rd_ptr)

Full condition:    wr_ptr - rd_ptr >= DEPTH
Empty condition:   wr_ptr == rd_ptr
```

10.5.2 Timing Diagram

10.5.3 Figure 2.5.2: Allocation and Release Timing





TODO: Replace with simulation-generated waveform showing actual flow control

---

10.6 Integration Example

```
// Source path: Pre-allocate SRAM space before AXI read
beats_alloc_ctrl #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_WR_MARGIN(8)
) u_src_alloc (
    .axi_aclk       (clk),
    .axi_aresetn    (rst_n),

    // Scheduler allocates space before issuing AXI read
    .wr_valid       (sched_rd_valid),
    .wr_size        (sched_rd_beats[7:0]),
    .wr_ready       (alloc_ready),

    // AXI read data arrival releases allocation
    .rd_valid       (m_axi_rvalid && m_axi_rready),
    .rd_ready       (),  // Not used for single-beat

    // Status
    .space_free     (sram_space_available),
    .wr_full        (sram_full),
    .wr_almost_full (sram_almost_full)
);
```

Last Updated: 2025-01-10

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11 Beats Drain Control Specification

Module: `beats_drain_ctrl.sv` **Location:** `projects/components/rapids/rtl/fub_beats/`
Status: Implemented **Last Updated:** 2025-01-10

11.1 Overview

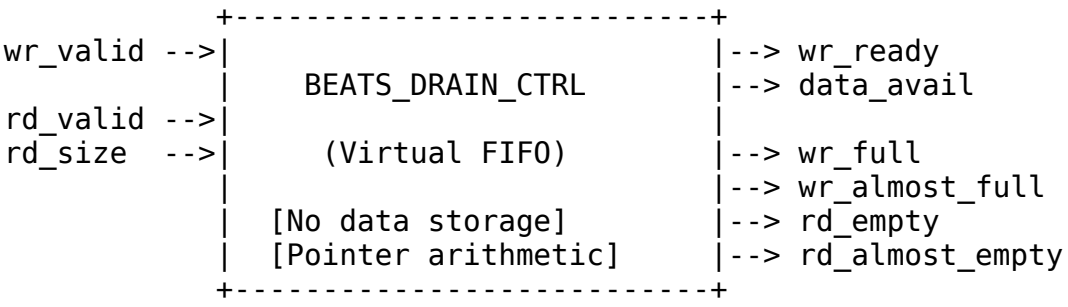
The Beats Drain Control is a "virtual FIFO" that tracks data availability without storing actual data. It complements beats_alloc_ctrl by tracking when data has been written to SRAM and is available for draining.

11.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Single-Beat Write:** Data arrivals tracked one beat at a time
- **Variable-Size Drain:** Supports multi-beat drain requests
- **Status Flags:** Full, almost full, empty, almost empty
- **Data Availability:** Reports beats available for drain

11.1.2 Block Diagram

11.1.3 Figure 2.6.1: Beats Drain Control Block Diagram



Source: 02_beats_drain_ctrl_block.mmd

11.2 Concept: Data Availability Tracking

The drain_ctrl tracks when data has been written to SRAM:

```
alloc_ctrl:  Tracks SPACE reservations (allocation -> actual write)
drain_ctrl:  Tracks DATA availability (data write -> drain request)
```

11.2.1 Complementary Roles

```
Source Path:
  1. Scheduler allocates space (alloc_ctrl.wr)
  2. AXI reads data into SRAM
  3. Data written to SRAM (drain_ctrl.wr)
  4. Drain consumer reads (drain_ctrl.rd)

Sink Path:
  1. Fill allocates space (alloc_ctrl.wr)
  2. Fill writes to SRAM (alloc_ctrl.rd)
  3. Data available in SRAM (drain_ctrl.wr)
  4. AXI write drains SRAM (drain_ctrl.rd)
```

11.3 Parameters

```
parameter int DEPTH = 512;                    // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;           // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;           // Almost empty margin
parameter int REGISTERED = 1;                 // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);             // Address width
```

: Table 2.6.1: Beats Drain Control Parameters

11.4 Port List

11.4.1 Clock and Reset

Table 2.6.2: Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low reset

11.4.2 Write Interface (Data Written to SRAM)

Table 2.6.3: Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	One beat

Signal	Direction	Width	Description
			written to SRAM
wr_ready	output	1	Can track more data

11.4.3 Read Interface (Drain Requests)

Table 2.6.4: Read Interface

Signal	Direction	Width	Description
rd_valid	input	1	Drain request
rd_size	input	8	Number of beats to drain
rd_ready	output	1	Data available for drain

11.4.4 Status Outputs

Table 2.6.5: Status Outputs

Signal	Direction	Width	Description
data_avail	output	AW+1	Beats available to drain
wr_full	output	1	Tracking full
wr_almost_full	output	1	Near full
rd_empty	output	1	No data to drain
rd_almost_empty	output	1	Near empty

11.5 Operation

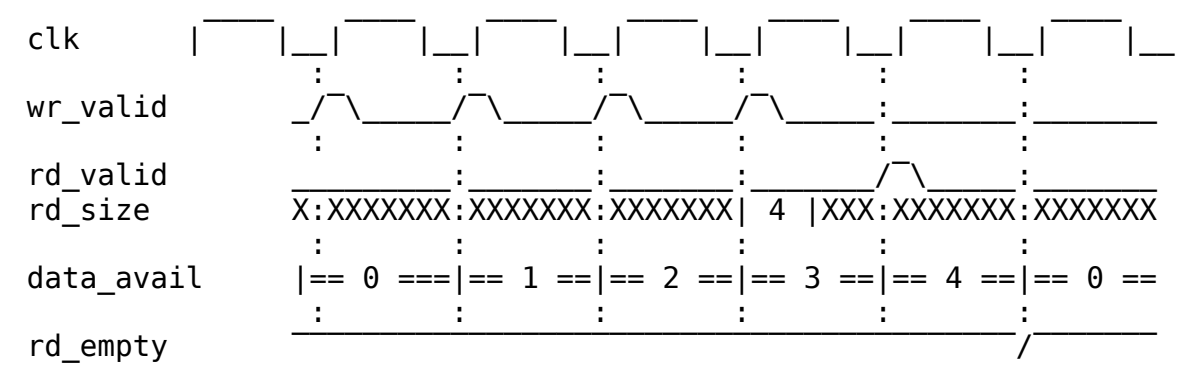
11.5.1 Pointer Arithmetic

```
data_avail = wr_ptr - rd_ptr

Full condition:    wr_ptr - rd_ptr >= DEPTH
Empty condition:   wr_ptr == rd_ptr
```

11.5.2 Timing Diagram

11.5.3 Figure 2.6.2: Data Arrival and Drain Timing



TODO: Replace with simulation-generated waveform showing actual flow control

---

11.6 Integration Example

```
// Sink path: Track data availability for AXI write engine
beats_drain_ctrl #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_RD_MARGIN(8)
) u_snk_drain (
    .axi_aclk       (clk),
    .axi_aresetn    (rst_n),

    // Fill writes data to SRAM
    .wr_valid       (fill_valid && fill_ready),
    .wr_ready       (),  // Single-beat, always ready

    // AXI write engine drains data
    .rd_valid       (axi_wr_drain_req),
    .rd_size        (axi_wr_burst_len),
    .rd_ready       (data_ready_for_drain),

    // Status
    .data_avail     (beats_to_write),
    .rd_empty       (no_data_to_write)
);
```

11.7 Relationship: alloc_ctrl vs drain_ctrl

Table 2.6.6: alloc_ctrl vs drain_ctrl Comparison

Aspect	beats_alloc_ctrl	beats_drain_ctrl
Tracks	Space reservations	Data availability
Write port	Multi-beat allocation	Single-beat arrival
Read port	Single-beat release	Multi-beat drain
Output	`space_free`	`data_avail`
Use case	Pre-allocate before fill	Know when to drain

Last Updated: 2025-01-10

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12 Beats Latency Bridge Specification

Module: `beats_latency_bridge.sv` **Location:** `projects/components/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

12.1 Overview

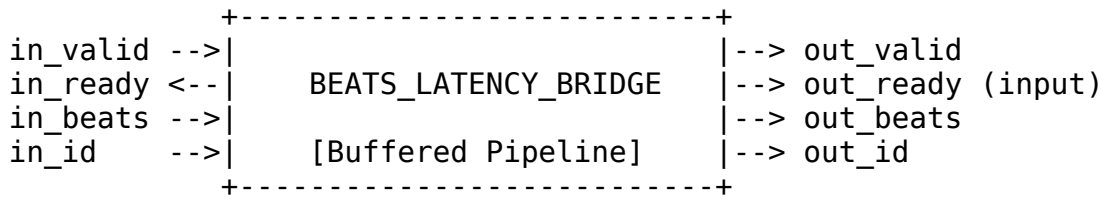
The Beats Latency Bridge provides buffering to compensate for pipeline latency between alloc_ctrl and drain_ctrl in the SRAM controller. It prevents race conditions where drain signals arrive before allocation is complete.

12.1.1 Key Features

- **Latency Compensation:** Buffers requests to match pipeline delays
- **Configurable Depth:** Matches expected pipeline latency
- **Flow Control:** Standard valid/ready handshaking
- **Pass-Through Data:** Preserves beat count and channel ID

12.1.2 Block Diagram

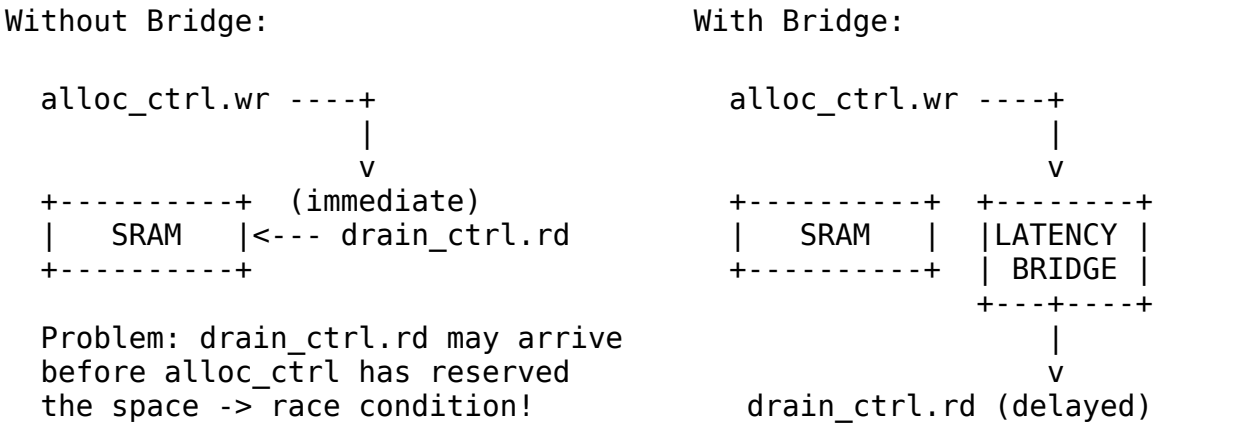
12.1.3 Figure 2.7.1: Beats Latency Bridge Block Diagram



Source: 02_beats_latency_bridge_block.mmd

12.2 Concept: Why Latency Compensation?

The SRAM controller has a timing challenge:



12.3 Parameters

```
parameter int DEPTH = 4;                    // Bridge FIFO depth
parameter int BEATS_WIDTH = 8;              // Beat count width
parameter int ID_WIDTH = 3;                 // Channel ID width
parameter int REGISTERED = 1;               // Registered outputs
```

: Table 2.7.1: Beats Latency Bridge Parameters

12.4 Port List

12.4.1 Clock and Reset

Table 2.7.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

12.4.2 Input Interface

Table 2.7.3: Input Interface

Signal	Direction	Width	Description
`in_valid`	input	1	Input request valid
`in_ready`	output	1	Bridge ready to accept
`in_beats`	input	BEATS_WIDTH	Beat count
`in_id`	input	ID_WIDTH	Channel ID

12.4.3 Output Interface

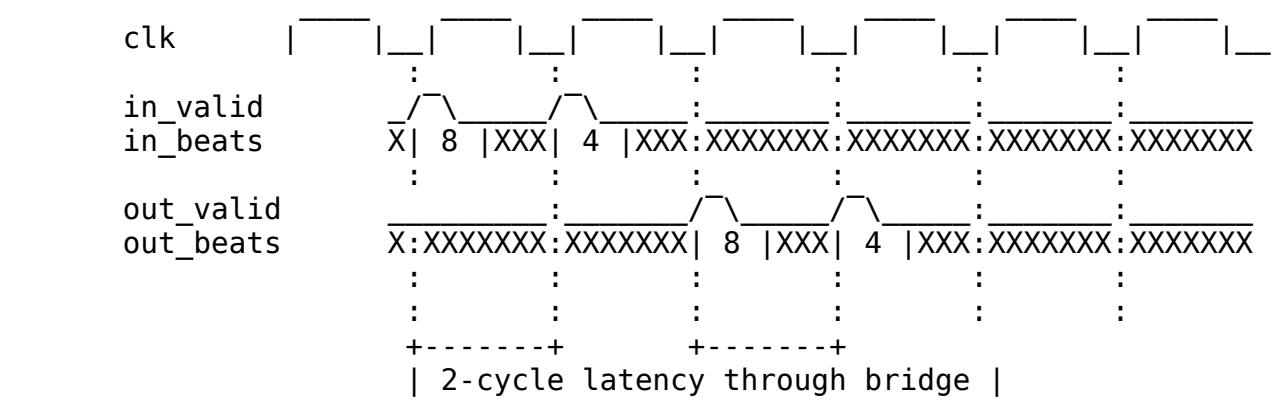
Table 2.7.4: Output Interface

Signal	Direction	Width	Description
`out_valid`	output	1	Output request valid
`out_ready`	input	1	Downstream ready
`out_beats`	output	BEATS_WIDTH	Beat count (delayed)
`out_id`	output	ID_WIDTH	Channel ID (delayed)

12.5 Operation

12.5.1 Timing Diagram

12.5.2 Figure 2.7.2: Latency Bridge Timing (Depth=2)



TODO: Replace with simulation-generated waveform

12.6 Integration Context

The latency bridge is used within the SRAM controller:

```
// In snk_sram_controller_unit
beats_alloc_ctrl u_alloc (
    // Allocation on fill requests
    .wr_valid       (fill_valid),
    .wr_size        (fill_size),
    // Release on actual data write
    .rd_valid       (sram_wr_complete)
);

beats_latency_bridge u_bridge (
    // Input from SRAM write path
    .in_valid       (sram_wr_complete),
    .in_beats       (beats_written),
    // Output to drain controller (delayed)
    .out_valid      (drain_trigger),
    .out_beats      (drain_beats)
);

beats_drain_ctrl u_drain (
    // Delayed notification of data available
    .wr_valid       (drain_trigger),
```

```
    // Drain requests from write engine
    .rd_valid       (axi_drain_req),
    .rd_size        (axi_drain_size)
);
```

12.7 Design Considerations

Table 2.7.5: Bridge Depth Selection

Depth	Use Case	Latency
2	Minimal pipeline	2 cycles
4	Standard pipeline	4 cycles
8	Deep pipeline	8 cycles

The bridge depth should match the maximum pipeline latency from alloc_ctrl write to SRAM data being valid for drain_ctrl.

Last Updated: 2025-01-10

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13 Control-Read Engine Specification

Module: `ctrlrd_engine.sv` **Location:** `projects/components/rapids/rtl/fub/` **Status:** Implemented **Last Updated:** 2026-07-12

13.1 Overview

The Control-Read Engine implements the `CTRL_READ` descriptor opcode (`DESC_OP_CTRL_READ = 2'b01`) as a **consumer gate**. When the scheduler decodes a `CTRL_READ` descriptor, it hands the engine a poll address, an expected value, and a compare mask. The engine issues an AXI4 read to the poll address, applies the mask to the returned data, and compares `(read_data & mask)` against the expected value. If they match, the gate is satisfied and the descriptor chain proceeds; if not, the engine retries. Retries are bounded so a gate that never matches cannot hang the channel.

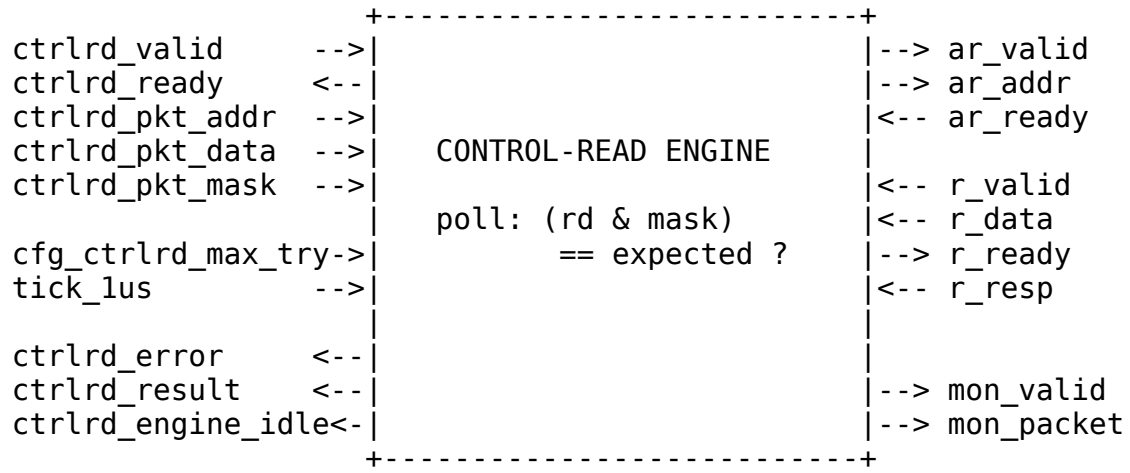
One Control-Read Engine is instantiated per channel in `scheduler_group_beats.sv` (`u_ctrlrd_engine`), configured for 32-bit AXI reads.

13.1.1 Key Features

- **Consumer Gate Semantics:** Polls memory until `(read_data & mask) == expected`
- **Masked Comparison:** Only the masked bits of the polled word participate in the match
- **Bounded Retry:** `cfg_ctrlrd_max_try` (0-511) caps the number of poll attempts
- **1 microsecond Retry Pacing:** Retries wait for `tick_1us` from the scheduler group
- **Error on Exhaustion:** Asserts `ctrlrd_error` if the gate never matches within budget
- **Null Address Fast-Path:** A poll address of zero completes immediately as a match
- **Channel Reset Support:** Drains cleanly on `cfg_channel_reset`
- **MonBus Integration:** Completion, retry, and error events

13.1.2 Block Diagram

13.1.3 Figure 2.8.1: Control-Read Engine Block Diagram



Source: 08_ctrlrd_engine_block.mmd

13.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 32;                 // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Channel index
width (derived)
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int AXI_DATA_WIDTH = 64;               // AXI read data
width (>= 32)
parameter int AXI_ID_WIDTH = 8;                  // AXI ID width (>=
CHAN_WIDTH)

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID  = 16'h0030; // Control-Read
Engine Agent ID
parameter logic [7:0]  MON_UNIT_ID   = 8'h02;    // Unit identifier
parameter logic [8:0]  MON_CHANNEL_ID = 9'h000;  // Base channel ID
```

: Table 2.8.1: Control-Read Engine Parameters

Parameter validation enforces `AXI_ID_WIDTH >= CHAN_WIDTH` and `AXI_DATA_WIDTH >= 32` (the engine consumes only the lower 32 bits of the read data). The per-channel instance in `scheduler_group_beats.sv` overrides `AXI_DATA_WIDTH` to 32.

13.3 Port List

13.3.1 Clock and Reset

Table 2.8.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

13.3.2 Scheduler Interface

Table 2.8.3: Scheduler Interface

Signal	Direction	Width	Description
`ctrlrd_valid`	input	1	Gate request valid from scheduler
`ctrlrd_ready`	output	1	Engine ready to accept a

Signal	Direction	Width	Description
			request
ctrlrd_pkt_addr	input	AW	Poll address (descriptor poll_addr[63:0])
ctrlrd_pkt_data	input	32	Expected value (descriptor expected[95:64])
ctrlrd_pkt_mask	input	32	Compare mask (descriptor mask[127:96])
ctrlrd_error	output	1	Gate failed (retries exhausted or AXI error)
ctrlrd_result	output	32	Last read value (lower 32 bits)
ctrlrd_engine_idle	output	1	Engine idle (post-issue completion indicator)

13.3.3 Configuration Interface

Table 2.8.4: Configuration Interface

Signal	Direction	Width	Description
cfg_ctrlrd_max_try	input	9	Poll retry budget, 0-511 (from CTRL_CONFIG.CTRLRD_MAX_TRY)
cfg_channel_reset	input	1	Per-channel reset request
tick_1us	input	1	1 microsecond tick that paces retries

13.3.4 AXI4 Read Master Interface

Table 2.8.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
ar_valid	output	1	Read address valid
ar_ready	input	1	Read address ready
ar_addr	output	AW	Read address (poll

Signal	Direction	Width	Description
			address)
ar_len	output	8	Burst length (8'h00, single beat)
ar_size	output	3	Burst size (3'b010, 4 bytes)
ar_burst	output	2	Burst type (2'b01, INCR)
ar_id	output	ID	Transaction ID (channel-derived)
ar_lock	output	1	Lock (1'b0, normal access)
ar_cache	output	4	Cache (4'b0010, non-cacheable bufferable)
ar_prot	output	3	Protection (3'b000)
ar_qos	output	4	Quality of service (4'h0)
ar_region	output	4	Region (4'h0)
r_valid	input	1	Read data valid (shared channel)
r_ready	output	1	Read data ready
r_data	input	ADW	Read data (lower 32 bits used)
r_id	input	ID	Read response ID (matched to our request)
r_resp	input	2	Read response
r_last	input	1	Read last beat

13.3.5 MonBus Interface

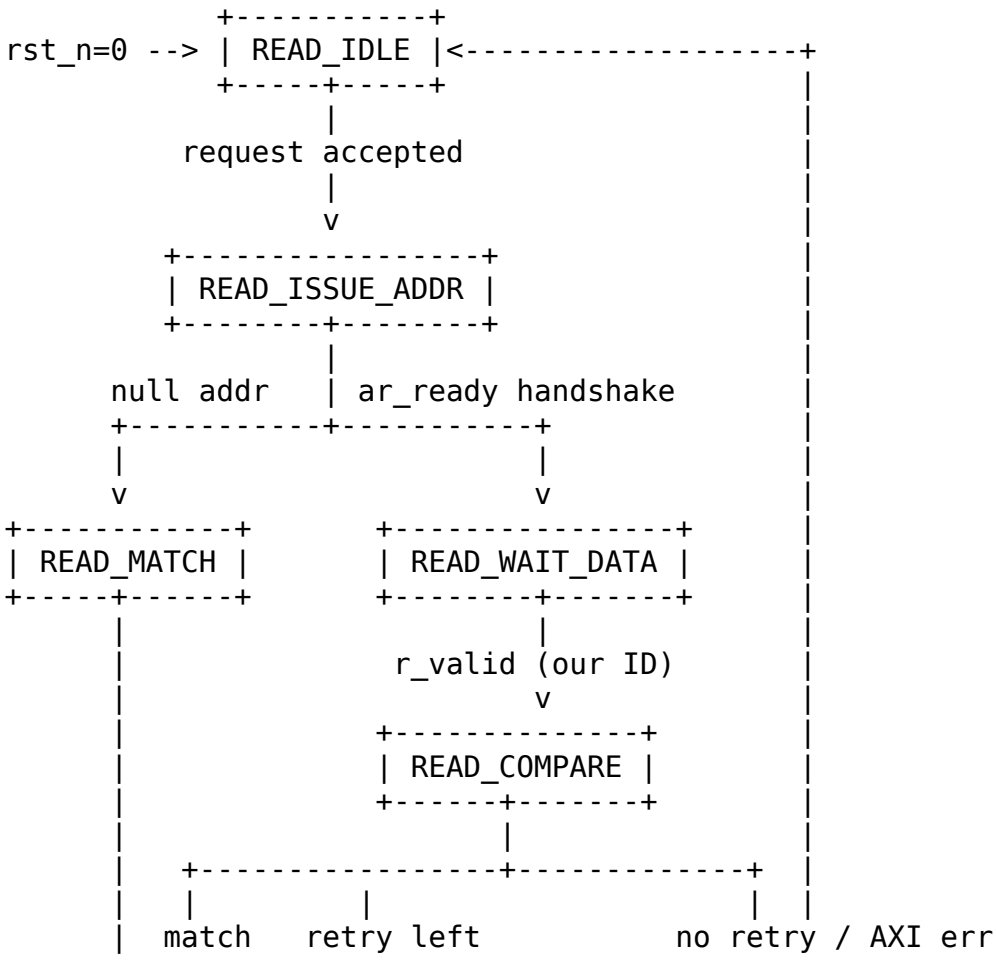
Table 2.8.6: MonBus Interface

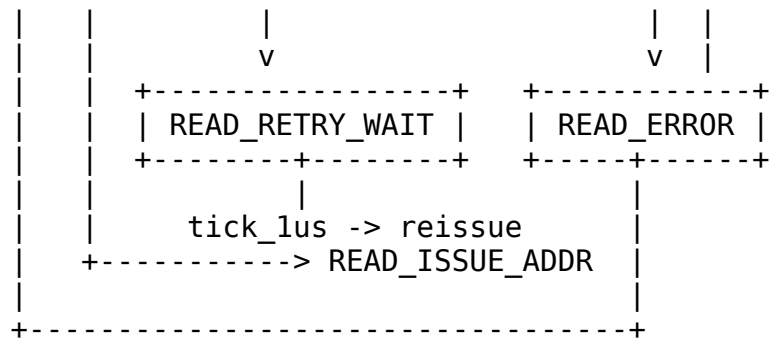
Signal	Direction	Width	Description
i_mon_time	input	64	Side-band monitor timestamp
mon_valid	output	1	Monitor packet

Signal	Direction	Width	Description
			valid
`mon_ready`	input	1	Monitor consumer ready
`mon_packet`	output	128	Monitor packet (`monitor_packet_t`)
`mon_timestamp`	output	64	Monitor packet timestamp

13.4 FSM States

13.4.1 Figure 2.8.2: Control-Read Engine FSM





Source: 08_ctrlrd_engine_fsm.mmd

State summary:

- **READ_IDLE:** Waits for a request from the scheduler (via a two-deep skid buffer). Latches poll address, expected value, mask, and the AXI ID; initializes the retry counter from `cfg_ctrlrd_max_try`.
- **READ_ISSUE_ADDR:** Drives `ar_valid`. A null address (`64'h0`) shortcuts to `READ_MATCH`; otherwise, once `ar_ready` handshakes, advances to `READ_WAIT_DATA`.
- **READ_WAIT_DATA:** Accepts the read response whose `r_id` matches the channel ID, capturing the lower 32 bits and the response code.
- **READ_COMPARE:** On an AXI error response, goes to `READ_ERROR`. On a masked match, goes to `READ_MATCH`. Otherwise, if retries remain, decrements the counter and goes to `READ_RETRY_WAIT`; if the budget is exhausted, goes to `READ_ERROR`.
- **READ_RETRY_WAIT:** Waits for `tick_1us`, then reissues the read.
- **READ_MATCH:** Gate satisfied. Clears the error flag and returns to `READ_IDLE`.
- **READ_ERROR:** Sets `ctrlrd_error` and returns to `READ_IDLE`.

13.5 Operation

13.5.1 Masked Compare

The engine gates on a masked equality check:

```
w_masked_expected = ctrlrd_pkt_data & ctrlrd_pkt_mask
w_masked_actual   = r_data[31:0]    & ctrlrd_pkt_mask
w_data_match      = (w_masked_expected == w_masked_actual)
```

Only the bits set in `mask` participate. A mask of `32'hFFFFFFFF` compares the full word; a mask of a single bit turns the gate into a flag-poll.

13.5.2 Retry Budget and Pacing

The retry counter is loaded from `cfg_ctrlrd_max_try` when the request is accepted. On each non-matching, non-error compare with retries remaining, the counter decrements and the engine parks in `READ_RETRY_WAIT` until the next `tick_1us`, which paces successive polls at roughly 1 microsecond intervals. When the counter reaches zero without a match, the engine transitions to `READ_ERROR` and asserts `ctrlrd_error`, so a never-matching gate cannot stall the channel indefinitely.

`cfg_ctrlrd_max_try` is driven by the `CTRL_CONFIG` register at offset `0x240`, field `CTRLRD_MAX_TRY[8:0]` (0-511, reset 16).

13.5.3 Null Address Fast-Path

If the latched poll address is `64'h0`, the engine treats the gate as already satisfied: `READ_ISSUE_ADDR` transitions directly to `READ_MATCH` without issuing an AXI read. This lets software emit an unconditional (no-op) gate.

13.5.4 AXI Read Transaction

Polls are single-beat, 4-byte INCR reads (`ar_len = 0`, `ar_size = 3'b010`, `ar_burst = 2'b01`). The `ar_id` is derived from `CHANNEL_ID`, and only responses whose `r_id` matches are consumed on the shared read data channel. `ctrlrd_result` exposes the last captured 32-bit value; in the per-channel instantiation the scheduler gates on match/error rather than the raw value, so `ctrlrd_result` is left unconnected there.

13.5.5 Channel Reset

`cfg_channel_reset` is registered into `r_channel_reset_active`. While asserted, the engine refuses new requests (`ctrlrd_ready` is forced low), any in-flight state is cleared, and the FSM is driven back to `READ_IDLE`. `ctrlrd_engine_idle` asserts only in `READ_IDLE` with no pending request and no active channel reset.

13.6 MonBus Reporting

The engine emits monitor packets on notable events:

Table 2.8.7: MonBus Events

Event	Packet Type	Code	Payload
Gate satisfied (non-null)	`PktTypeCompletion`	`CORE_COMPL_CTRLRD_COMPLETED`	Poll address

Event	Packet Type	Code	Payload
Retry attempt	PktTypePerf	CORE_PERF_CTRLRD_RETRY	Remaining retry count
Retries exhausted	PktTypeError	CORE_ERR_CTRLRD_MAX_RETRIES	Poll address
AXI error response	PktTypeError	AXI_ERR_RESP_SLVERR / AXI_ERR_RESP_DECERR	Response code

All packets carry MON_AGENT_ID, MON_UNIT_ID, and MON_CHANNEL_ID, and are timestamped from i_mon_time.

13.7 Descriptor Field Mapping

The CTRL_READ opcode reinterprets the shared 256-bit descriptor (the descriptor engine's existing field extraction is reused):

Table 2.8.8: CTRL_READ Descriptor Field Mapping

Descriptor Field	Bits	Engine Port
poll_addr	[63:0]	ctrlrd_pkt_addr
expected	[95:64]	ctrlrd_pkt_data
mask	[127:96]	ctrlrd_pkt_mask
max_try	[143:128]	Per-descriptor budget (0 = use cfg_ctrlrd_max_try default)

Last Updated: 2026-07-12

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14 Control-Write Engine Specification

Module: ctrlwr_engine.sv **Location:** projects/components/rapids/rtl/fub/ **Status:** Implemented **Last Updated:** 2025-01-10

14.1 Overview

The Control-Write Engine executes a CTRL_WRITE control descriptor (DESC_OP_CTRL_WRITE = 2'b10). A CTRL_WRITE is a producer doorbell: the engine performs a single-beat AXI4 write of the descriptor's 32-bit wr_data value to its 64-bit wr_addr target, then completes so the descriptor chain continues. One instance is dedicated per channel.

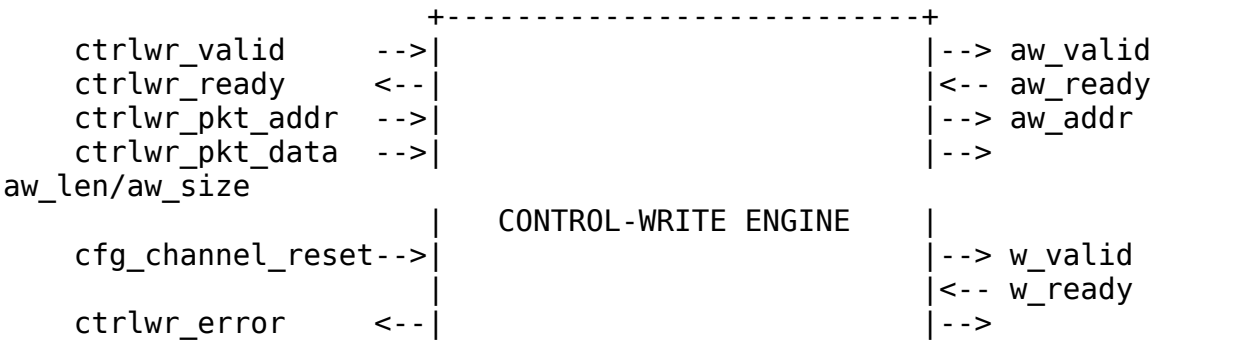
The write target address must be 4-byte aligned because the transfer is a 32-bit (aw_size = 3'b010) single-beat write. A null address (wr_addr == 0) is treated as a no-op and the request completes without issuing an AXI transaction.

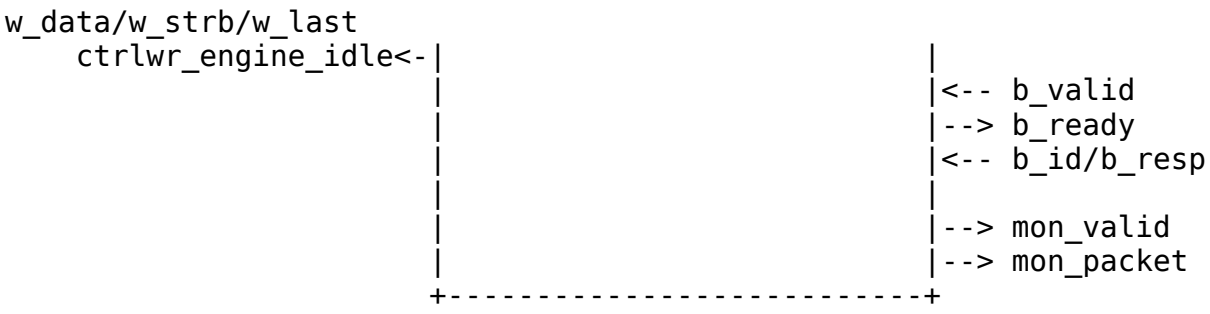
14.1.1 Key Features

- **Single-Beat AXI4 Write:** One 32-bit doorbell write per request (aw_len = 0, w_last = 1)
- **Producer Doorbell Semantics:** Write value to target, then release the chain (no poll, no retry)
- **Address Alignment Check:** Requires 4-byte-aligned wr_addr; misalignment raises an error
- **Null-Address Skip:** wr_addr == 0 completes as a no-op
- **Channel Reset Support:** cfg_channel_reset drains and forces the FSM back to idle
- **MonBus Integration:** Completion and error event reporting

14.1.2 Block Diagram

14.1.3 Figure 2.9.1: Control-Write Engine Block Diagram





Source: 09_ctrlwr_engine_block.mmd

14.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 32;                 // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Derived channel
index width
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int AXI_ID_WIDTH = 8;                  // AXI transaction ID
width

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID  = 16'h0020; // Ctrlwr Engine
Agent ID
parameter logic [7:0]  MON_UNIT_ID   = 8'h01;    // Unit identifier
parameter logic [8:0]  MON_CHANNEL_ID = 9'h000;  // Base channel ID
```

: Table 2.9.1: Control-Write Engine Parameters

Note: `AXI_ID_WIDTH` must be `>= CHAN_WIDTH` (checked by an elaboration-time `$fatal`). When instantiated in `scheduler_group_beats.sv` as `u_ctrlwr_engine`, `MON_AGENT_ID` is overridden to `16'h0021` (`CTRLWR_MON_AGENT_ID`) to distinguish it from the Control-Read Engine's `16'h0020`.

14.3 Port List

14.3.1 Clock and Reset

Table 2.9.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low

Signal	Direction	Width	Description
			asynchronous reset

14.3.2 Configuration Interface

Table 2.9.3: Configuration Interface

Signal	Direction	Width	Description
`cfg_channel_reset`	input	1	Per-channel reset: blocks new requests, drains transaction, forces idle

14.3.3 Scheduler Interface

Table 2.9.4: Scheduler Interface

Signal	Direction	Width	Description
`ctrlwr_valid`	input	1	Request valid (doorbell write requested)
`ctrlwr_ready`	output	1	Ready to accept request
`ctrlwr_pkt_addr`	input	AW	Doorbell target address (`wr_addr`, 4-byte aligned)
`ctrlwr_pkt_data`	input	32	Doorbell write value (`wr_data`)
`ctrlwr_error`	output	1	Error flag (alignment or AXI response error)
`ctrlwr_engine_idle`	output	1	Engine idle (in `WRITE_IDLE`, no active transaction, not in channel reset)

14.3.4 AXI4 Write Master Interface

Table 2.9.5: AXI4 Write Master Interface

Signal	Direction	Width	Description
`aw_valid`	output	1	Write address valid
`aw_ready`	input	1	Write address ready
`aw_addr`	output	AW	Write address (=

Signal	Direction	Width	Description
			wr_addr)
aw_len	output	8	Burst length (0 = single beat)
aw_size	output	3	Transfer size (3'b010 = 4 bytes / 32-bit)
aw_burst	output	2	Burst type (2'b01 = INCR)
aw_id	output	AXI_ID_WIDTH	Transaction ID (derived from CHANNEL_ID)
aw_lock	output	1	Lock type (0 = normal)
aw_cache	output	4	Cache type (4'b0010 = non-cacheable bufferable)
aw_prot	output	3	Protection type (3'b000)
aw_qos	output	4	Quality of service (0)
aw_region	output	4	Region identifier (0)
w_valid	output	1	Write data valid
w_ready	input	1	Write data ready
w_data	output	32	Write data (= wr_data)
w_strb	output	4	Write strobes (4'b1111 = all bytes)
w_last	output	1	Last beat (1 = single beat)
b_valid	input	1	Write response valid
b_ready	output	1	Write response ready
b_id	input	AXI_ID_WIDTH	Response ID (matched against expected AXI ID)
b_resp	input	2	Write response code

14.3.5 MonBus Interface

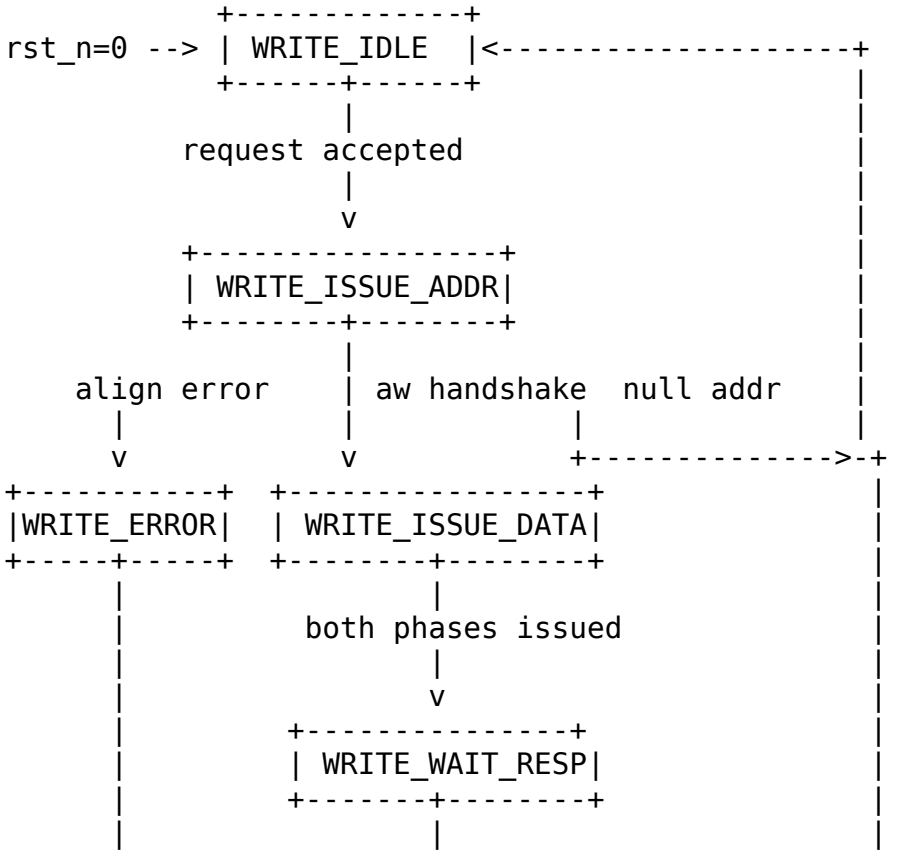
Table 2.9.6: MonBus Interface

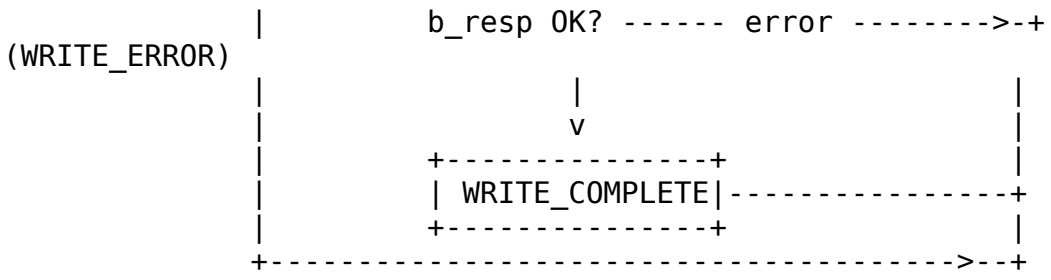
Signal	Direction	Width	Description
i_mon_time	input	64	Monitor timestamp

Signal	Direction	Width	Description
			side-band input
`mon_valid`	output	1	Monitor packet valid
`mon_ready`	input	1	Monitor packet ready
`mon_packet`	output	128	Monitor packet (`monitor_packet_t`)
`mon_timestamp`	output	64	Monitor packet timestamp

14.4 FSM States

14.4.1 Figure 2.9.2: Control-Write Engine FSM





Source: 09_ctrlwr_engine_fsm.mmd

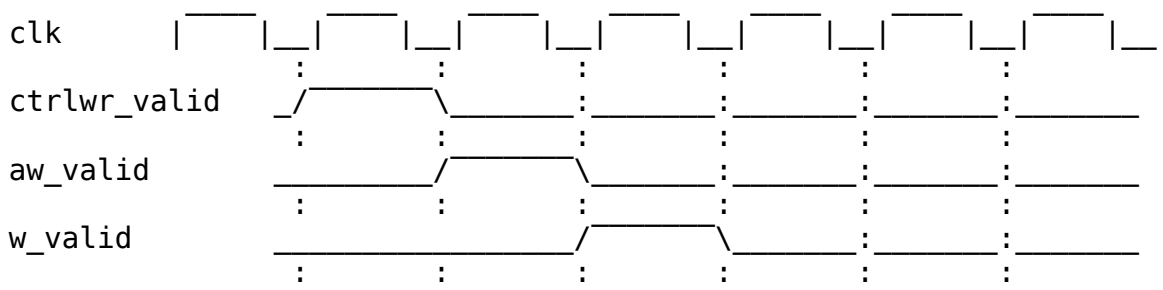
State summary:

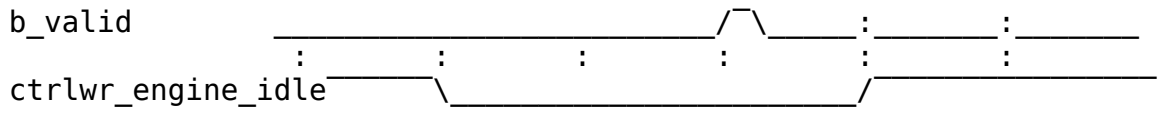
- **WRITE_IDLE:** Wait for a request from the request skid buffer; latch `wr_addr`/`wr_data` and the expected AXI ID.
- **WRITE_ISSUE_ADDR:** Drive `aw_valid` until `aw_ready`. A misaligned address routes to `WRITE_ERROR`; a null address returns to `WRITE_IDLE` (no-op skip).
- **WRITE_ISSUE_DATA:** Drive `w_valid` until `w_ready`; advance once both address and data phases have been issued.
- **WRITE_WAIT_RESP:** Accept the matching `b`-channel response; a non-OKAY `b_resp` routes to `WRITE_ERROR`, otherwise to `WRITE_COMPLETE`.
- **WRITE_COMPLETE:** Emit a completion MonBus packet (unless null address) and return to `WRITE_IDLE`.
- **WRITE_ERROR:** Latch `ctrlwr_error`, emit an error MonBus packet, and return to `WRITE_IDLE`. The error flag persists until reset or `cfg_channel_reset`.

14.5 Operation

14.5.1 Doorbell Write Sequence

14.5.2 Figure 2.9.3: Control-Write Doorbell Timing





TODO: Replace with simulation-generated waveform

14.5.3 Request Handshake and Skid Buffer

Incoming requests are captured through a depth-2 `gaxi_skid_buffer`. The engine accepts a request (`ctrlwr_ready`) only when not in an active channel reset. The FSM pops the buffered request in `WRITE_IDLE`, latching `{wr_addr, wr_data}` and computing the expected AXI ID from `CHANNEL_ID`:

```
aw_id = {{(AXI_ID_WIDTH-CHAN_WIDTH){1'b0}}, CHANNEL_ID[CHAN_WIDTH-
1:0]};
```

The b-channel response is matched against this expected ID (`b_id == r_expected_axi_id`) so a shared write master can be demultiplexed per channel.

14.5.4 Address Validation

```
Null (skip)  : wr_addr == 64'h0
Align error  : wr_addr[1:0] != 2'b00  (and not null)
```

A null address completes immediately as a no-op with no AXI activity and no completion packet. A misaligned address never issues on AXI; it transitions to `WRITE_ERROR`, asserts `ctrlwr_error`, and emits an error MonBus packet.

14.5.5 Error Reporting

`ctrlwr_error` is a registered, sticky flag. It is set on an alignment error or on a non-OKAY AXI write response (`b_resp != 2'b00`), and it persists until `rst_n` or `cfg_channel_reset`. The MonBus error packet carries either the offending address (alignment error) or the captured `b_resp` code (AXI response error).

14.5.6 Channel Reset

Asserting `cfg_channel_reset` registers `r_channel_reset_active`, which blocks new requests, clears in-flight transaction tracking, and forces the FSM back to `WRITE_IDLE`. `ctrlwr_engine_idle` is only asserted in `WRITE_IDLE` when the skid buffer is empty and no channel reset is active.

14.6 Contrast with the Control-Read Engine

The Control-Write Engine is the producer counterpart to the Control-Read Engine (Section 2.8). Both are per-channel control-descriptor executors, but they implement opposite half of the producer/consumer handshake:

Aspect	Control-Read Engine (2.8)	Control-Write Engine (2.9)
Opcode	`DESC_OP_CTRL_READ` (`2'b01`)	`DESC_OP_CTRL_WRITE` (`2'b10`)
Role	Consumer gate	Producer doorbell
AXI direction	Read (poll)	Single-beat write
Semantics	Poll until `(rd & mask) == expected`	Write `wr_data` to `wr_addr`, then continue
Retry budget	`max_try` retry budget (config default)	None (single unconditional write)
Chain effect	Holds off the chain until the gate passes	Releases the chain after the write completes
Descriptor fields	`poll_addr` / `expected` / `mask` / `max_try`	`wr_addr` / `wr_data`

A `CTRL_WRITE` therefore has no maximum-try configuration: it is one unconditional 32-bit write, whereas `CTRL_READ` polls with a bounded retry budget before either passing the gate or timing out.

Last Updated: 2026-07-12

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15 Beats Scheduler Group Specification

Module: `beats_scheduler_group.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

15.1 Overview

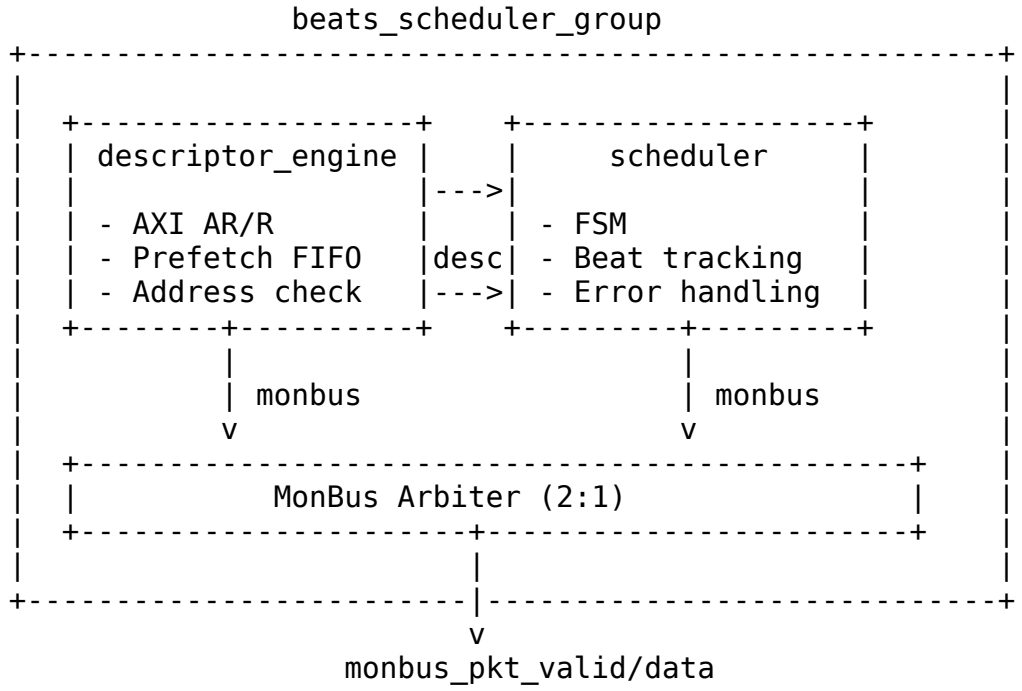
The Beats Scheduler Group wraps a single channel's scheduler and descriptor engine together with MonBus aggregation. It provides a clean integration point for the scheduler_group_array.

15.1.1 Key Features

- **Single Channel Integration:** One scheduler + one descriptor engine
- **MonBus Aggregation:** Combines scheduler and descriptor engine MonBus outputs
- **Configuration Pass-Through:** Routes configuration to both sub-modules
- **Status Aggregation:** Combined idle and error status

15.1.2 Block Diagram

15.1.3 Figure 3.1.1: Beats Scheduler Group Block Diagram



Source: 03_beats_scheduler_group_block.mmd

15.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 8;                  // Total channels
parameter int ADDR_WIDTH = 64;                   // Address bus width
```

```
parameter int DATA_WIDTH = 512;                    // Data bus width

// Monitor Bus Parameters
parameter int DESC_MON_AGENT_ID = 16 + CHANNEL_ID;  // Descriptor
engine agent
parameter int SCHED_MON_AGENT_ID = 48 + CHANNEL_ID; // Scheduler agent
parameter int MON_UNIT_ID = 1;
```

: Table 3.1.1: Beats Scheduler Group Parameters

15.3 Port List

15.3.1 Clock and Reset

Table 3.1.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

15.3.2 APB Programming Interface

Table 3.1.3: APB Programming Interface

Signal	Direction	Width	Description
`apb_valid`	input	1	Channel kick-off
`apb_ready`	output	1	Ready for kick-off
`apb_addr`	input	AW	First descriptor address

15.3.3 Configuration Interface

Table 3.1.4: Configuration Interface

Signal	Direction	Width	Description
`cfg_channel_enable`	input	1	Enable channel
`cfg_channel_reset`	input	1	Soft reset

Signal	Direction	Width	Description
cfg_sched_timeout_cycles	input	32	Write-progress timeout window (cycles)
cfg_sched_timeout_limit	input	8	Consecutive-timeout windows before fatal escalation (0 = never)
cfg_sched_timeout_enable	input	1	Enable timeout
cfg_desceng_enable	input	1	Enable descriptor engine
cfg_desceng_prefetch	input	1	Enable prefetch chaining
cfg_desceng_fifo_thresh	input	4	Prefetch threshold
cfg_desceng_addr0_base	input	AW	Address range 0 base
cfg_desceng_addr0_limit	input	AW	Address range 0 limit

15.3.4 Descriptor AXI Master Interface

Table 3.1.5: Descriptor AXI Master Interface

Signal	Direction	Width	Description
m_axi_arvalid	output	1	AR channel valid
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	AR address
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	256	R data (descriptor)
m_axi_rresp	input	2	R response

15.3.5 Scheduler Data Interfaces

Table 3.1.6: Scheduler Data Interfaces

Signal	Direction	Width	Description
`sched_rd_valid`	output	1	Read request
`sched_rd_addr`	output	AW	Read address
`sched_rd_beats`	output	32	Read beats
`sched_rd_done_strobe`	input	1	Read complete
`sched_wr_valid`	output	1	Write request
`sched_wr_addr`	output	AW	Write address
`sched_wr_beats`	output	32	Write beats
`sched_wr_done_strobe`	input	1	Write AW-issue strobe
`sched_wr_commit_strobe`	input	1	Write COMMIT strobe (B response)
`sched_wr_commit_beats`	input	32	Beats committed this strobe

15.3.6 Status

Table 3.1.7: Status Interface

Signal	Direction	Width	Description
`scheduler_idle`	output	1	Scheduler idle
`descriptor_engine_idle`	output	1	Descriptor engine idle
`scheduler_state`	output	7	FSM state
`sched_error`	output	1	Error flag

15.3.7 MonBus Interface

Table 3.1.8: MonBus Interface

Signal	Direction	Width	Description
`monbus_pkt_valid`	output	1	Packet valid
`monbus_pkt_ready`	input	1	Consumer ready
`monbus_pkt_data`	output	64	Packet data

15.4 Internal Architecture

```
Internal Signal Flow:

apb_valid/addr -----> descriptor_engine -----> scheduler
                            |                       |
                      desc_valid/packet           |
                            |                       |
                            +--> descriptor_ready <--+

MonBus Aggregation:

desc_engine.monbus_pkt ----+
                           |---> round_robin_arbiter ---> monbus_out
scheduler.monbus_pkt   ----+
```

Last Updated: 2026-07-02

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16 Beats Scheduler Group Array Specification

Module: `beats_scheduler_group_array.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

16.1 Overview

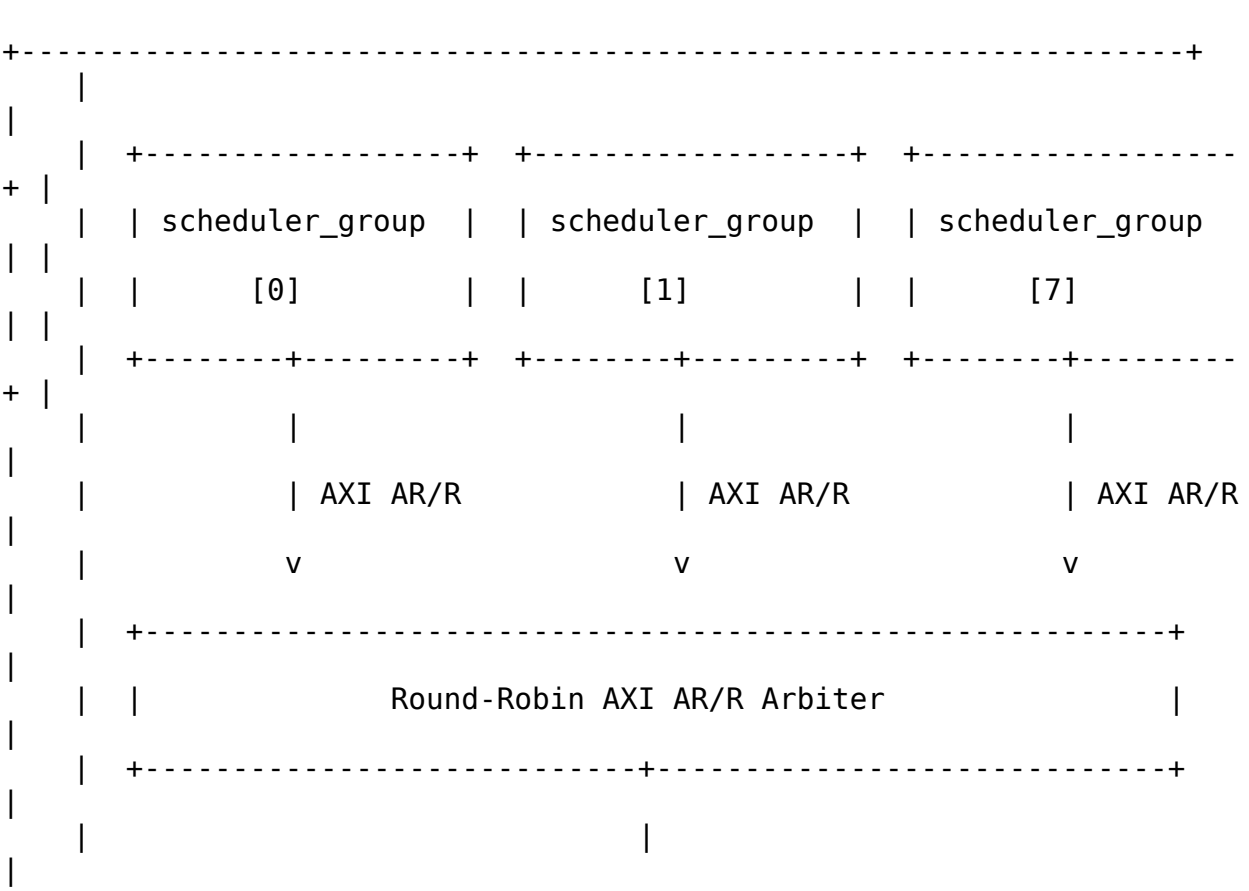
The Beats Scheduler Group Array instantiates 8 scheduler groups with a shared descriptor AXI master and unified MonBus output. It provides the complete scheduler infrastructure for all RAPIDS channels.

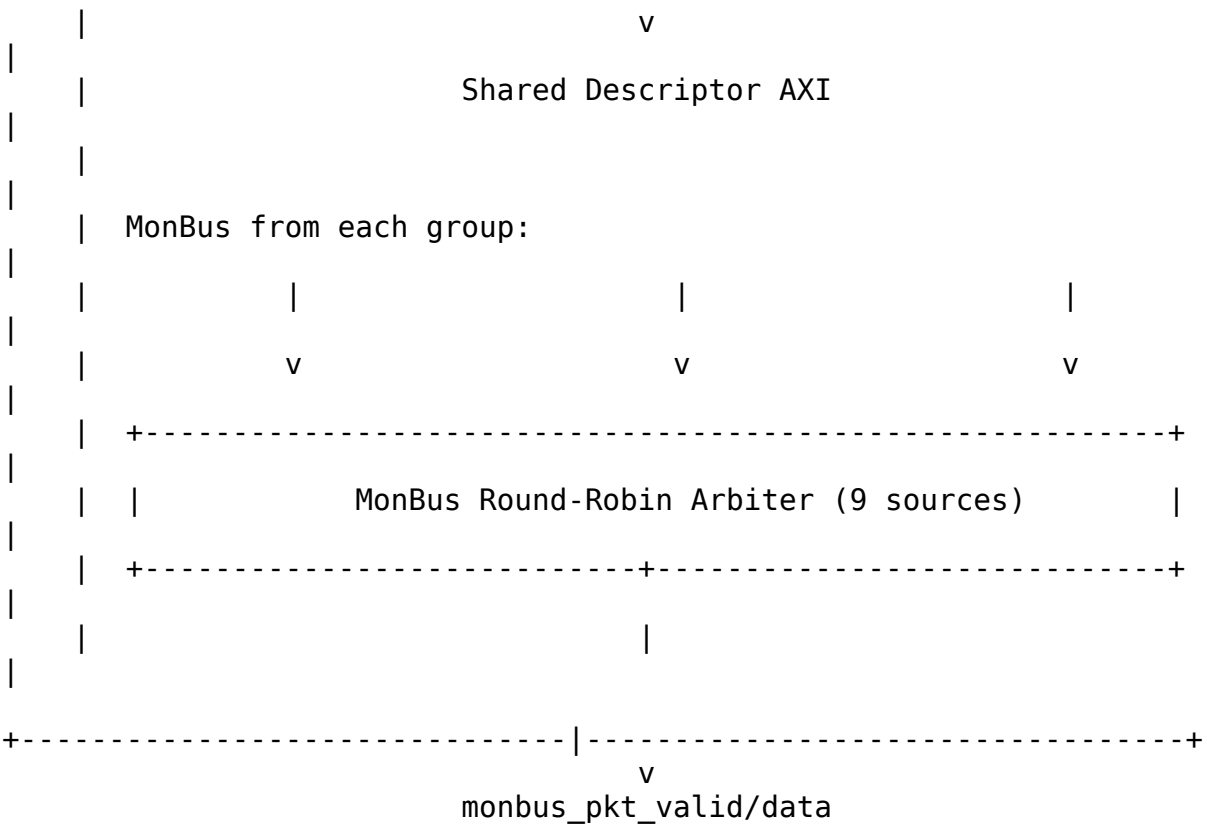
16.1.1 Key Features

- **8-Channel Array:** One scheduler_group per channel
- **Shared Descriptor AXI:** Round-robin arbitration for descriptor fetches
- **Unified MonBus:** Aggregates 9 MonBus sources (8 scheduler + 1 arbiter)
- **Per-Channel Configuration:** Individual enable/reset per channel
- **Aggregate Status:** Combined idle and error flags

16.1.2 Block Diagram

16.1.3 Figure 3.2.1: Beats Scheduler Group Array Block Diagram





Source: 03_beats_scheduler_group_array_block.mmd

16.2 Parameters

```
parameter int NUM_CHANNELS = 8;                  // Number of channels
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int DATA_WIDTH = 512;                  // Data bus width
parameter int DESC_DATA_WIDTH = 256;             // Descriptor width

// AXI ID Management
parameter int AXI_ID_WIDTH = 8;                  // ID field width
parameter int MAX_OUTSTANDING = 8;               // Outstanding
descriptors

// Monitor Bus Parameters
parameter int MON_UNIT_ID = 1;                   // Unit ID for MonBus
```

: Table 3.2.1: Beats Scheduler Group Array Parameters

16.3 Port List

16.3.1 Clock and Reset

Table 3.2.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

16.3.2 Per-Channel APB Programming

Table 3.2.3: APB Programming Interface

Signal	Direction	Width	Description
`apb_valid`	input	NC	Channel kick-off (per-channel)
`apb_ready`	output	NC	Ready for kick-off
`apb_addr`	input	NC*AW	First descriptor address

16.3.3 Per-Channel Configuration

Table 3.2.4: Per-Channel Configuration

Signal	Direction	Width	Description
`cfg_channel_enable`	input	NC	Enable per channel
`cfg_channel_reset`	input	NC	Soft reset per channel
`cfg_sched_timeout_cycles`	input	NC*16	Timeout threshold
`cfg_sched_timeout_enable`	input	NC	Enable timeout
`cfg_desceng_enable`	input	NC	Enable descriptor engine
`cfg_desceng_p`	input	NC	Enable

Signal	Direction	Width	Description
refetch			prefetching
cfg_desceng_fifo_thresh	input	NC*4	Prefetch threshold
cfg_desceng_addr0_base	input	NC*AW	Address range 0 base
cfg_desceng_addr0_limit	input	NC*AW	Address range 0 limit

16.3.4 Shared Descriptor AXI Master Interface

Table 3.2.5: Shared Descriptor AXI Master Interface

Signal	Direction	Width	Description
m_axi_arvalid	output	1	AR channel valid
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	AR address
m_axi_arid	output	ID_W	Transaction ID
m_axi_arlen	output	8	Burst length
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	256	R data
m_axi_rid	input	ID_W	Response ID
m_axi_rresp	input	2	R response
m_axi_rlast	input	1	Last beat

16.3.5 Per-Channel Scheduler Data Interfaces

Table 3.2.6: Per-Channel Scheduler Data Interfaces

Signal	Direction	Width	Description
sched_rd_valid	output	NC	Read request per channel
sched_rd_addr	output	NC*AW	Read address
sched_rd_beats	output	NC*32	Read beats
sched_rd_done_strobe	input	NC	Read complete
sched_wr_valid	output	NC	Write request per channel
sched_wr_addr	output	NC*AW	Write address
sched_wr_beats	output	NC*32	Write beats
sched_wr_done_strobe	input	NC	Write complete

16.3.6 Aggregate Status

Table 3.2.7: Aggregate Status

Signal	Direction	Width	Description
all_channels_idle	output	1	All channels idle
scheduler_idle	output	NC	Per-channel scheduler idle
descriptor_engine_idle	output	NC	Per-channel desc eng idle
scheduler_state	output	NC*7	FSM states
sched_error	output	NC	Error flags

16.3.7 Unified MonBus Interface

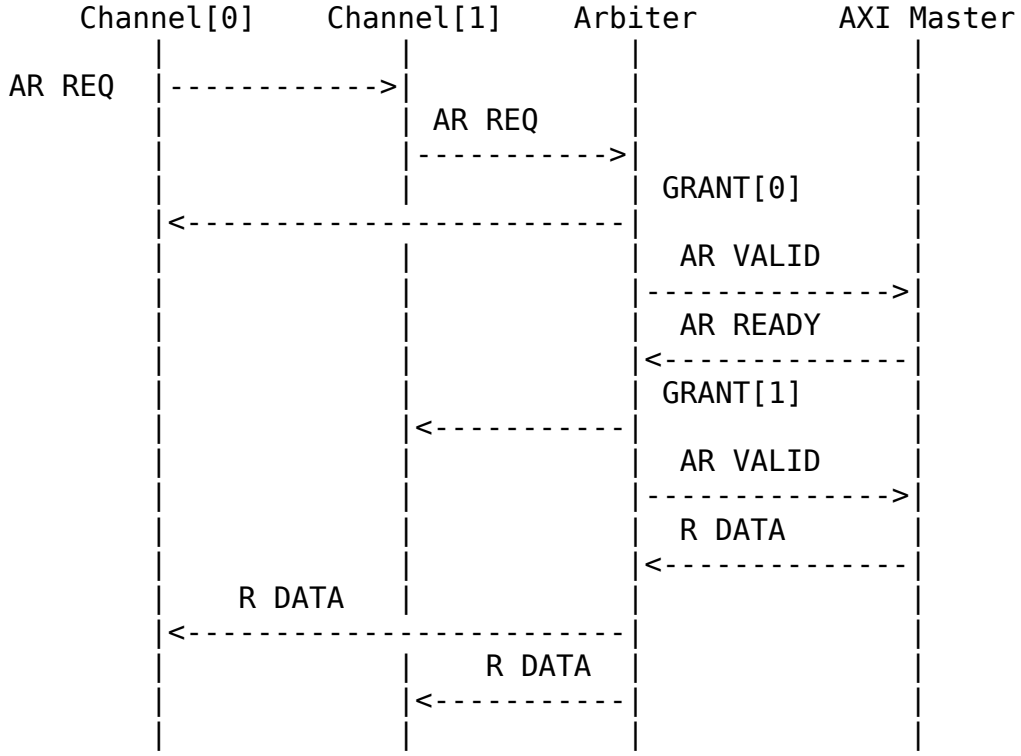
Table 3.2.8: Unified MonBus Interface

Signal	Direction	Width	Description
monbus_pkt_valid	output	1	Packet valid

Signal	Direction	Width	Description
monbus_pkt_ready	input	1	Consumer ready
monbus_pkt_data	output	64	Packet data

16.4 AXI Arbitration

16.4.1 Figure 3.2.2: Descriptor AXI Arbitration Sequence



Source: 03_scheduler_group_array_arbitration.mmd

16.5 MonBus Aggregation

The unified MonBus output aggregates 9 sources using round-robin arbitration:

Table 3.2.9: MonBus Source Assignment

Source ID	Origin	Description
0-7	scheduler_group[0:7]	Per-channel

Source ID	Origin	Description
		aggregated MonBus
8	AXI Arbiter	Arbitration events

Each scheduler_group provides a single MonBus output that combines both scheduler and descriptor engine packets (2:1 internal arbitration).

16.6 Integration Context

```
beats_scheduler_group_array #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512)
) u_scheduler_array (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // APB programming (per-channel)
    .apb_valid              (apb_kick_valid),
    .apb_ready              (apb_kick_ready),
    .apb_addr               (apb_kick_addr),

    // Configuration (per-channel)
    .cfg_channel_enable     (cfg_ch_enable),
    .cfg_sched_timeout_cycles(cfg_timeout_cycles),
    // ... additional config ...

    // Shared descriptor AXI
    .m_axi_arvalid          (desc_axi_arvalid),
    .m_axi_arready          (desc_axi_arready),
    .m_axi_araddr           (desc_axi_araddr),
    .m_axi_rvalid           (desc_axi_rvalid),
    .m_axi_rready           (desc_axi_rready),
    .m_axi_rdata            (desc_axi_rdata),

    // Per-channel scheduler outputs
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),
    .sched_rd_done_strobe   (sched_rd_done_strobe),
    .sched_wr_valid         (sched_wr_valid),
    .sched_wr_addr          (sched_wr_addr),
    .sched_wr_beats         (sched_wr_beats),
    .sched_wr_done_strobe   (sched_wr_done_strobe),
```

```
    // Status
    .all_channels_idle      (all_schedulers_idle),

    // Unified MonBus
    .monbus_pkt_valid       (sched_array_monbus_valid),
    .monbus_pkt_ready       (sched_array_monbus_ready),
    .monbus_pkt_data        (sched_array_monbus_data)
);
```

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17 Sink Data Path Specification

Module: `sink_data_path.sv` **Location:** `projects/components/rapids/rtl/macro_beats/`
Status: Implemented **Last Updated:** 2025-01-10

17.1 Overview

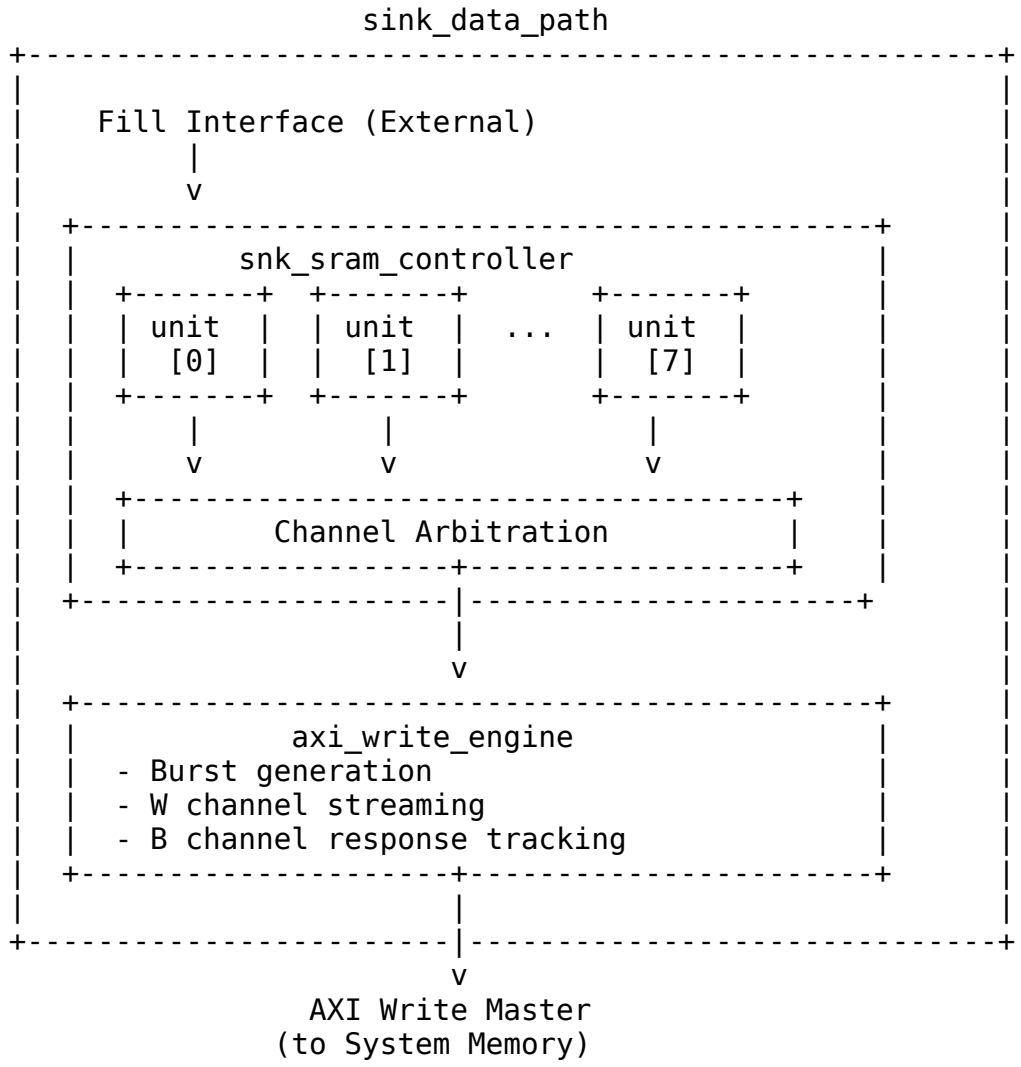
The Sink Data Path integrates the SRAM controller and AXI write engine for network-to-memory transfers. Data enters through the fill interface, is buffered in SRAM, and is written to system memory via AXI4.

17.1.1 Key Features

- **8-Channel SRAM Controller:** Per-channel buffering with flow control
- **AXI Write Engine:** Streaming burst writes to system memory
- **Beat-Level Tracking:** Precise flow control at beat granularity
- **Scheduler Integration:** Receives transfer requests from scheduler

17.1.2 Block Diagram

17.1.3 Figure 3.3.1: Sink Data Path Block Diagram



Source: 03_sink_data_path_block.mmd

17.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;
parameter int AW_MAX_OUTSTANDING = 8;
```

```
parameter int W_PHASE_FIFO_DEPTH = 64;
parameter int B_PHASE_FIFO_DEPTH = 16;
```

: Table 3.3.1: Sink Data Path Parameters

17.3 Data Flow

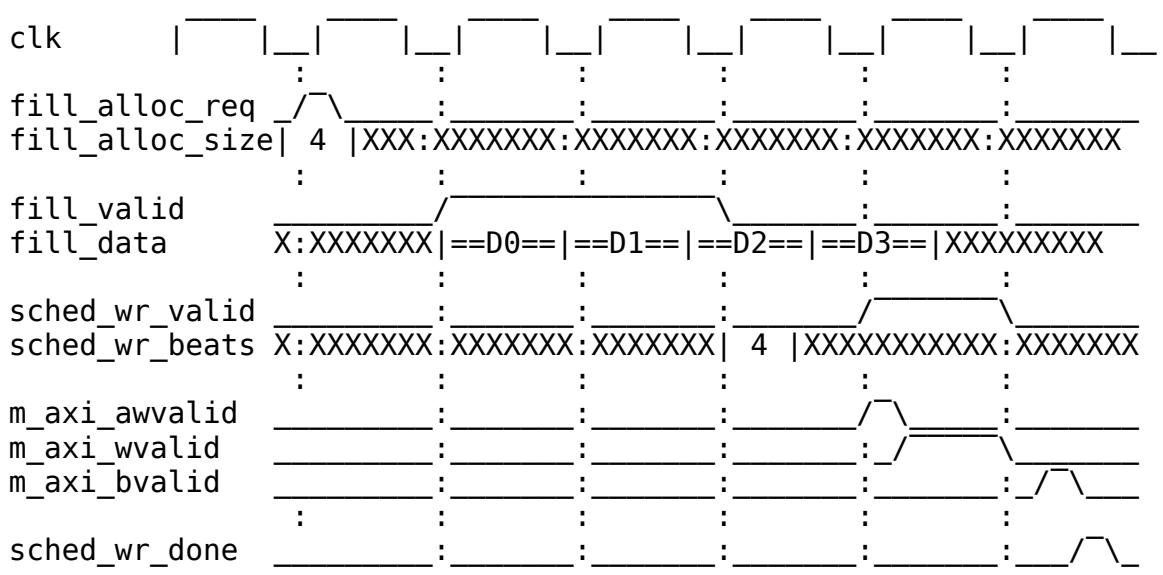
17.3.1 Figure 3.3.2: Sink Path Data Flow Sequence

```
1. Fill Request
   - snk_fill_alloc_req = 1
   - snk_fill_alloc_size = N (beats to allocate)
   - snk_fill_alloc_id = channel
        |
        v
2. Space Check (beats_alloc_ctrl)
   - Check snk_fill_space_free[channel] >= N
   - Allocate N beats of space
        |
        v
3. Fill Data Transfer
   - snk_fill_valid = 1
   - snk_fill_data = data beats
   - snk_fill_id = channel
        |
        v
4. SRAM Write
   - Data written to channel's SRAM partition
   - beats_drain_ctrl notified of data availability
        |
        v
5. AXI Write Request
   - Scheduler requests write: sched_wr_valid
   - Address and beat count provided
        |
        v
6. SRAM Read + AXI Write
   - axi_write_engine reads from SRAM
   - Issues AXI bursts to memory
        |
        v
7. Completion
   - B channel response received
   - sched_wr_done_strobe = 1
```

Source: 03_sink_data_flow.mmd

17.4 Timing Diagram

17.4.1 Figure 3.3.3: Sink Path Transfer Timing



TODO: Replace with simulation-generated waveform showing complete sink transfer

17.5 Interface Summary

17.5.1 Fill Interface (Input)

Table 3.3.2: Fill Interface

Signal	Direction	Description
snk_fill_alloc_req	input	Allocation request
snk_fill_alloc_size	input	Beats to allocate
snk_fill_alloc_id	input	Channel ID
snk_fill_space_free	output	Available space per channel
snk_fill_valid	input	Data valid
snk_fill_ready	output	Ready for data
snk_fill_data	input	Fill data

Signal	Direction	Description
snk_fill_last	input	Last beat
snk_fill_id	input	Channel ID

17.5.2 Scheduler Interface

Table 3.3.3: Scheduler Interface

Signal	Direction	Description
sched_wr_valid	input	Write request
sched_wr_addr	input	Destination address
sched_wr_beats	input	Beats to write
sched_wr_id	input	Channel ID
sched_wr_done_strobe	output	Write complete
sched_wr_beats_done	output	Beats completed
sched_wr_error	output	Error flag

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18 Sink Data Path AXIS Wrapper Specification

Module: sink_data_path_axis.sv **Location:** projects/components/rapids/rtl/macro_beats/ **Status:** Implemented **Last Updated:** 2025-01-10

18.1 Overview

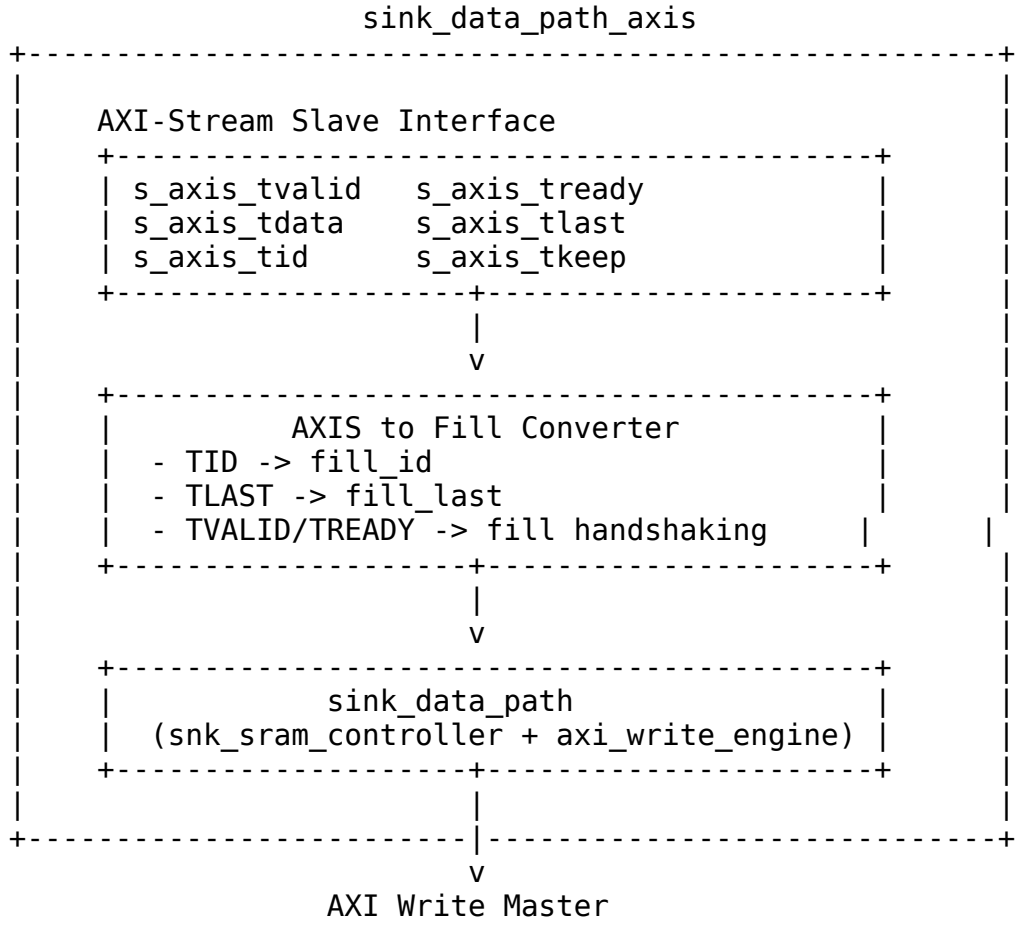
The Sink Data Path AXIS wrapper adds an AXI-Stream slave interface to the sink data path, enabling standard AXIS ingress for network-to-memory transfers.

18.1.1 Key Features

- **AXI-Stream Slave Interface:** Standard AXIS for data ingress
- **Per-Channel TID Mapping:** AXIS TID maps to fill channel ID
- **TLAST to Last Mapping:** Packet boundary tracking
- **Core Sink Integration:** Wraps sink_data_path module

18.1.2 Block Diagram

18.1.3 Figure 3.4.1: Sink Data Path AXIS Block Diagram



Source: 03_sink_data_path_axis_block.mmd

18.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
```

```
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;
parameter int TID_WIDTH = 3;                    // log2(NUM_CHANNELS)
parameter int TDEST_WIDTH = 1;
parameter int TUSER_WIDTH = 1;
```

: Table 3.4.1: Sink Data Path AXIS Parameters

18.3 Port List

18.3.1 Clock and Reset

Table 3.4.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

18.3.2 AXI-Stream Slave Interface

Table 3.4.3: AXI-Stream Slave Interface

Signal	Direction	Width	Description
`s_axis_tvalid`	input	1	Data valid
`s_axis_tready`	output	1	Ready for data
`s_axis_tdata`	input	DATA_WIDTH	Data payload
`s_axis_tkeep`	input	DATA_WIDTH/8	Byte enables
`s_axis_tlast`	input	1	Last beat of packet
`s_axis_tid`	input	TID_WIDTH	Stream ID (channel)
`s_axis_tdest`	input	TDEST_WIDTH	Destination
`s_axis_tuser`	input	TUSER_WIDTH	User sideband

18.3.3 Space Allocation Interface

Table 3.4.4: Space Allocation Interface

Signal	Direction	Width	Description
snk_alloc_req	input	1	Allocation request
snk_alloc_size	input	16	Beats to allocate
snk_alloc_id	input	3	Channel ID
snk_space_free	output	NC*16	Available space per channel

18.3.4 Scheduler Interface

Table 3.4.5: Scheduler Interface

Signal	Direction	Width	Description
sched_wr_valid	input	1	Write request
sched_wr_addr	input	AW	Destination address
sched_wr_beats	input	32	Beats to write
sched_wr_id	input	3	Channel ID
sched_wr_done_strobe	output	1	Write complete
sched_wr_beats_done	output	32	Beats completed
sched_wr_error	output	1	Error flag

18.3.5 AXI Write Master Interface

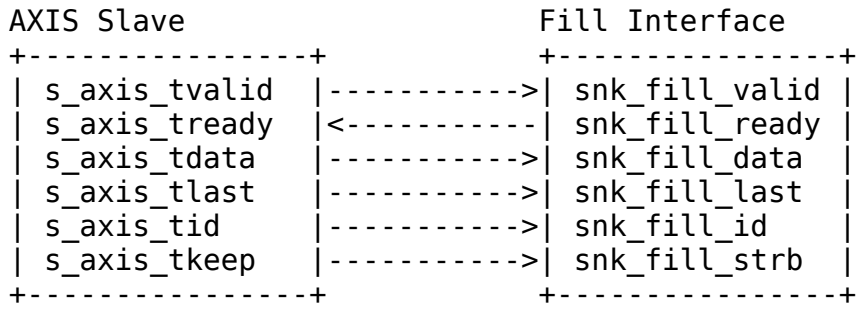
Table 3.4.6: AXI Write Master Interface

Signal	Direction	Width	Description
m_axi_awvalid	output	1	AW channel valid
m_axi_awready	input	1	AW channel

Signal	Direction	Width	Description
			ready
m_axi_awaddr	output	AW	Write address
m_axi_awlen	output	8	Burst length
m_axi_awsize	output	3	Burst size
m_axi_awburst	output	2	Burst type
m_axi_awid	output	ID_W	Transaction ID
m_axi_wvalid	output	1	W channel valid
m_axi_wready	input	1	W channel ready
m_axi_wdata	output	DW	Write data
m_axi_wstrb	output	DW/8	Write strobes
m_axi_wlast	output	1	Last beat
m_axi_bvalid	input	1	B channel valid
m_axi_bready	output	1	B channel ready
m_axi_bresp	input	2	Write response
m_axi_bid	input	ID_W	Response ID

18.4 Signal Mapping

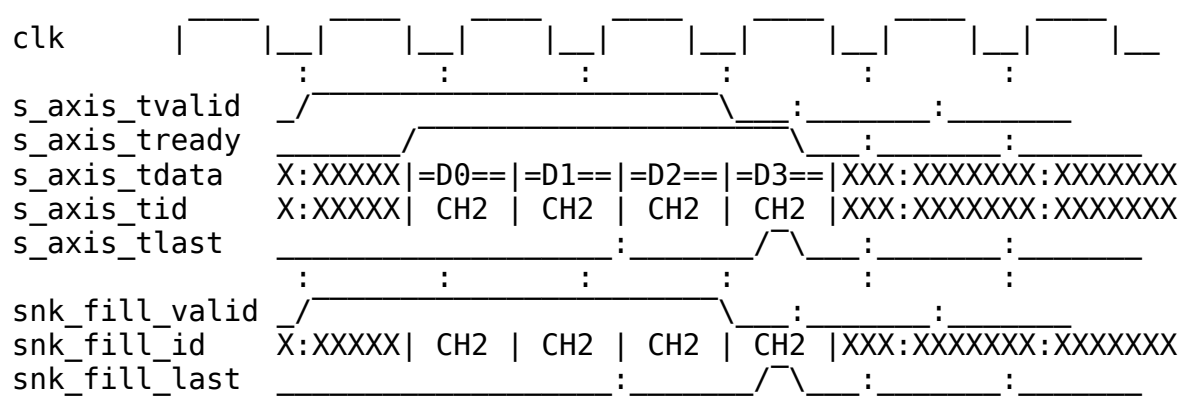
18.4.1 Figure 3.4.2: AXIS to Fill Interface Mapping



Source: 03_sink_axis_mapping.mmd

18.5 Timing Diagram

18.5.1 Figure 3.4.3: AXIS Ingress Timing



TODO: Replace with simulation-generated waveform showing AXIS packet reception

18.6 Usage Notes

1. **TID Usage:** AXIS TID directly maps to channel ID for multi-channel routing
2. **Allocation Required:** Space must be allocated via `snk_alloc_*` before AXIS data arrives
3. **Backpressure:** `s_axis_tready` deasserts when SRAM space exhausted
4. **Packet Boundaries:** TLAST marks end of AXIS packets, mapped to fill_last

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19 Sink SRAM Controller Specification

Module: `snk_sram_controller.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

19.1 Overview

The Sink SRAM Controller manages 8 SRAM controller units, providing per-channel buffering with channel arbitration for the shared AXI write engine.

19.1.1 Key Features

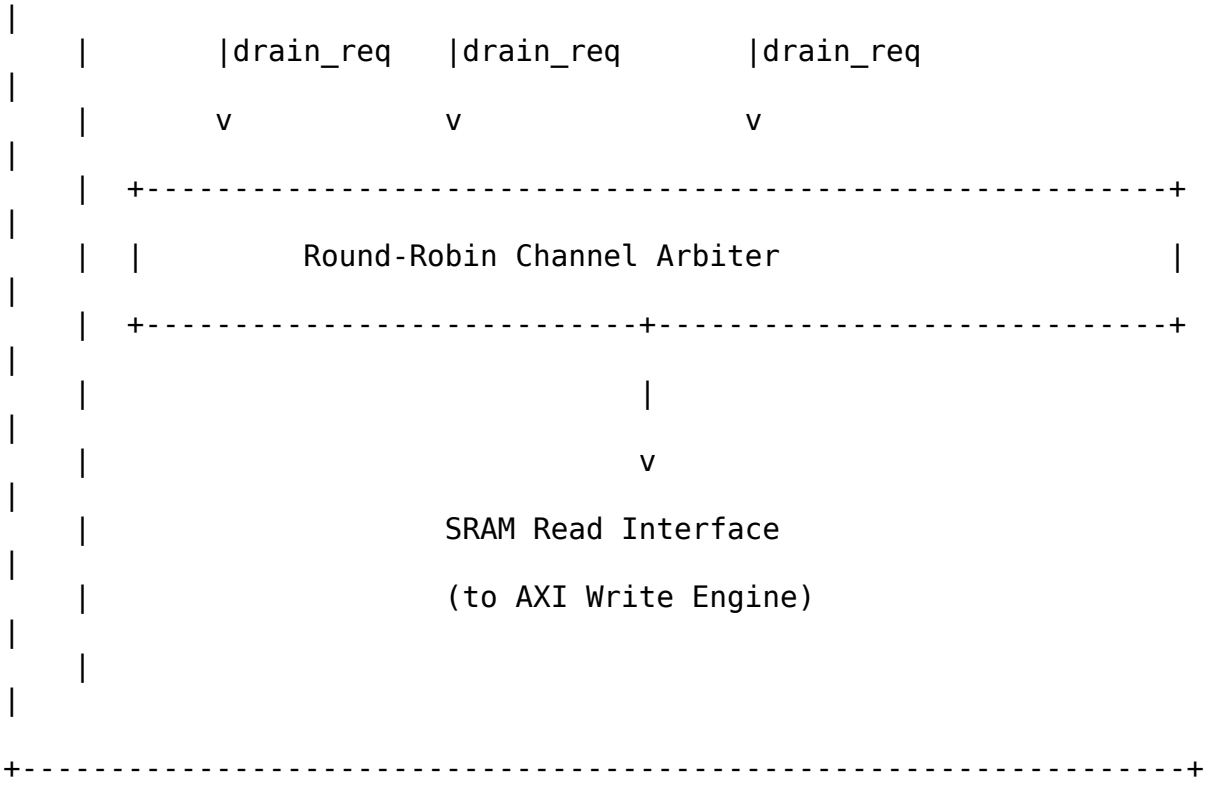
- **8-Channel SRAM Array:** Instantiates 8 `snk_sram_controller_unit` modules
- **Channel Arbitration:** Round-robin access to shared AXI write engine
- **Flow Control Integration:** Per-channel alloc_ctrl and drain_ctrl
- **Space Tracking:** Reports available space per channel

19.1.2 Block Diagram

19.1.3 Figure 3.5.1: Sink SRAM Controller Block Diagram

```
                    snk_sram_controller

+-------------------------------------------------------------------+
    |
|
    |  Fill Interface (per-channel)
|
    |         |         |                   |
|
    |         v         v                   v
|
    |  +----------+ +----------+     +----------+
|
    |  |snk_sram_ | |snk_sram_ | ... |snk_sram_ |
|
    |  |ctrl_unit | |ctrl_unit |     |ctrl_unit |
|
    |  |   [0]    | |   [1]    |     |   [7]    |
|
    |  +----+-----+ +----+-----+     +----+-----+
|
    |       |            |                |
```



Source: 03_snk_sram_controller_block.mmd

19.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;                    // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
```

: Table 3.5.1: Sink SRAM Controller Parameters

19.3 Port List

19.3.1 Clock and Reset

Table 3.5.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock

Signal	Direction	Width	Description
rst_n	input	1	Active-low reset

19.3.2 Per-Channel Fill Interface

Table 3.5.3: Per-Channel Fill Interface

Signal	Direction	Width	Description
snk_fill_alloc_req	input	NC	Allocation request per channel
snk_fill_alloc_size	input	NC*16	Beats to allocate
snk_fill_space_free	output	NC*16	Available space per channel
snk_fill_valid	input	NC	Fill data valid
snk_fill_ready	output	NC	Ready for fill data
snk_fill_data	input	NC*DW	Fill data
snk_fill_last	input	NC	Last beat marker

19.3.3 Arbitrated Drain Interface (to AXI Write Engine)

Table 3.5.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
drain_req	output	1	Drain request (any channel)
drain_gnt	input	1	Drain grant
drain_id	output	3	Granted channel ID
drain_beats	output	16	Beats available to drain
sram_rd_en	input	1	SRAM read enable

Signal	Direction	Width	Description
`sram_rd_addr`	input	AW	SRAM read address
`sram_rd_data`	output	DW	SRAM read data

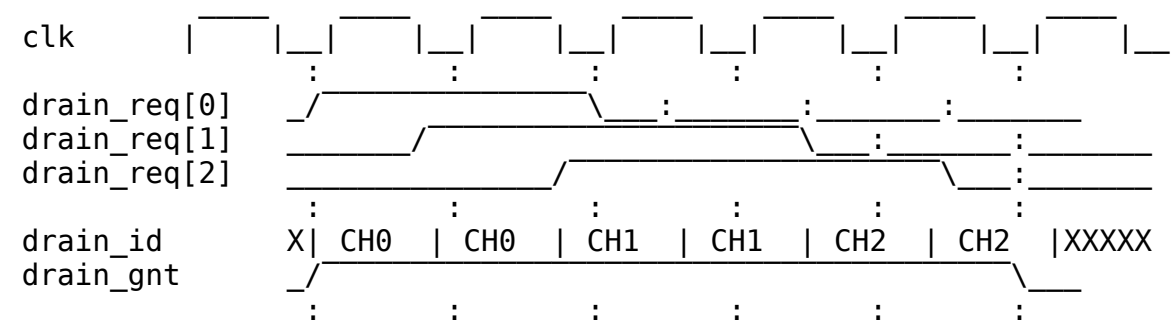
19.3.4 Per-Channel Status

Table 3.5.5: Per-Channel Status

Signal	Direction	Width	Description
`channel_data_avail`	output	NC	Data available per channel
`channel_empty`	output	NC	Channel empty flags
`channel_full`	output	NC	Channel full flags

19.4 Arbitration Logic

19.4.1 Figure 3.5.2: Channel Arbitration Timing



TODO: Replace with simulation-generated waveform showing round-robin arbitration

---

19.5 Integration with AXI Write Engine

```
snk_sram_controller #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
```

```
) u_snk_sram_ctrl (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Per-channel fill interface
    .snk_fill_alloc_req     (fill_alloc_req),
    .snk_fill_alloc_size    (fill_alloc_size),
    .snk_fill_space_free    (fill_space_free),
    .snk_fill_valid         (fill_valid),
    .snk_fill_ready         (fill_ready),
    .snk_fill_data          (fill_data),
    .snk_fill_last          (fill_last),

    // Arbitrated drain interface
    .drain_req              (sram_drain_req),
    .drain_gnt              (sram_drain_gnt),
    .drain_id               (sram_drain_id),
    .drain_beats            (sram_drain_beats),
    .sram_rd_en             (sram_rd_en),
    .sram_rd_addr           (sram_rd_addr),
    .sram_rd_data           (sram_rd_data),

    // Status
    .channel_data_avail     (channel_data_avail),
    .channel_empty          (channel_empty),
    .channel_full           (channel_full)
);
```

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20 Sink SRAM Controller Unit Specification

Module: `snk_sram_controller_unit.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

20.1 Overview

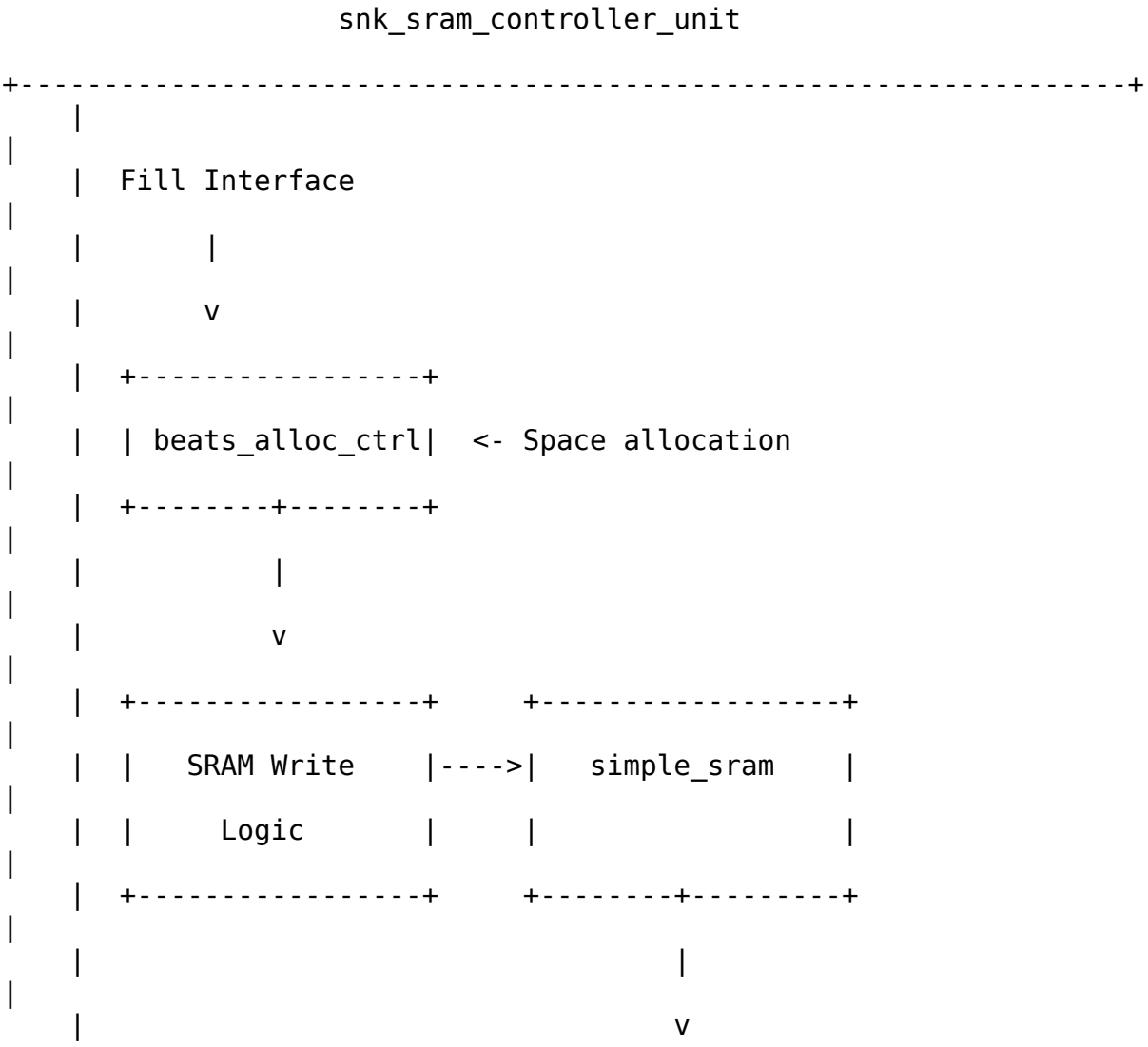
The Sink SRAM Controller Unit manages a single channel's SRAM buffer with beat-level flow control using virtual FIFOs (alloc_ctrl and drain_ctrl).

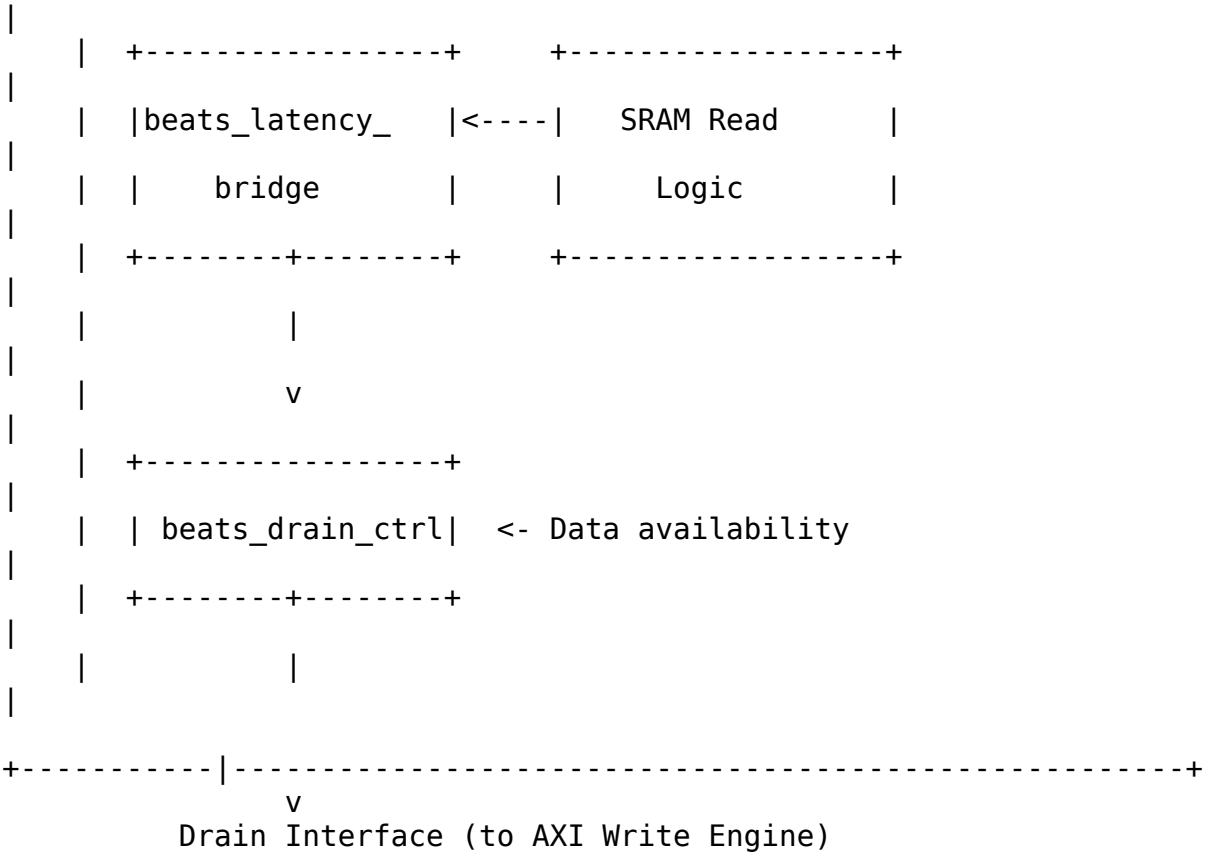
20.1.1 Key Features

- **Single-Channel SRAM:** Dedicated SRAM buffer for one channel
- **Virtual FIFO Flow Control:** beats_alloc_ctrl + beats_drain_ctrl
- **Latency Bridge:** Compensates for pipeline timing
- **Simple SRAM:** No reset on memory, only on control logic

20.1.2 Block Diagram

20.1.3 Figure 3.6.1: Sink SRAM Controller Unit Block Diagram





Source: 03_snk_sram_controller_unit_block.mmd

20.2 Parameters

```
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
parameter int LATENCY_BRIDGE_DEPTH = 4;
```

: Table 3.6.1: Sink SRAM Controller Unit Parameters

20.3 Port List

20.3.1 Clock and Reset

Table 3.6.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

20.3.2 Fill Interface (from Network)

Table 3.6.3: Fill Interface

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Allocate space
`fill_alloc_size`	input	16	Beats to allocate
`fill_space_free`	output	16	Available space
`fill_valid`	input	1	Fill data valid
`fill_ready`	output	1	Ready for fill data
`fill_data`	input	DW	Fill data
`fill_last`	input	1	Last beat marker

20.3.3 Drain Interface (to AXI Write Engine)

Table 3.6.4: Drain Interface

Signal	Direction	Width	Description
`drain_req`	output	1	Data available to drain
`drain_gnt`	input	1	Drain grant
`drain_beats`	output	16	Beats available
`sram_rd_en`	input	1	SRAM read enable

Signal	Direction	Width	Description
sram_rd_addr	input	AW	SRAM read address
sram_rd_data	output	DW	SRAM read data

20.3.4 Status

Table 3.6.5: Status Interface

Signal	Direction	Width	Description
data_avail	output	1	Data available flag
empty	output	1	Buffer empty
full	output	1	Buffer full

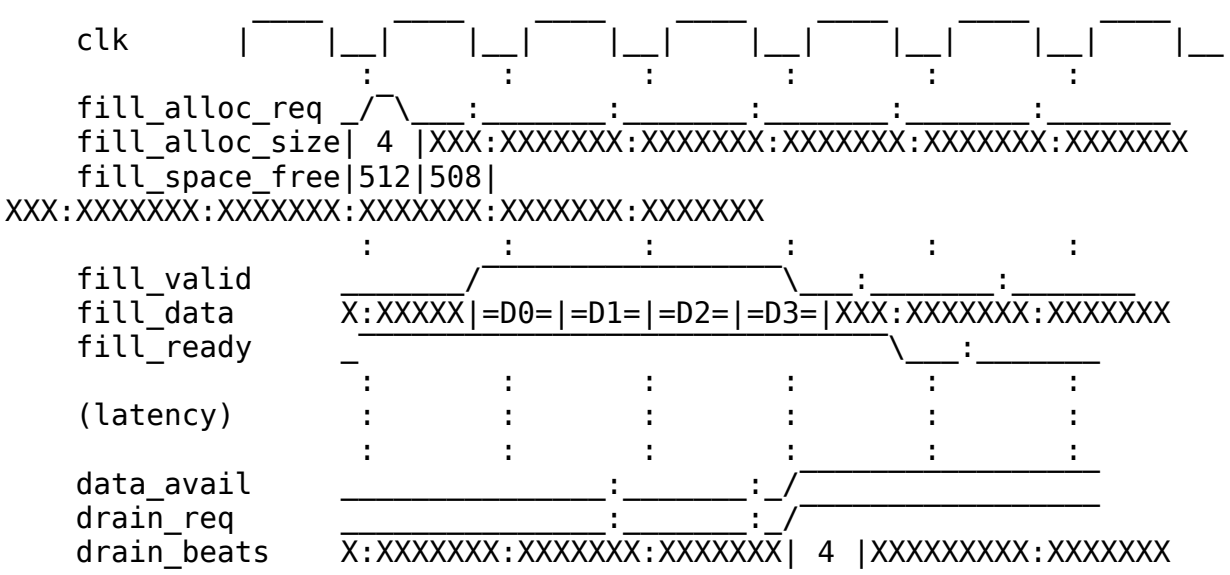
20.4 Internal Architecture

20.4.1 Figure 3.6.2: Internal Data Flow

```
Fill Path:

   fill_alloc_req -----> beats_alloc_ctrl -----> wr_ptr management
                                 |
   fill_valid/data ---------> | -----> SRAM write
                                 |
   fill_ready <-------------- | <----- space_free > 0


Drain Path:

   SRAM read <--------------- | <----- sram_rd_en
        |                     |
        v                     |
   beats_latency_bridge ------+-----> timing compensation
        |                     |
        v                     |
   beats_drain_ctrl ----------+-----> data_avail tracking
        |
        v
   drain_req ----------------> channel arbiter
```

20.5 Timing Diagram

20.5.1 Figure 3.6.3: Fill and Drain Timing

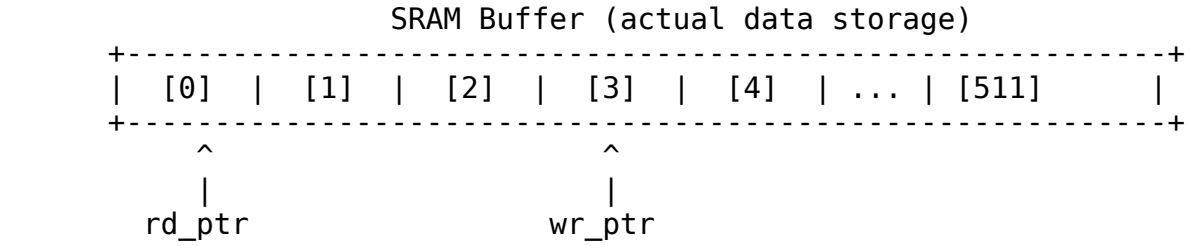


TODO: Replace with simulation-generated waveform showing complete fill-to-drain flow

20.6 Virtual FIFO Concept

The controller uses “virtual FIFOs” - pointer-based flow control without duplicating data storage:

20.6.1 Figure 3.6.4: Virtual FIFO Operation



20.7 Integration Example

```
snk_sram_controller_unit #(
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512),
    .LATENCY_BRIDGE_DEPTH(4)
) u_snk_sram_unit (
    .clk                (clk),
    .rst_n              (rst_n),

    // Fill interface
    .fill_alloc_req     (fill_alloc_req[ch]),
    .fill_alloc_size    (fill_alloc_size[ch]),
    .fill_space_free    (fill_space_free[ch]),
    .fill_valid         (fill_valid[ch]),
    .fill_ready         (fill_ready[ch]),
    .fill_data          (fill_data[ch]),
    .fill_last          (fill_last[ch]),

    // Drain interface
    .drain_req          (drain_req[ch]),
    .drain_gnt          (drain_gnt[ch]),
    .drain_beats        (drain_beats[ch]),
    .sram_rd_en         (sram_rd_en[ch]),
    .sram_rd_addr       (sram_rd_addr[ch]),
    .sram_rd_data       (sram_rd_data[ch]),

    // Status
    .data_avail         (data_avail[ch]),
    .empty              (empty[ch]),
    .full               (full[ch])
);
```

Last Updated: 2025-01-10

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21 Source Data Path Specification

Module: `source_data_path.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

21.1 Overview

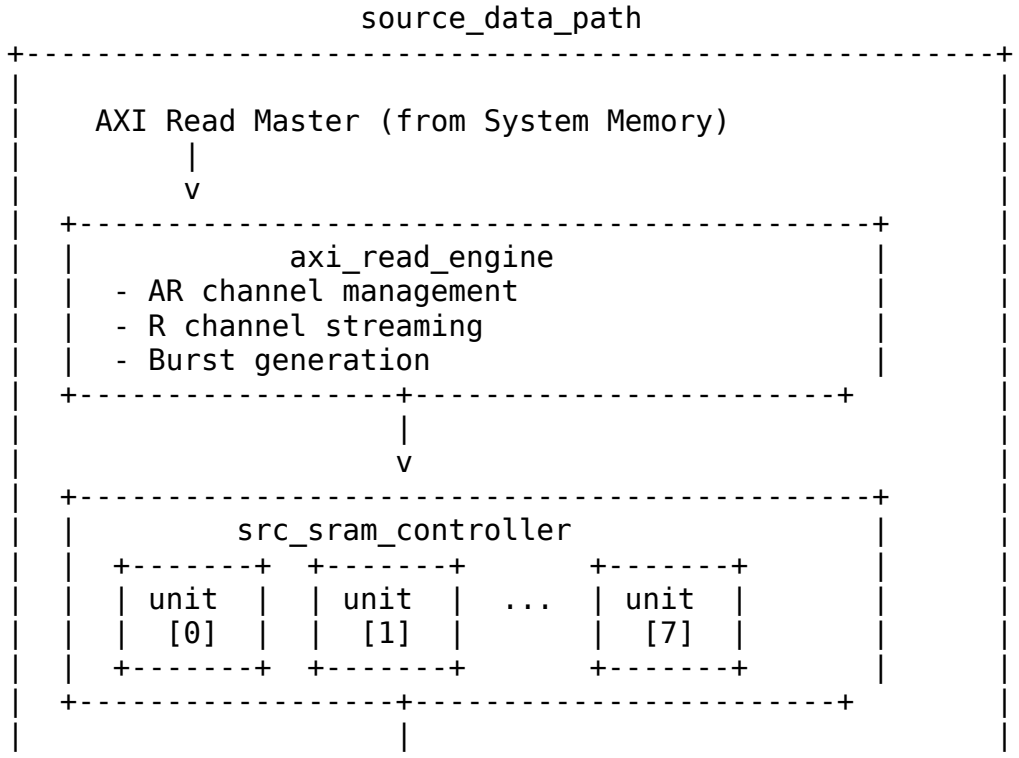
The Source Data Path integrates the AXI read engine and SRAM controller for memory-to-network transfers. Data is read from system memory via AXI4, buffered in SRAM, and delivered through the drain interface to the network.

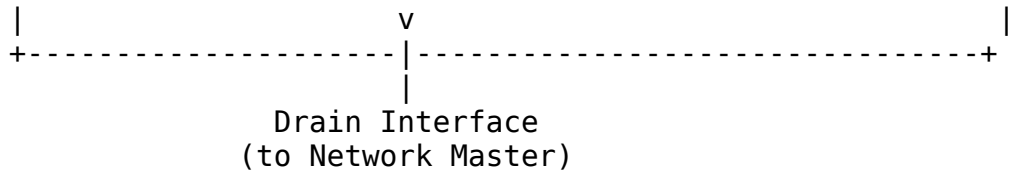
21.1.1 Key Features

- **AXI Read Engine:** Streaming burst reads from system memory
- **8-Channel SRAM Controller:** Per-channel buffering with flow control
- **Beat-Level Tracking:** Precise flow control at beat granularity
- **Scheduler Integration:** Receives transfer requests from scheduler

21.1.2 Block Diagram

21.1.3 Figure 3.7.1: Source Data Path Block Diagram





Source: 03_source_data_path_block.mmd

21.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;
parameter int AR_MAX_OUTSTANDING = 8;
parameter int R_PHASE_FIFO_DEPTH = 64;
```

: Table 3.7.1: Source Data Path Parameters

21.3 Data Flow

21.3.1 Figure 3.7.2: Source Path Data Flow Sequence

```
1. Scheduler Read Request
   - sched_rd_valid = 1
   - sched_rd_addr = source address
   - sched_rd_beats = N
   - sched_rd_id = channel
       |
       v
2. Space Check (beats_alloc_ctrl)
   - Check src_drain_space_free[channel] >= N
   - Allocate N beats of space
       |
       v
3. AXI Read Transaction
   - axi_read_engine issues AR
   - m_axi_araddr = sched_rd_addr
   - m_axi_arlen = calculated from beats
       |
       v
4. R Channel Data
   - Data arrives on m_axi_rdata
```

```
   - Written to channel's SRAM partition
        |
        v
5. SRAM Write Complete
   - beats_drain_ctrl notified via latency bridge
   - Data available for drain
        |
        v
6. Drain Interface
   - src_drain_valid = 1
   - Data delivered to network master
        |
        v
7. Completion
   - sched_rd_done_strobe = 1
   - Scheduler notified of completion
```

Source: 03_source_data_flow.mmd

21.4 Port List

21.4.1 Clock and Reset

Table 3.7.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

21.4.2 Scheduler Interface

Table 3.7.3: Scheduler Interface

Signal	Direction	Width	Description
`sched_rd_valid`	input	1	Read request
`sched_rd_addr`	input	AW	Source address
`sched_rd_beats`	input	32	Beats to read
`sched_rd_id`	input	3	Channel ID
`sched_rd_done_strobe`	output	1	Read complete

Signal	Direction	Width	Description
sched_rd_beats_done	output	32	Beats completed
sched_rd_error	output	1	Error flag

21.4.3 AXI Read Master Interface

Table 3.7.4: AXI Read Master Interface

Signal	Direction	Width	Description
m_axi_arvalid	output	1	AR channel valid
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	Read address
m_axi_arlen	output	8	Burst length
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type
m_axi_arid	output	ID_W	Transaction ID
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Read response
m_axi_rid	input	ID_W	Response ID
m_axi_rlast	input	1	Last beat

21.4.4 Drain Interface (Output to Network)

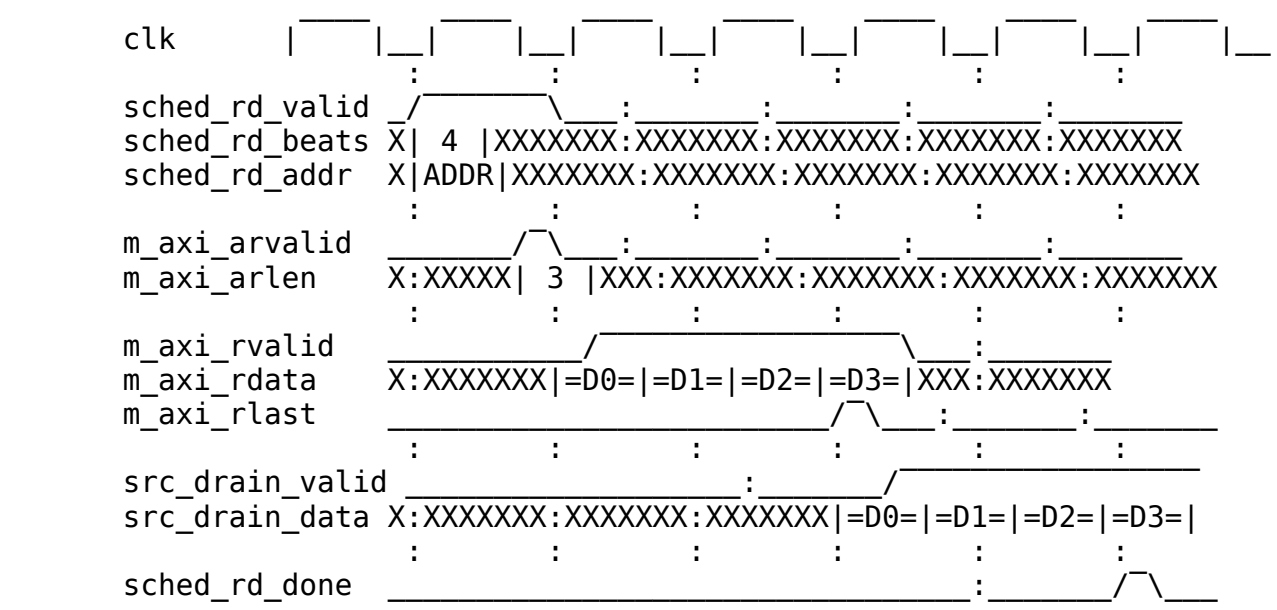
Table 3.7.5: Drain Interface

Signal	Direction	Width	Description
src_drain_valid	output	1	Drain data valid
src_drain_ready	input	1	Ready for drain data

Signal	Direction	Width	Description
src_drain_data	output	DW	Drain data
src_drain_last	output	1	Last beat marker
src_drain_id	output	3	Channel ID
src_drain_data_avail	output	NC*16	Available data per channel

21.5 Timing Diagram

21.5.1 Figure 3.7.3: Source Path Transfer Timing



TODO: Replace with simulation-generated waveform showing complete source transfer

---

21.6 Integration Example

```
source_data_path #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_source_path (
```

```
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Scheduler interface
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),
    .sched_rd_id            (sched_rd_id),
    .sched_rd_done_strobe   (sched_rd_done_strobe),
    .sched_rd_beats_done    (sched_rd_beats_done),
    .sched_rd_error         (sched_rd_error),

    // AXI read master
    .m_axi_arvalid          (src_axi_arvalid),
    .m_axi_arready          (src_axi_arready),
    .m_axi_araddr           (src_axi_araddr),
    .m_axi_arlen            (src_axi_arlen),
    .m_axi_rvalid           (src_axi_rvalid),
    .m_axi_rready           (src_axi_rready),
    .m_axi_rdata            (src_axi_rdata),
    .m_axi_rlast            (src_axi_rlast),

    // Drain interface
    .src_drain_valid        (src_drain_valid),
    .src_drain_ready        (src_drain_ready),
    .src_drain_data         (src_drain_data),
    .src_drain_last         (src_drain_last),
    .src_drain_id           (src_drain_id),
    .src_drain_data_avail   (src_data_avail)
);
```

Last Updated: 2025-01-10

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22 Source Data Path AXIS Wrapper Specification

Module: `source_data_path_axis.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

22.1 Overview

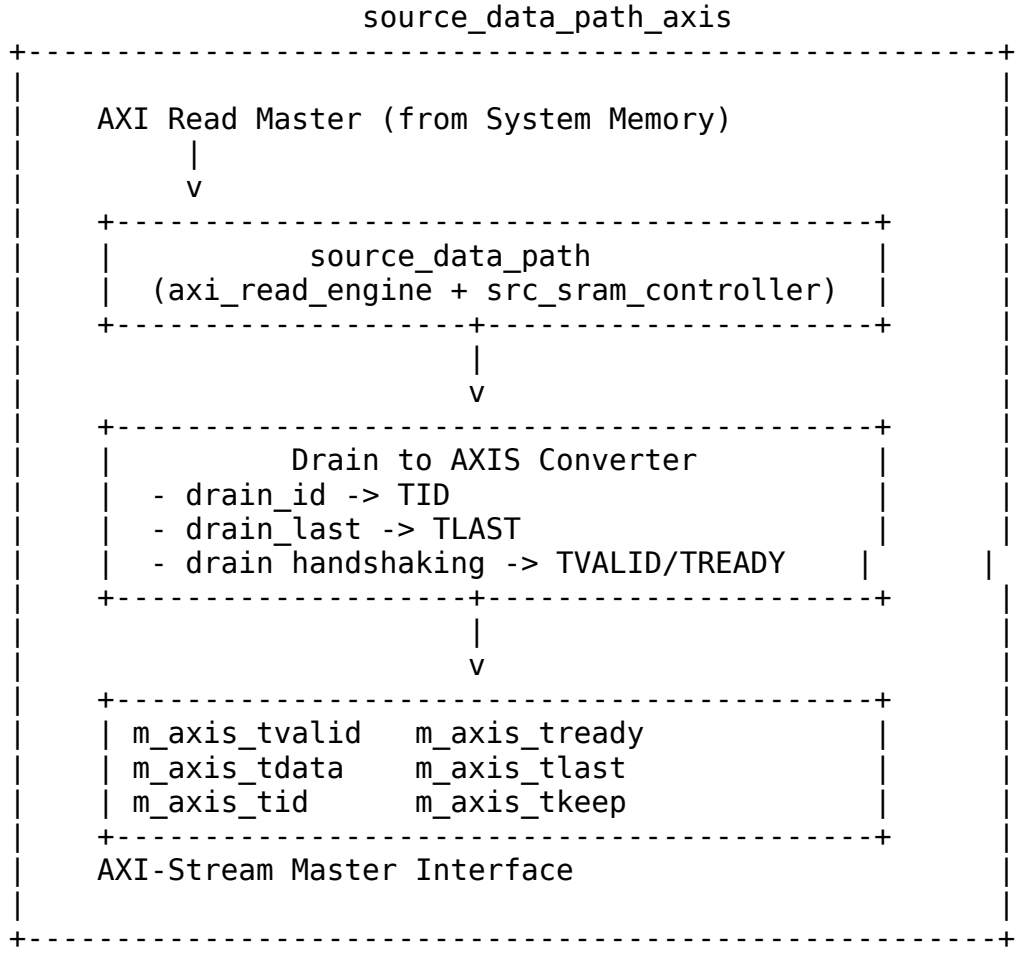
The Source Data Path AXIS wrapper adds an AXI-Stream master interface to the source data path, enabling standard AXIS egress for memory-to-network transfers.

22.1.1 Key Features

- **AXI-Stream Master Interface:** Standard AXIS for data egress
- **Per-Channel TID Mapping:** Drain channel ID maps to AXIS TID
- **TLAST Generation:** Packet boundary marking
- **Core Source Integration:** Wraps source_data_path module

22.1.2 Block Diagram

22.1.3 Figure 3.8.1: Source Data Path AXIS Block Diagram



Source: 03_source_data_path_axis_block.mmd

22.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;
parameter int TID_WIDTH = 3;                    // log2(NUM_CHANNELS)
parameter int TDEST_WIDTH = 1;
parameter int TUSER_WIDTH = 1;
```

: Table 3.8.1: Source Data Path AXIS Parameters

22.3 Port List

22.3.1 Clock and Reset

Table 3.8.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low reset

22.3.2 Scheduler Interface

Table 3.8.3: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_valid	input	1	Read request
sched_rd_addr	input	AW	Source address
sched_rd_beats	input	32	Beats to read
sched_rd_id	input	3	Channel ID
sched_rd_done_strobe	output	1	Read complete
sched_rd_beats_done	output	32	Beats

Signal	Direction	Width	Description
			completed
sched_rd_error	output	1	Error flag

22.3.3 AXI Read Master Interface

Table 3.8.4: AXI Read Master Interface

Signal	Direction	Width	Description
m_axi_arvalid	output	1	AR channel valid
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	Read address
m_axi_arlen	output	8	Burst length
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type
m_axi_arid	output	ID_W	Transaction ID
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Read response
m_axi_rid	input	ID_W	Response ID
m_axi_rlast	input	1	Last beat

22.3.4 AXI-Stream Master Interface

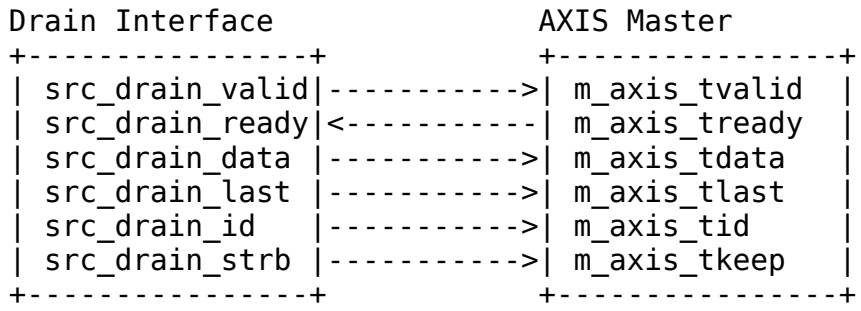
Table 3.8.5: AXI-Stream Master Interface

Signal	Direction	Width	Description
m_axis_tvalid	output	1	Data valid
m_axis_tready	input	1	Ready for data
m_axis_tdata	output	DATA_WIDTH	Data payload
m_axis_tkeep	output	DATA_WIDTH/	Byte enables

Signal	Direction	Width	Description
		8	
m_axis_tlast	output	1	Last beat of packet
m_axis_tid	output	TID_WIDTH	Stream ID (channel)
m_axis_tdest	output	TDEST_WIDTH	Destination
m_axis_tuser	output	TUSER_WIDTH	User sideband

22.4 Signal Mapping

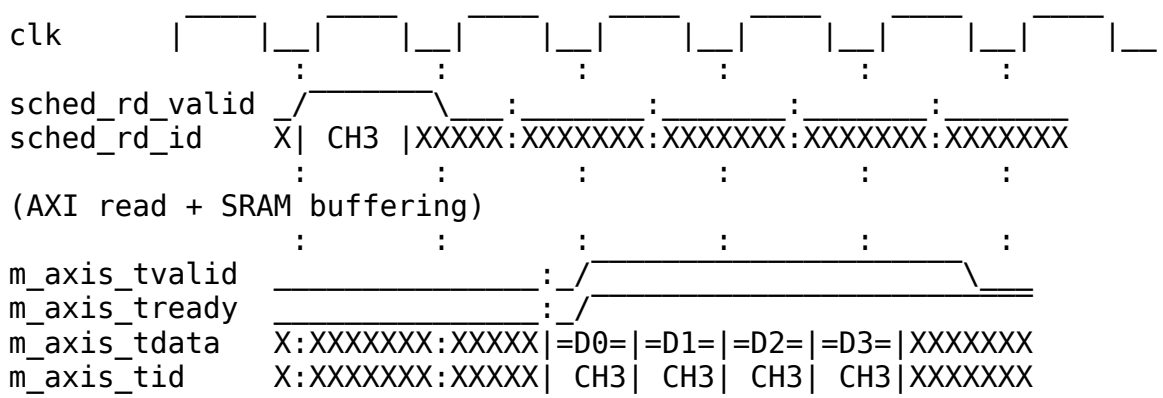
22.4.1 Figure 3.8.2: Drain to AXIS Interface Mapping

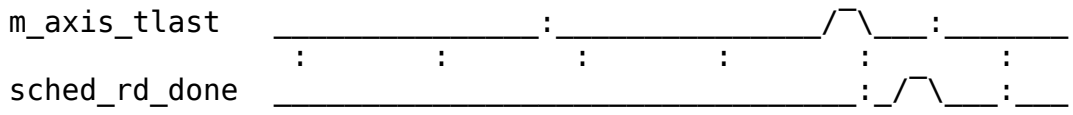


Source: 03_source_axis_mapping.mmd

22.5 Timing Diagram

22.5.1 Figure 3.8.3: AXIS Egress Timing





TODO: Replace with simulation-generated waveform showing AXIS packet transmission

---

22.6 Usage Notes

1. **TID Generation:** Drain channel ID directly maps to AXIS TID
2. **TLAST Generation:** Drain last signal maps to AXIS TLAST
3. **Backpressure:** Network TREADY backpressure propagates to SRAM drain
4. **Packet Boundaries:** TLAST marks end of AXIS packets

22.7 Integration Example

```
source_data_path_axis #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512),
    .TID_WIDTH(3)
) u_source_axis (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Scheduler interface
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),
    .sched_rd_id            (sched_rd_id),
    .sched_rd_done_strobe   (sched_rd_done_strobe),

    // AXI read master
    .m_axi_arvalid          (src_axi_arvalid),
    .m_axi_arready          (src_axi_arready),
    .m_axi_araddr           (src_axi_araddr),
    .m_axi_rvalid           (src_axi_rvalid),
    .m_axi_rready           (src_axi_rready),
    .m_axi_rdata            (src_axi_rdata),

    // AXIS master
    .m_axis_tvalid          (network_tvalid),
    .m_axis_tready          (network_tready),
    .m_axis_tdata           (network_tdata),
```

```
    .m_axis_tlast           (network_tlast),
    .m_axis_tid             (network_tid)
);
```

Last Updated: 2025-01-10

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23 Source SRAM Controller Specification

Module: `src_sram_controller.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

23.1 Overview

The Source SRAM Controller manages 8 SRAM controller units for the source data path, providing per-channel buffering with channel arbitration for data delivery to the network.

23.1.1 Key Features

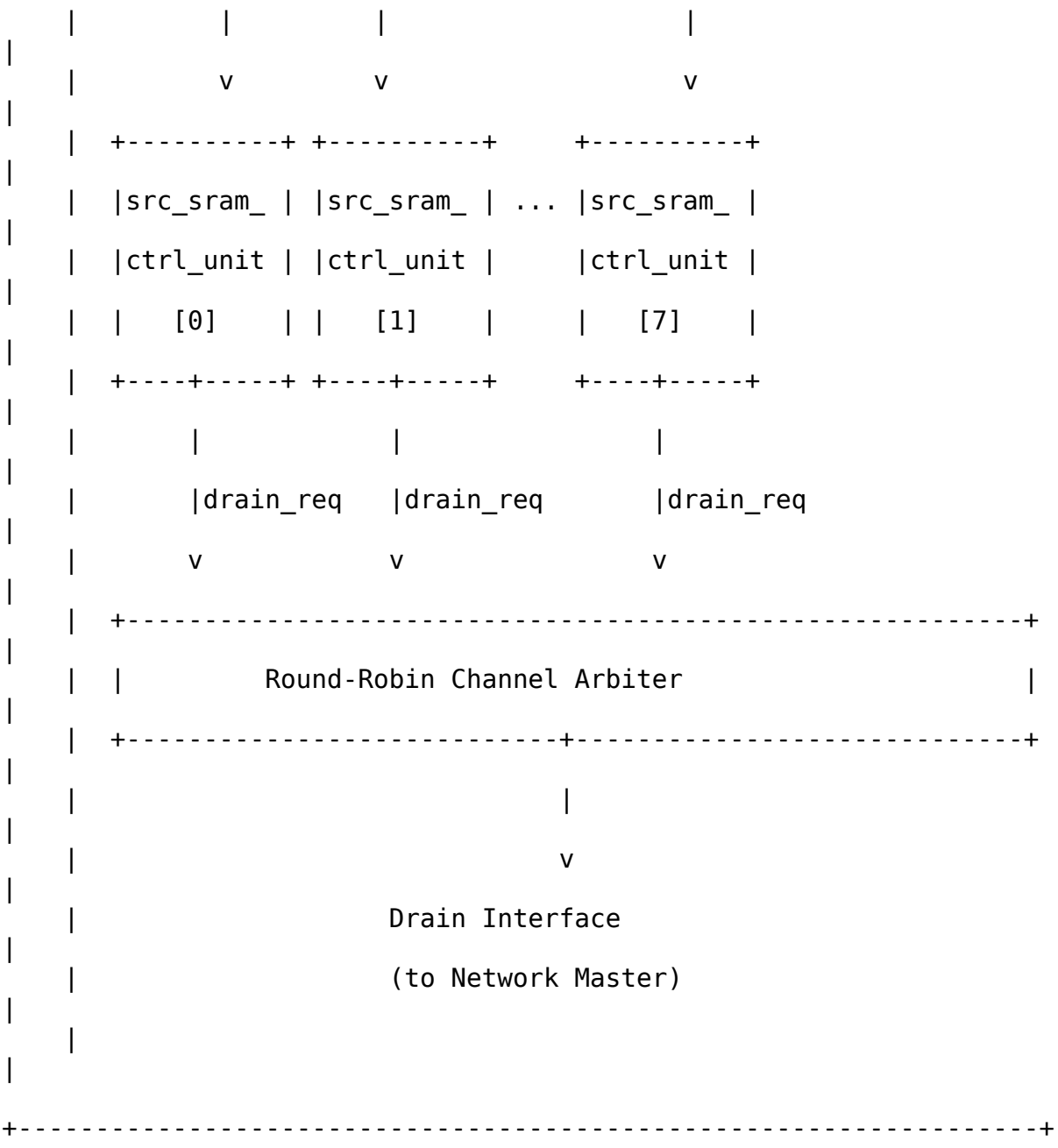
- **8-Channel SRAM Array:** Instantiates 8 `src_sram_controller_unit` modules
- **Channel Arbitration:** Round-robin access for drain requests
- **Flow Control Integration:** Per-channel alloc_ctrl and drain_ctrl
- **Data Availability Tracking:** Reports available data per channel

23.1.2 Block Diagram

23.1.3 Figure 3.9.1: Source SRAM Controller Block Diagram

```
                    src_sram_controller

+-------------------------------------------------------------------+
    |
|
    |    SRAM Write Interface (from AXI Read Engine)
|
```



Source: 03_src_sram_controller_block.mmd

23.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
```

```
parameter int SRAM_DEPTH = 512;                    // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
```

: Table 3.9.1: Source SRAM Controller Parameters

23.3 Port List

23.3.1 Clock and Reset

Table 3.9.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

23.3.2 Per-Channel Fill Interface (from AXI Read Engine)

Table 3.9.3: Per-Channel Fill Interface

Signal	Direction	Width	Description
`src_fill_alloc_req`	input	NC	Allocation request per channel
`src_fill_alloc_size`	input	NC*16	Beats to allocate
`src_fill_space_free`	output	NC*16	Available space per channel
`src_fill_valid`	input	NC	Fill data valid
`src_fill_ready`	output	NC	Ready for fill data
`src_fill_data`	input	NC*DW	Fill data
`src_fill_last`	input	NC	Last beat marker

23.3.3 Arbitrated Drain Interface (to Network)

Table 3.9.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
`drain_valid`	output	1	Drain data valid
`drain_ready`	input	1	Ready for drain
`drain_data`	output	DW	Drain data
`drain_id`	output	3	Source channel ID
`drain_last`	output	1	Last beat of transfer

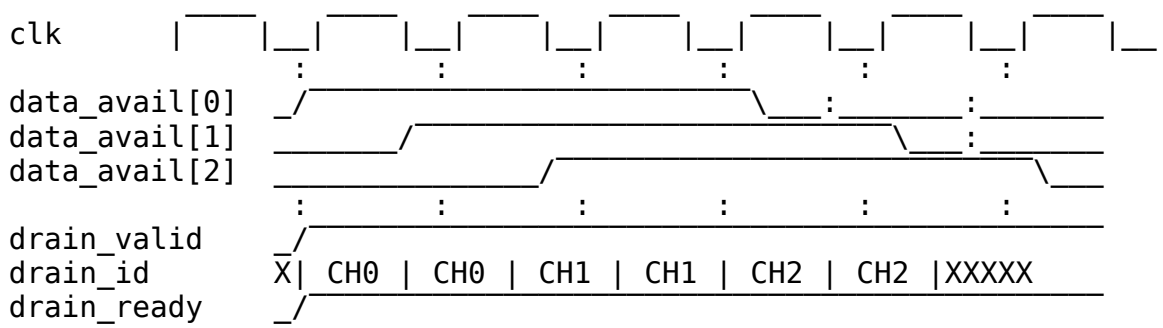
23.3.4 Per-Channel Status

Table 3.9.5: Per-Channel Status

Signal	Direction	Width	Description
`channel_data_avail`	output	NC*16	Data available per channel
`channel_empty`	output	NC	Channel empty flags
`channel_full`	output	NC	Channel full flags

23.4 Arbitration Logic

23.4.1 Figure 3.9.2: Source Channel Arbitration



TODO: Replace with simulation-generated waveform showing round-robin drain

---

23.5 Comparison: Sink vs Source SRAM Controller

Table 3.9.6: Sink vs Source Controller Comparison

Aspect	Sink Controller	Source Controller
Fill Source	Network ingress	AXI read data
Drain Destination	AXI write engine	Network egress
Fill Interface	Per-channel from network	Per-channel from AXI R
Drain Interface	Arbitrated SRAM read	Arbitrated drain to network
Primary Use	Network -> Memory	Memory -> Network

23.6 Integration Example

```
src_sram_controller #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_src_sram_ctrl (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Per-channel fill interface (from AXI read engine)
    .src_fill_alloc_req     (fill_alloc_req),
    .src_fill_alloc_size    (fill_alloc_size),
    .src_fill_space_free    (fill_space_free),
    .src_fill_valid         (fill_valid),
    .src_fill_ready         (fill_ready),
    .src_fill_data          (fill_data),
    .src_fill_last          (fill_last),

    // Arbitrated drain interface (to network)
    .drain_valid            (src_drain_valid),
    .drain_ready            (src_drain_ready),
    .drain_data             (src_drain_data),
    .drain_id               (src_drain_id),
    .drain_last             (src_drain_last),
```

```
    // Status
    .channel_data_avail     (channel_data_avail),
    .channel_empty          (channel_empty),
    .channel_full           (channel_full)
);
```

Last Updated: 2025-01-10

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24 Source SRAM Controller Unit Specification

Module: `src_sram_controller_unit.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

24.1 Overview

The Source SRAM Controller Unit manages a single channel's SRAM buffer for the source data path, with beat-level flow control using virtual FIFOs.

24.1.1 Key Features

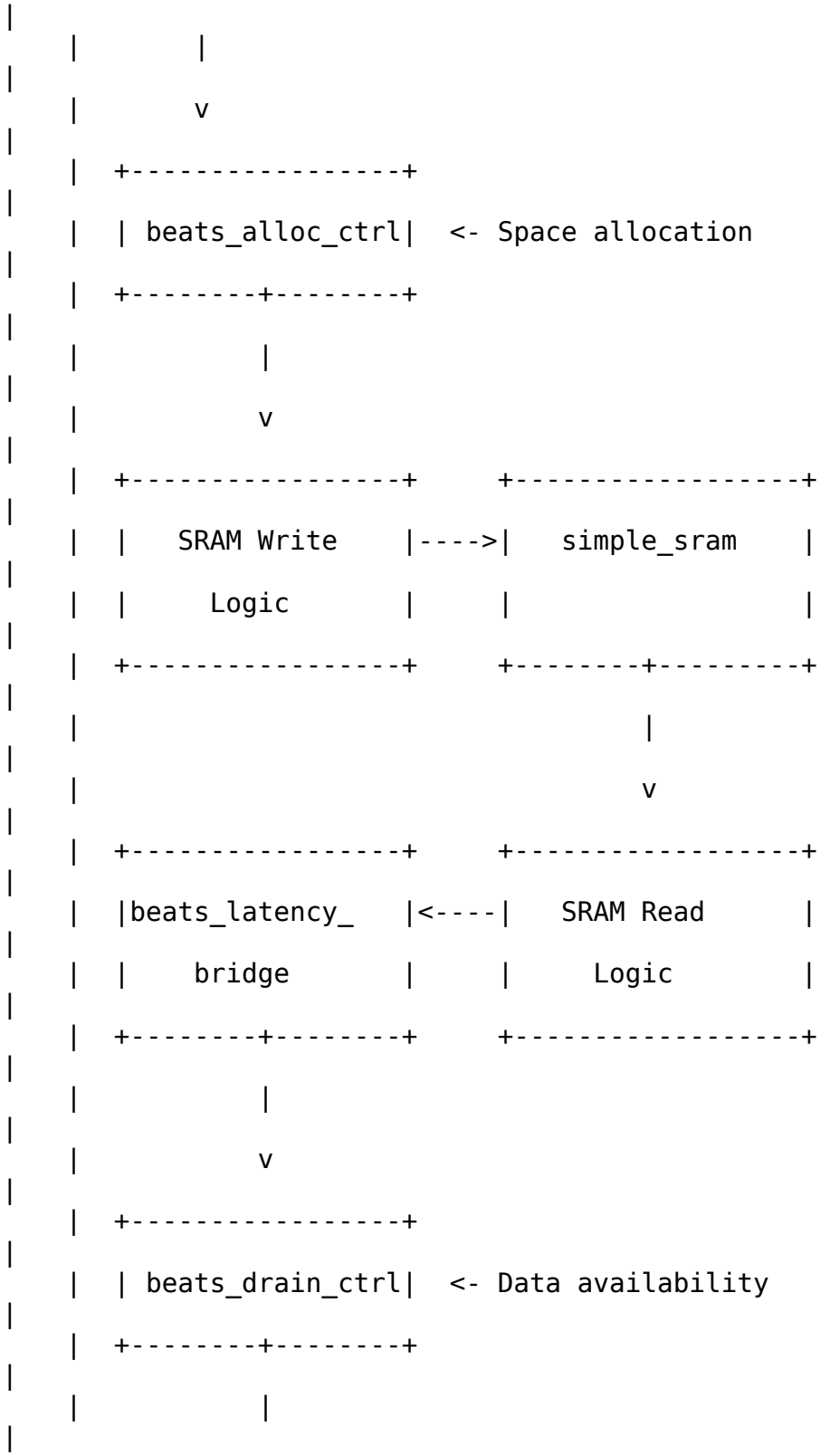
- **Single-Channel SRAM:** Dedicated SRAM buffer for one channel
- **Virtual FIFO Flow Control:** beats_alloc_ctrl + beats_drain_ctrl
- **Latency Bridge:** Compensates for pipeline timing
- **Simple SRAM:** No reset on memory, only on control logic

24.1.2 Block Diagram

24.1.3 Figure 3.10.1: Source SRAM Controller Unit Block Diagram

```
                    src_sram_controller_unit

+-------------------------------------------------------------------+
    |
|
    |  Fill Interface (from AXI Read Engine)
```


beats_alloc_ctrl
Space allocation
SRAM Write Logic
simple_sram
beats_latency_ bridge
SRAM Read Logic
beats_drain_ctrl
Data availability

```
+-----------|-----------------------------------------------------+
                v
          Drain Interface (to Network)
```

Source: 03_src_sram_controller_unit_block.mmd

24.2 Parameters

```
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
parameter int LATENCY_BRIDGE_DEPTH = 4;
```

: Table 3.10.1: Source SRAM Controller Unit Parameters

24.3 Port List

24.3.1 Clock and Reset

Table 3.10.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

24.3.2 Fill Interface (from AXI Read Engine)

Table 3.10.3: Fill Interface

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Allocate space
`fill_alloc_size`	input	16	Beats to allocate
`fill_space_free`	output	16	Available space
`fill_valid`	input	1	Fill data valid
`fill_ready`	output	1	Ready for fill data

Signal	Direction	Width	Description
fill_data	input	DW	Fill data
fill_last	input	1	Last beat marker

24.3.3 Drain Interface (to Network)

Table 3.10.4: Drain Interface

Signal	Direction	Width	Description
drain_req	output	1	Data available to drain
drain_gnt	input	1	Drain grant from arbiter
drain_beats	output	16	Beats available
drain_valid	output	1	Drain data valid
drain_ready	input	1	Network ready
drain_data	output	DW	Drain data
drain_last	output	1	Last beat marker

24.3.4 Status

Table 3.10.5: Status Interface

Signal	Direction	Width	Description
data_avail	output	16	Data available count
empty	output	1	Buffer empty
full	output	1	Buffer full

24.4 Internal Architecture

24.4.1 Figure 3.10.2: Internal Data Flow

```
Fill Path (from AXI Read Engine):

   fill_alloc_req -----> beats_alloc_ctrl -----> wr_ptr management
```

```
                                |
   fill_valid/data ---------->  | -----> SRAM write
                                |
   fill_ready <--------------   | <----- space_free > 0


Drain Path (to Network):

   SRAM read <---------------   | <----- drain_gnt && drain_ready
        |                       |
        v                       |
   beats_latency_bridge ------+-----> timing compensation
        |                       |
        v                       |
   beats_drain_ctrl ----------+-----> data_avail tracking
        |
        v
   drain_req ----------------->  channel arbiter
   drain_valid/data ---------->  network interface
```

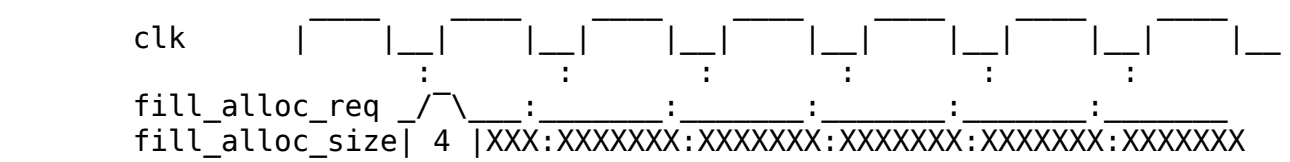
24.5 Comparison: Sink vs Source Unit

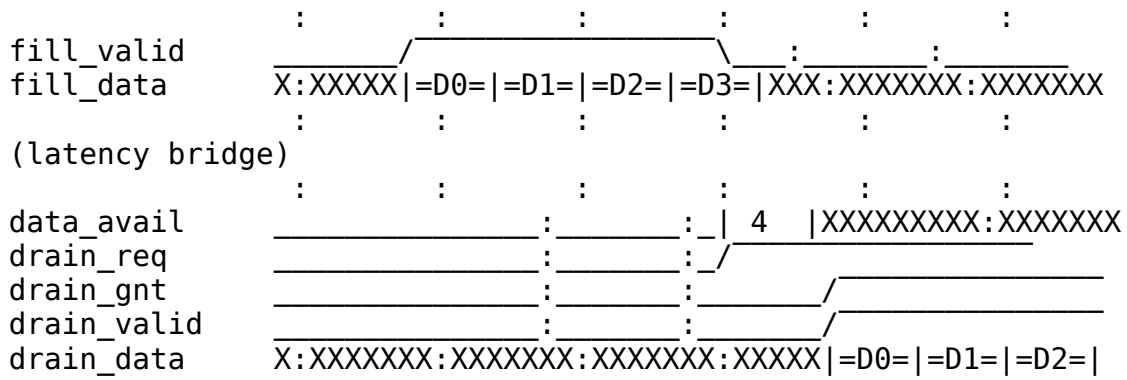
Table 3.10.6: Sink vs Source Unit Comparison

Aspect	Sink Unit	Source Unit
Fill Source	Network	AXI Read Engine
Drain Destination	AXI Write Engine	Network
Fill Handshake	fill_valid/ready	fill_valid/ready
Drain Handshake	SRAM read interface	drain_valid/ready
Primary Direction	Network -> SRAM -> AXI	AXI -> SRAM -> Network

24.6 Timing Diagram

24.6.1 Figure 3.10.3: Source Unit Fill and Drain Timing





TODO: Replace with simulation-generated waveform showing complete flow

---

24.7 Integration Example

```
src_sram_controller_unit #(
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512),
    .LATENCY_BRIDGE_DEPTH(4)
) u_src_sram_unit (
    .clk                (clk),
    .rst_n              (rst_n),

    // Fill interface (from AXI read engine)
    .fill_alloc_req     (fill_alloc_req[ch]),
    .fill_alloc_size    (fill_alloc_size[ch]),
    .fill_space_free    (fill_space_free[ch]),
    .fill_valid         (fill_valid[ch]),
    .fill_ready         (fill_ready[ch]),
    .fill_data          (fill_data[ch]),
    .fill_last          (fill_last[ch]),

    // Drain interface (to network)
    .drain_req          (drain_req[ch]),
    .drain_gnt          (drain_gnt[ch]),
    .drain_beats        (drain_beats[ch]),
    .drain_valid        (drain_valid[ch]),
    .drain_ready        (drain_ready[ch]),
    .drain_data         (drain_data[ch]),
    .drain_last         (drain_last[ch]),

    // Status
    .data_avail         (data_avail[ch]),
    .empty              (empty[ch]),
```

```
    .full                (full[ch])
);
```

Last Updated: 2025-01-10

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25 RAPIDS Core Beats Specification

Module: `rapids_core_beats.sv` **Location:** `projects/components/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

25.1 Overview

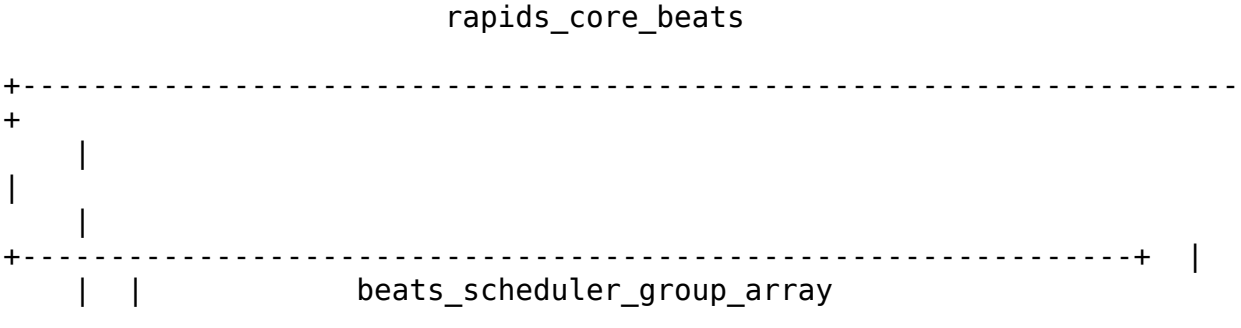
The RAPIDS Core Beats module is the top-level integration of the "beats" architecture, combining the scheduler group array with sink and source data paths.

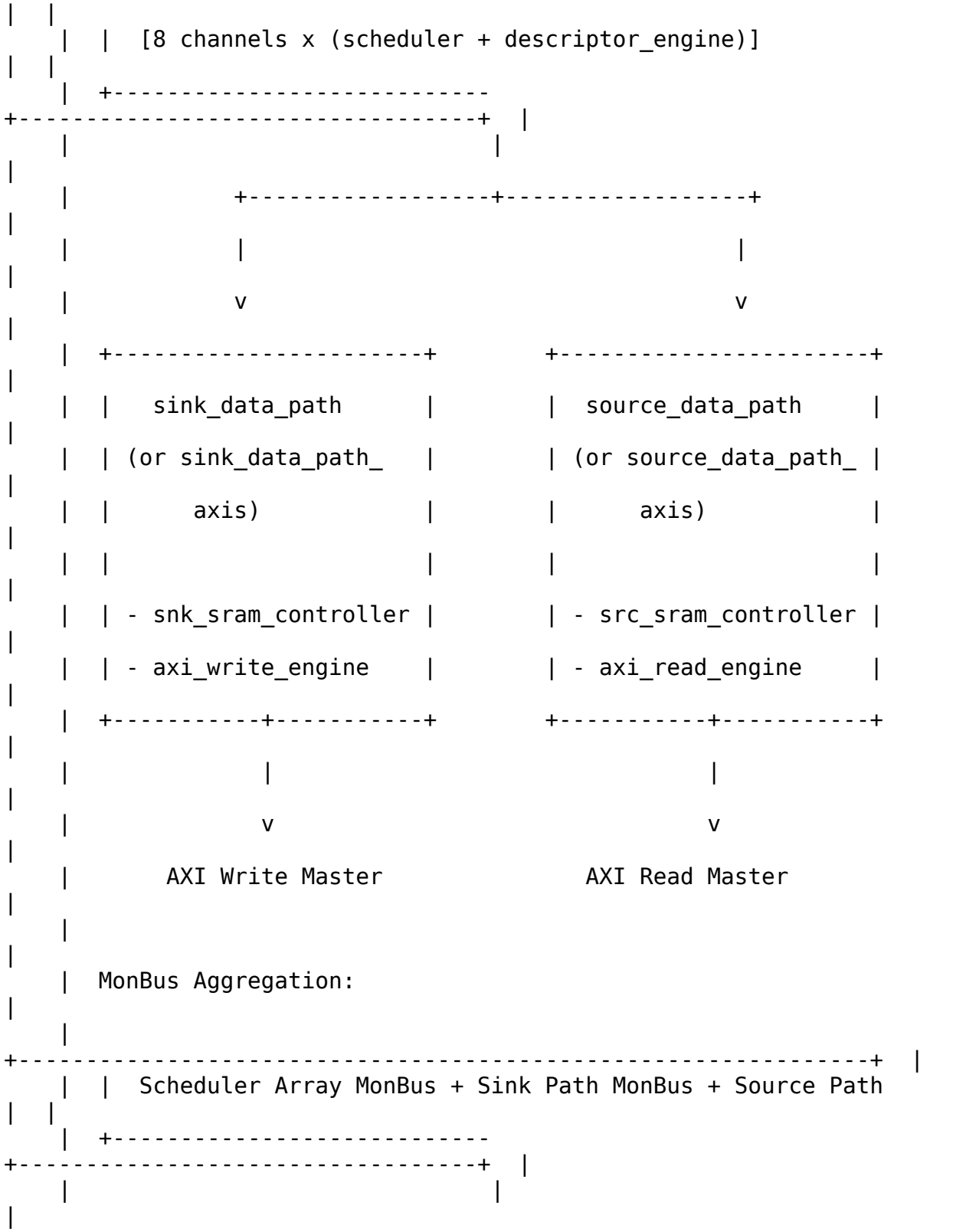
25.1.1 Key Features

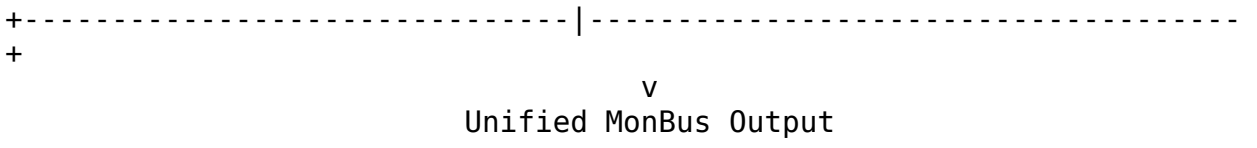
- **Complete RAPIDS Core:** Scheduler array + sink path + source path
- **8-Channel Architecture:** Full multi-channel support
- **Unified MonBus:** Aggregated monitoring from all subsystems
- **Configurable Parameters:** Data width, address width, SRAM depths

25.1.2 Block Diagram

25.1.3 Figure 3.11.1: RAPIDS Core Beats Block Diagram




[8 channels x (scheduler + descriptor_engine)]
sink_data_path
(or sink_data_path_
axis)
- snk_sram_controller
- axi_write_engine
source_data_path
(or source_data_path_
axis)
- src_sram_controller
- axi_read_engine
AXI Write Master
AXI Read Master
MonBus Aggregation:
Scheduler Array MonBus + Sink Path MonBus + Source Path



Source: 03_rapids_core_beats_block.mmd

25.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
parameter int DESC_DATA_WIDTH = 256;
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;

// AXI Parameters
parameter int AR_MAX_OUTSTANDING = 8;
parameter int AW_MAX_OUTSTANDING = 8;
parameter int R_PHASE_FIFO_DEPTH = 64;
parameter int W_PHASE_FIFO_DEPTH = 64;
parameter int B_PHASE_FIFO_DEPTH = 16;

// MonBus Parameters
parameter int MON_UNIT_ID = 1;

// Feature Enables
parameter bit ENABLE_AXIS_WRAPPERS = 0;          // Use AXIS
interfaces
```

: Table 3.11.1: RAPIDS Core Beats Parameters

25.3 Port List

25.3.1 Clock and Reset

Table 3.11.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

25.3.2 Per-Channel APB Programming

Table 3.11.3: APB Programming Interface

Signal	Direction	Width	Description
apb_valid	input	NC	Channel kick-off (per-channel)
apb_ready	output	NC	Ready for kick-off
apb_addr	input	NC*AW	First descriptor address

25.3.3 Per-Channel Configuration

Table 3.11.4: Per-Channel Configuration

Signal	Direction	Width	Description
cfg_channel_enable	input	NC	Enable per channel
cfg_channel_reset	input	NC	Soft reset per channel
cfg_sched_timeout_cycles	input	32	Write-progress timeout window (cycles)
cfg_sched_timeout_limit	input	8	Consecutive-timeout windows before fatal escalation (0 = never)
cfg_sched_timeout_enable	input	NC	Enable timeout
cfg_desceng_enable	input	NC	Enable descriptor engine
cfg_desceng_prefetch	input	NC	Enable prefetch chaining

`cfg_sched_timeout_limit` is passed straight through to the `beats_scheduler_group_array` instance (recoverable-timeout escalation, see the Scheduler FUB). The write-side COMMIT strobes (`sched_wr_commit_strobe` / `sched_wr_commit_beats`) generated by the sink data path's `axi_write_engine_beats` are routed internally from the sink data path up into the scheduler array so completion is gated on B responses.

25.3.4 Descriptor AXI Master Interface

Table 3.11.5: Descriptor AXI Master Interface

Signal	Direction	Width	Description
`desc_m_axi_arvalid`	output	1	AR channel valid
`desc_m_axi_arready`	input	1	AR channel ready
`desc_m_axi_araddr`	output	AW	AR address
`desc_m_axi_rvalid`	input	1	R channel valid
`desc_m_axi_rready`	output	1	R channel ready
`desc_m_axi_rdata`	input	256	R data

25.3.5 Sink AXI Write Master Interface

Table 3.11.6: Sink AXI Write Master Interface

Signal	Direction	Width	Description
`snk_m_axi_awvalid`	output	1	AW channel valid
`snk_m_axi_awready`	input	1	AW channel ready
`snk_m_axi_awaddr`	output	AW	Write address
`snk_m_axi_awlen`	output	8	Burst length
`snk_m_axi_wvalid`	output	1	W channel valid
`snk_m_axi_wready`	input	1	W channel ready
`snk_m_axi_wdata`	output	DW	Write data
`snk_m_axi_wlast`	output	1	Last beat
`snk_m_axi_bvalid`	input	1	B channel

Signal	Direction	Width	Description
			valid
snk_m_axi_bready	output	1	B channel ready
snk_m_axi_bresp	input	2	Write response

25.3.6 Source AXI Read Master Interface

Table 3.11.7: Source AXI Read Master Interface

Signal	Direction	Width	Description
src_m_axi_arvalid	output	1	AR channel valid
src_m_axi_arready	input	1	AR channel ready
src_m_axi_araddr	output	AW	Read address
src_m_axi_arlen	output	8	Burst length
src_m_axi_rvalid	input	1	R channel valid
src_m_axi_rready	output	1	R channel ready
src_m_axi_rdata	input	DW	Read data
src_m_axi_rlast	input	1	Last beat

25.3.7 Sink Fill Interface (or AXIS Slave)

Table 3.11.8: Sink Fill Interface

Signal	Direction	Width	Description
snk_fill_valid	input	1	Fill data valid
snk_fill_ready	output	1	Ready for fill
snk_fill_data	input	DW	Fill data
snk_fill_id	input	3	Channel ID
snk_fill_last	input	1	Last beat

25.3.8 Source Drain Interface (or AXIS Master)

Table 3.11.9: Source Drain Interface

Signal	Direction	Width	Description
src_drain_valid	output	1	Drain data valid
src_drain_ready	input	1	Ready for drain
src_drain_data	output	DW	Drain data
src_drain_id	output	3	Channel ID
src_drain_last	output	1	Last beat

25.3.9 Unified MonBus Interface

Table 3.11.10: Unified MonBus Interface

Signal	Direction	Width	Description
monbus_pkt_valid	output	1	Packet valid
monbus_pkt_ready	input	1	Consumer ready
monbus_pkt_data	output	64	Packet data

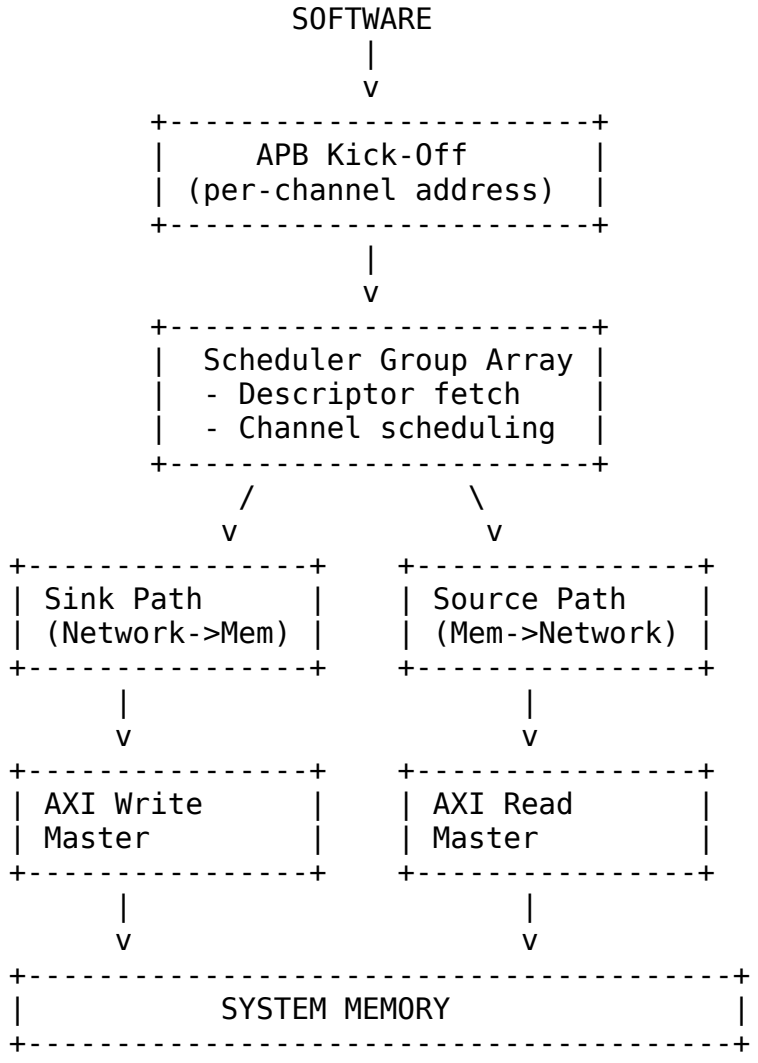
25.3.10 Aggregate Status

Table 3.11.11: Aggregate Status

Signal	Direction	Width	Description
all_channels_idle	output	1	All channels idle
scheduler_idle	output	NC	Per-channel scheduler idle
sink_idle	output	1	Sink path idle
source_idle	output	1	Source path idle
error_flags	output	32	Combined error flags

25.4 Data Flow

25.4.1 Figure 3.11.2: RAPIDS Core Complete Data Flow



25.5 MonBus Aggregation

The unified MonBus output aggregates sources from all subsystems:

Table 3.11.12: MonBus Source Assignment

Source Range	Origin	Description
0-15	Scheduler Array	Per-channel scheduler + desc engine
16-23	Sink Data Path	Sink SRAM controllers
24-31	Source Data Path	Source SRAM controllers
32	AXI Write Engine	Write completions
33	AXI Read Engine	Read completions

25.6 Integration Example

```
rapids_core_beats #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512),
    .ENABLE_AXIS_WRAPPERS(0)
) u_rapids_core (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // APB kick-off
    .apb_valid              (apb_kick_valid),
    .apb_ready              (apb_kick_ready),
    .apb_addr               (apb_kick_addr),

    // Configuration
    .cfg_channel_enable     (cfg_ch_enable),
    .cfg_sched_timeout_cycles(cfg_timeout),

    // Descriptor AXI
    .desc_m_axi_arvalid     (desc_arvalid),
    .desc_m_axi_arready     (desc_arready),
    .desc_m_axi_araddr      (desc_araddr),
    .desc_m_axi_rvalid      (desc_rvalid),
    .desc_m_axi_rready      (desc_rready),
    .desc_m_axi_rdata       (desc_rdata),
```

```
    // Sink AXI write
    .snk_m_axi_awvalid      (snk_awvalid),
    .snk_m_axi_awready      (snk_awready),
    .snk_m_axi_awaddr       (snk_awaddr),
    .snk_m_axi_wvalid       (snk_wvalid),
    .snk_m_axi_wready       (snk_wready),
    .snk_m_axi_wdata        (snk_wdata),
    .snk_m_axi_bvalid       (snk_bvalid),
    .snk_m_axi_bready       (snk_bready),

    // Source AXI read
    .src_m_axi_arvalid      (src_arvalid),
    .src_m_axi_arready      (src_arready),
    .src_m_axi_araddr       (src_araddr),
    .src_m_axi_rvalid       (src_rvalid),
    .src_m_axi_rready       (src_rready),
    .src_m_axi_rdata        (src_rdata),

    // Sink fill interface
    .snk_fill_valid         (network_rx_valid),
    .snk_fill_ready         (network_rx_ready),
    .snk_fill_data          (network_rx_data),
    .snk_fill_id            (network_rx_id),

    // Source drain interface
    .src_drain_valid        (network_tx_valid),
    .src_drain_ready        (network_tx_ready),
    .src_drain_data         (network_tx_data),
    .src_drain_id           (network_tx_id),

    // MonBus
    .monbus_pkt_valid       (mon_valid),
    .monbus_pkt_ready       (mon_ready),
    .monbus_pkt_data        (mon_data),

    // Status
    .all_channels_idle      (rapids_idle)
);
```

Note: `rapids_core_beats` is a core-level block. The register interface, configuration mapping, and top-level integration (APB slave, descriptor kick-off, AXI monitors, and the MonBus AXI-Lite group) are provided by the `rapids_regs`, `rapids_config_block`, and `rapids_beats_top` modules – see RAPIDS Registers, RAPIDS Config Block, and RAPIDS Beats Top.

Last Updated: 2026-07-02

26 RAPIDS Registers Specification

Module: `rapids_regs.sv` (generated) **Source:** `projects/components/rapids/rtl/macro_beats/rapids_regs.rdl`, `rapids_mon_regs.rdl` **Generated:** `projects/components/rapids/regs/generated/rtl/rapids_regs.sv` **Status:** Implemented (PeakRDL-generated)

26.1 Overview

`rapids_regs` is the PeakRDL-generated register block for the RAPIDS Beats top-level. It exposes configuration and status through a single command/response CPU interface and drives the hardware via a `hwif_out` structure that `rapids_config_block` maps onto the core/monitor `cfg_*` signals.

A single addrmap (one APB slave) contains two regfiles:

- **Base config/status** registers at `0x100-0x3FF`.
- A nested **`rapids_mon_regs`** monitor regfile at base `0x1000`, decoded under `hwif_out.MON.*` (all DAXMON / RDMON / WRMON / MON_FIFO and per-monitor performance registers).

Because the monitor regfile lives at `0x1000`, the register-block address decode requires at least **13 bits** (`cpuif_addr` is a 13-bit compare in the generated RTL). The APB address bus feeding the block (`APB_ADDR_WIDTH`) must therefore be >= 13 to reach the monitor regfile.

26.2 Base Register Map (0x100-0x3FF)

Table 3.12.1: Base Register Map

Offset	Name	Access	Description
0x100	GLOBAL_CTRL	RW	`GLOBAL_EN[0]`, `GLOBAL_RST[1]` (self-clearing)

Offset	Name	Access	Description
0x104	GLOBAL_STATUS	RO	SYSTEM_IDLE[0] (all channels idle)
0x108	VERSION	RO	MINOR[7:0]=0x5A, MAJOR[15:8], NUM_CHANNELS[23:16]=0x08
0x120	CHANNEL_ENABLE	RW	CH_EN[7:0] per-channel enable
0x124	CHANNEL_RESET	RW	CH_RST[7:0] per-channel reset (self-clearing)
0x140	CHANNEL_IDLE	RO	CH_IDLE[7:0] per-channel idle
0x144	DESC_ENGINE_IDLE	RO	Per-channel descriptor-engine idle
0x148	SCHEDULER_IDLE	RO	Per-channel scheduler idle (excludes CH_ERROR)
0x150-0x16C	CH_STATE[0..7].STATE	RO	Per-channel FSM state (stride 0x4)
0x170	SCHED_ERROR	RO	Per-channel sticky scheduler error
0x174	AXI_RD_COMPLETE	RO	Per-channel read-complete flags
0x178	AXI_WR_COMPLETE	RO	Per-channel write-complete flags
0x200	SCHED_TIMEOUT_CYCLES	RW	TIMEOUT_CYCLES[31:0] write-progress window (reset 1000)
0x204	SCHED_CONFIG	RW	SCHED_EN, TIMEOUT_EN, ERR_EN, COMPL_EN, PERF_EN
0x208	SCHED_TIMEOUT_LIMIT	RW	LIMIT[7:0] consecutive-timeout escalation limit (reset 4, 0 = never)
0x220	DESCENG_CONFIG	RW	DESCENG_EN[0], PREFETCH_EN[1], FIFO_THRESH[5:2] (reset 8)

Offset	Name	Access	Description
0x224	DESCENG_ADDR0_BASE	RW	Descriptor address range 0 base [31:0]
0x228	DESCENG_ADDR0_LIMIT	RW	Descriptor address range 0 limit [31:0]
0x22C	DESCENG_ADDR1_BASE	RW	Descriptor address range 1 base [31:0]
0x230	DESCENG_ADDR1_LIMIT	RW	Descriptor address range 1 limit [31:0]
0x240	CTRL_CONFIG	RW	CTRLRD_MAX_TRY[8:0] control-read poll retry budget (0-511, reset 16)
0x2A0	AXI_XFER_CONFIG	RW	AXI transfer sizing configuration
0x2B0	PERF_CONFIG	RW	Performance profiler configuration
0x2C0	OBS_CTRL	RW	Observation mux control (channel/category select)
0x2C4	OBS_FLAGS	RO	Observation flags
0x2C8	OBS_DATA0	RO	Observation data word 0
0x2CC	OBS_DATA1	RO	Observation data word 1
0x35C	PERF_CH_SEL	RW	Performance channel select
0x378	HIST_SEL	RW	Histogram bucket select
0x37C	HIST_DATA	RO	Histogram bucket data
0x380	HIST_TOTAL	RO	Histogram total count

26.2.1 Key Register Fields

SCHED_TIMEOUT_LIMIT (0x208) – `LIMIT[7:0]` sets the number of consecutive write-progress timeout windows a channel tolerates before the recoverable timeout escalates to a fatal, sticky `CH_ERROR`. `0` = never escalate (pure soft timeout). Total time to escalate is approximately `LIMIT` x `TIMEOUT_CYCLES`. Reset value is 4.

DESCENG_CONFIG (0x220) – `PREFETCH_EN[1]` and `FIFO_THRESH[5:2]` control the now-functional descriptor prefetch: prefetch off = on-demand chaining (~1 ahead), prefetch on = buffer up to `FIFO_THRESH` descriptors ahead.

CTRL_CONFIG (0x240) – `CTRLRD_MAX_TRY[8:0]` bounds the control-read poll retry budget (0-511, reset 16) fed to every channel's `ctrlrd_engine`. A `CTRL_READ` descriptor polls its gate address once per retry; if the masked value never matches within `CTRLRD_MAX_TRY` attempts the engine raises `ctrlrd_error` so a never-satisfied gate cannot hang the channel. See the Control-Read / Control-Write Engine specs (Sections 2.8 / 2.9) and the Scheduler control interface (Section 2.1).

26.3 Monitor Register Map (rapids_mon_regs @ 0x1000)

The monitor regfile is instantiated at base `0x1000` under `hwif_out.MON.*`. It holds the MonBus FIFO status, the three AXI-monitor configuration groups (DAXMON = descriptor monitor, RDMON = read monitor, WRMON = write monitor), and per-monitor performance counters.

Table 3.12.2: Monitor Register Map (base 0x1000)

Offset	Name	Access	Description
0x1000	`MON.MON_FIFO_STATUS`	RO	MonBus capture/error FIFO status
0x1004	`MON.MON_FIFO_COUNT`	RO	MonBus FIFO occupancy
0x10C0	`MON.DAXMON_ENABLE`	RW	Descriptor-monitor enables (incl. `COMPRESS_EN`)
0x10C4	`MON.DAXMON_TIMEOUT`	RW	Descriptor-monitor timeout cycles
0x10C8	`MON.DAXMON_LATENCY_THRESH`	RW	Descriptor-monitor latency threshold
0x10CC	`MON.DAXMON_PKT_MASK`	RW	Descriptor-monitor packet mask
0x10D0	`MON.DAXMON_ERR_CFG`	RW	Descriptor-monitor error select/mask
0x10D4-0x10DC	`MON.DAXMON_MASK1/2/3`	RW	Descriptor-monitor category masks
0x10E0-0x10FC	`MON.RDMON_*`	RW	Read-monitor config (same layout as DAXMON)
0x1100-0x111C	`MON.WRMON_*`	RW	Write-monitor config (same layout as DAXMON)
0x1150-0x1178	`MON.DAXMON`	RO/RW	Descriptor-monitor

Offset	Name	Access	Description
	_PERF_*		performance counters
0x1180-0x11A8	MON.RDMON_PERF_*	RO/RW	Read-monitor performance counters
0x11B0-0x11D8	MON.WRMON_PERF_*	RO/RW	Write-monitor performance counters
0x11E0-0x11EC	MON.{RD,WR}MON_PERF_CH_*	RO	Per-channel producer/backpressure/starve/idle
0x11F0	MON.RDMON_PERF_CH_OVERFLOW	RO	Read-monitor per-channel overflow
0x11F4	MON.WRMON_PERF_CH_OVERFLOW	RO	Write-monitor per-channel overflow

Each *_ENABLE register carries MON_EN, ERR_EN, COMPL_EN, TIMEOUT_EN (and COMPRESS_EN on the write monitor); each *_PERF_* group provides window, producer, backpressure, starve, idle, beat, byte (lo/hi), and burst counters.

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27 RAPIDS Config Block Specification

Module: rapids_config_block.sv **Location:** projects/components/rapids/rtl/macro_beats/ **Status:** Implemented

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27.1 Overview

rapids_config_block is a combinational adapter that maps the register block's hwif_out structure onto the flat cfg_* signals consumed by rapids_core_beats and the AXI monitors. It

contains no state; every output is a direct assignment (with a small number of width expansions and global-enable gates).

The monitor registers arrive under `hwif_out.MON.*`; base configuration and status arrive at the top level of `hwif_out`.

27.2 Mapping Groups

27.2.1 Base Configuration

Routes scheduler, channel, and descriptor-engine configuration to the core:

- `cfg_channel_enable = CHANNEL_ENABLE.CH_EN & {8{GLOBAL_CTRL.GLOBAL_EN}}` (per-channel enable gated by the global master enable).
- `cfg_sched_timeout_cycles = SCHED_TIMEOUT_CYCLES.TIMEOUT_CYCLES` (32-bit).
- `cfg_sched_timeout_limit  = SCHED_TIMEOUT_LIMIT.LIMIT` (8-bit, direct).
- `cfg_desceng_*` from `DESCENG_CONFIG` (enable / prefetch / FIFO threshold).
- Descriptor address ranges (`cfg_desceng_addr{0,1}_{base,limit}`) are zero-extended from the 32-bit register fields to the 64-bit `ADDR_WIDTH`.

27.2.2 Monitor Configuration (hwif_out.MON.*)

Three parallel groups map to the descriptor, read, and write AXI monitors:

Table 3.13.1: Monitor Register-to-Config Mapping

Register group	cfg_* target
`MON.DAXMON_*`	`cfg_desc_mon_*` (descriptor monitor)
`MON.RDMON_*`	`cfg_rdeng_mon_*` (read monitor)
`MON.WRMON_*`	`cfg_wreng_mon_*` (write monitor)

Within each group the enable/timeout/latency/mask fields map 1:1, e.g. `cfg_desc_mon_enable = MON.DAXMON_ENABLE.MON_EN & GLOBAL_CTRL.GLOBAL_EN`, `cfg_desc_mon_timeout_cycles = MON.DAXMON_TIMEOUT.TIMEOUT_CYCLES`, and the category masks (`timeout/compl/thresh/perf/addr/debug`) pass through directly. Key enables are gated by `GLOBAL_CTRL.GLOBAL_EN`; masks pass through ungated.

27.2.3 Performance and Observation

`PERF_CONFIG`, `PERF_CH_SEL`, and `OBS_CTRL` map to the profiler and channel-observation-mux `cfg_*` signals.

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28 RAPIDS Beats Top Specification

Module: `rapids_beats_top.sv` **Location:** `projects/components/rapids/rtl/top_beats/`
Status: Implemented

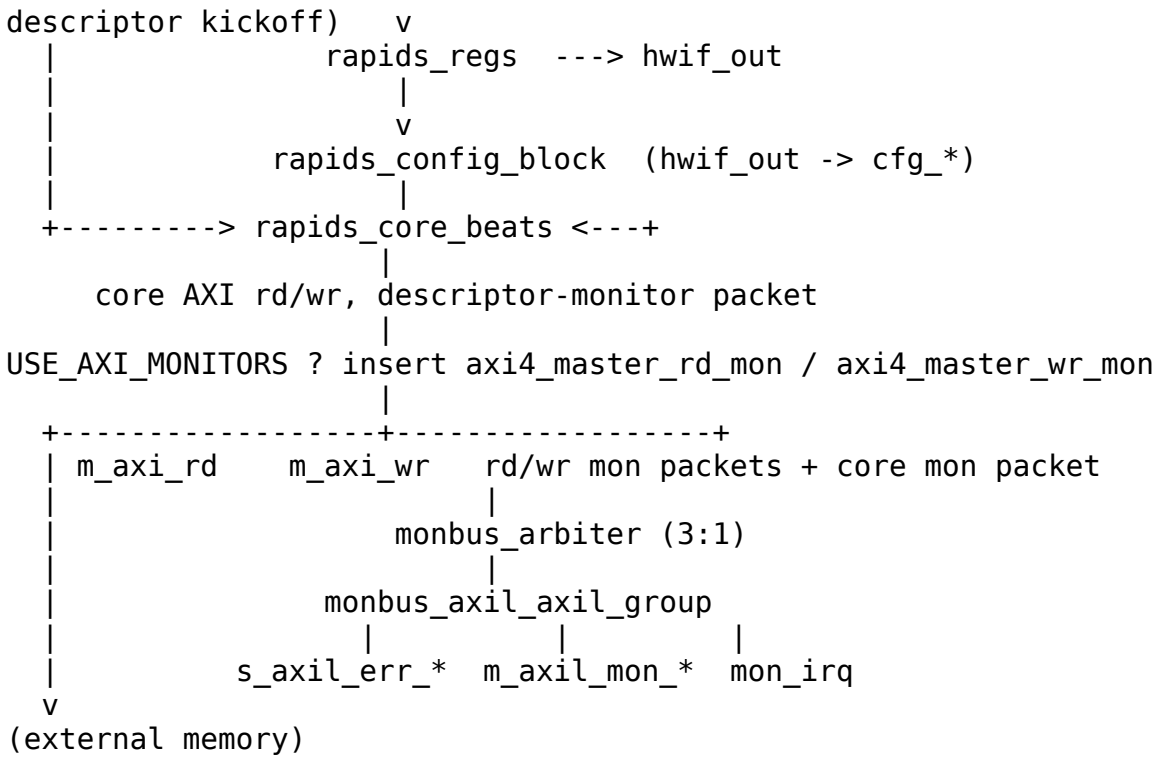
28.1 Overview

`rapids_beats_top` is the synthesizable top-level integration of the RAPIDS Beats accelerator. It wraps the register block, config-block adapter, and core with a single APB slave for all software access, and adds optional AXI transaction monitors whose events are merged with the core's descriptor-monitor packet and delivered through a MonBus AXI-Lite group (error-drain slave, capture master, and interrupt).

28.2 Integration Datapath

28.2.1 Figure 3.14.1: RAPIDS Beats Top Integration

```
   s_apb_* (APB4 slave)
        |
     apb_slave  (APB -> cmd/rsp)
        |
   cmdrsp_router (address decode)
     |                          |
     | 0x000-0x03F              | 0x100+ (and 0x1000 MON)
     v                          v
  apbtodescr              peakrdl_to_cmdrsp
  (per-channel             |
```



28.2.2 Address Decode

Table 3.14.1: Top-Level Address Decode

Range	Target	Purpose
0x000-0x03F	`apbtodescr`	Per-channel descriptor kick-off
0x100-0x3FF	`rapids_regs` (base)	Configuration / status registers
0x1000+	`rapids_regs` (MON regfile)	Monitor configuration / performance

Because the monitor regfile is at `0x1000`, the APB address bus must be at least 13 bits wide (`APB_ADDR_WIDTH >= 13`) to reach it.

28.3 AXI Monitors (USE_AXI_MONITORS)

When `USE_AXI_MONITORS = 1`, an `axi4_master_rd_mon` is inserted on the read master (`m_axi_rd`) and an `axi4_master_wr_mon` on the write master (`m_axi_wr`). Their `monitor_packet_t` outputs are combined with the core's descriptor-monitor packet (zero-

extended to 128 bits) by a 3-input `monbus_arbiter` (round-robin, with input/output skid buffers). The combined stream feeds `monbus_axil_axil_group`, which provides:

- `s_axil_err_*` – AXI-Lite (32-bit) **error-drain slave**: CPU reads captured error events from the error FIFO.
- `m_axil_mon_*` – AXI-Lite (64-bit) **capture master**: bulk-writes the MonBus trace to system memory (base/limit/watermark from `cfg_mon_*`).
- `mon_irq` – interrupt on error/threshold events.

When `USE_AXI_MONITORS = 0` the monitor taps are bypassed (core AXI passes straight through to `m_axi_rd`/`m_axi_wr`), the core MonBus is dropped (always-ready), and the AXI-Lite group outputs are tied off (`s_axil_err` read-inactive, `m_axil_mon` write-inactive, `mon_irq = 0`).

28.4 Parameters

```
parameter int NUM_CHANNELS         = 8;
parameter int DATA_WIDTH           = 512;
parameter int ADDR_WIDTH           = 64;
parameter int AXI_ID_WIDTH         = 8;
parameter int SRAM_DEPTH           = 4096;
parameter int APB_ADDR_WIDTH       = 12;   // must be >= 13 to reach
MON regfile @ 0x1000
parameter int APB_DATA_WIDTH       = 32;
parameter bit USE_AXI_MONITORS     = 0;    // 1 = insert rd/wr monitors
+ MonBus group
parameter int MON_MAX_TRANSACTIONS = 16;
parameter int AR_MAX_OUTSTANDING   = 8;
parameter int AW_MAX_OUTSTANDING   = 8;
```

: Table 3.14.2: RAPIDS Beats Top Parameters

28.5 Top-Level Interfaces

Table 3.14.3: Top-Level Interfaces

Interface	Signals	Notes
Clock / reset	`aclk`, `aresetn`, `cam_clear`	Single clock domain; `cam_clear` for monitor CAMs
APB slave	`s_apb_paddr` (APB_ADDR_WIDTH),	32-bit data

Interface	Signals	Notes
	s_apb_psel/penable/pwrite/pwdata/pstrb, s_apb_prdata/pready/pslverr	
Descriptor AXI (master)	m_axi_desc_ar*, m_axi_desc_r*	256-bit read data
Data read AXI (master)	m_axi_rd_ar*, m_axi_rd_r*	DATA_WIDTH; monitored when enabled
Data write AXI (master)	m_axi_wr_aw*, m_axi_wr_w*, m_axi_wr_b*	DATA_WIDTH; monitored when enabled
Sink fill	snk_fill_alloc_*, snk_fill_valid/ready/id/data, snk_fill_space_free	Network ingress
Source drain	src_drain_data_avail/req/size, src_drain_valid/read/id/data	Network egress
MonBus error-drain slave	s_axil_err_ar*, s_axil_err_r*	AXI-Lite, 32-bit read
MonBus capture master	m_axil_mon_aw*, m_axil_mon_w* (64-bit wdata), m_axil_mon_b*	AXI-Lite write
MonBus interrupt	mon_irq	Error/threshold interrupt
MonBus config	cfg_mon_base_addr, cfg_mon_limit_addr, cfg_mon_flush_watermark	Capture region + flush watermark
Status	system_idle, sched_error[NC-1:0]	Aggregate status

Last Updated: 2026-07-02

29 Interface Overview

Last Updated: 2025-01-10

29.1 External Interfaces

The RAPIDS Beats architecture exposes the following external interfaces:

29.1.1 AXI4 Master Interfaces

Table 4.0.1: AXI4 Master Interfaces

Interface	Purpose	Data Width
Descriptor AXI	Fetch descriptors from memory	256-bit
Sink AXI Write	Write sink data to memory	512-bit (param)
Source AXI Read	Read source data from memory	512-bit (param)

29.1.2 Streaming Interfaces

Table 4.0.2: Streaming Interfaces

Interface	Type	Purpose
Fill Interface	Custom	Network data ingress (to SRAM)
Drain Interface	Custom	Network data egress (from SRAM)
AXIS Sink	AXI-Stream Slave	Optional AXIS wrapper for fill
AXIS Source	AXI-Stream Master	Optional AXIS wrapper for drain

29.1.3 Control/Monitor Interfaces

Table 4.0.3: Control and Monitor Interfaces

Interface	Type	Purpose
APB	APB slave	Per-channel kick-off
Configuration	Custom	Global/per-channel config
Status	Custom	Idle, state, error flags
MonBus	Custom 64-bit	Monitor packet output

29.2 Interface Documentation

- AXI4 Interface Specification - Descriptor, sink write, source read
- AXIS Interface Specification - Optional streaming wrappers
- MonBus Interface Specification - Monitor packet format

29.3 Interface Signal Naming Conventions

Table 4.0.4: Signal Naming Conventions

Prefix	Meaning	Example
`m_axi_`	AXI master port	`m_axi_arvalid`
`s_axi_`	AXI slave port	`s_axi_awready`
`m_axis_`	AXIS master port	`m_axis_tvalid`
`s_axis_`	AXIS slave port	`s_axis_tready`
`cfg_`	Configuration input	`cfg_enable`
`snk_`	Sink data path	`snk_fill_valid`
`src_`	Source data path	`src_drain_ready`
`sched_`	Scheduler interface	`sched_rd_valid`
`monbus_`	Monitor bus	`monbus_pkt_data`

Last Updated: 2025-01-10

30 AXI4 Interface Specification

Last Updated: 2025-01-10

30.1 Overview

The RAPIDS Beats architecture uses three AXI4 master interfaces: 1. **Descriptor AXI:** Fetches descriptors from memory (256-bit data) 2. **Sink AXI Write:** Writes sink data to memory (512-bit data, configurable) 3. **Source AXI Read:** Reads source data from memory (512-bit data, configurable)

30.2 Descriptor AXI Master Interface

30.2.1 Purpose

Fetches descriptor packets from system memory. Shared across all 8 channels with round-robin arbitration.

30.2.2 Signal Table

Table 4.1.1: Descriptor AXI Master Interface Signals

Signal	Direction	Width	Description
`m_axi_arvalid`	output	1	AR channel valid
`m_axi_arready`	input	1	AR channel ready
`m_axi_araddr`	output	AW	Read address
`m_axi_arid`	output	ID_W	Transaction ID
`m_axi_arlen`	output	8	Burst length (fixed: 0 = 1 beat)
`m_axi_arsize`	output	3	Burst size

Signal	Direction	Width	Description
			(fixed: 5 = 32 bytes)
m_axi_arburst	output	2	Burst type (fixed: INCR)
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	256	Read data (descriptor)
m_axi_rid	input	ID_W	Response ID
m_axi_rresp	input	2	Read response
m_axi_rlast	input	1	Last beat

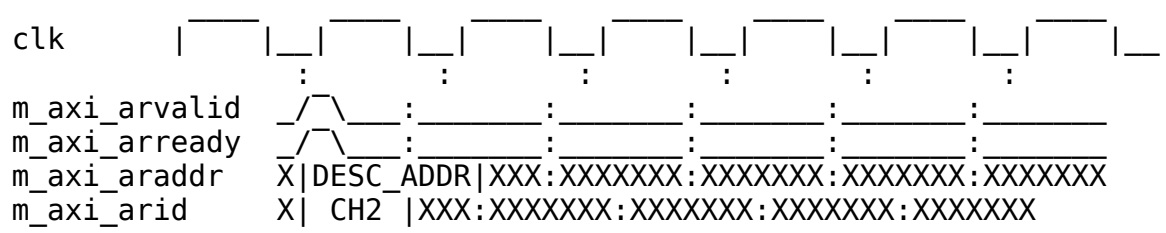
30.2.3 Transaction Characteristics

Table 4.1.2: Descriptor AXI Transaction Characteristics

Parameter	Value	Description
Data Width	256 bits	Fixed descriptor size
Burst Length	1 beat	Single descriptor per transaction
Burst Type	INCR	Incrementing burst
Max Outstanding	8	Configurable via parameter
ID Usage	Per-channel	ID encodes source channel

30.2.4 Timing Diagram

30.2.5 Figure 4.1.1: Descriptor Fetch Timing



```
                  :       :       :       :       :       :
m_axi_rvalid   _______:_/‾\___:_______:_______:_______:_______
m_axi_rready   _______:_/‾\___:_______:_______:_______:_______
m_axi_rdata    X:XXXXXXX|=DESCRIPTOR=|XXX:XXXXXXX:XXXXXXX
m_axi_rid      X:XXXXXXX|  CH2  |XXX:XXXXXXX:XXXXXXX:XXXXXXX
m_axi_rlast    _______:_/‾\___:_______:_______:_______:_______
```

TODO: Replace with simulation-generated waveform

30.3 Sink AXI Write Master Interface

30.3.1 Purpose

Writes sink data from SRAM to system memory. Supports burst transactions for efficient memory bandwidth.

30.3.2 Signal Table

Table 4.1.3: Sink AXI Write Master Interface Signals

Signal	Direction	Width	Description
`m_axi_awvalid`	output	1	AW channel valid
`m_axi_awready`	input	1	AW channel ready
`m_axi_awaddr`	output	AW	Write address
`m_axi_awid`	output	ID_W	Transaction ID
`m_axi_awlen`	output	8	Burst length (0-255)
`m_axi_awsize`	output	3	Burst size
`m_axi_awburst`	output	2	Burst type
`m_axi_wvalid`	output	1	W channel valid
`m_axi_wready`	input	1	W channel ready
`m_axi_wdata`	output	DW	Write data
`m_axi_wstrb`	output	DW/8	Write strobes
`m_axi_wlast`	output	1	Last beat

Signal	Direction	Width	Description
m_axi_bvalid	input	1	B channel valid
m_axi_bready	output	1	B channel ready
m_axi_bid	input	ID_W	Response ID
m_axi_bresp	input	2	Write response

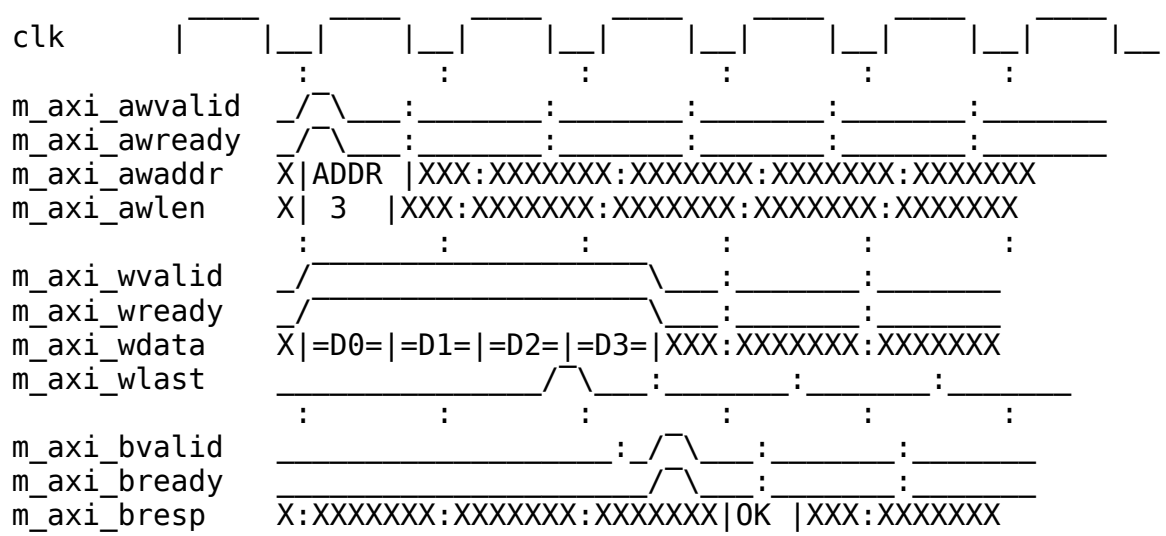
30.3.3 Transaction Characteristics

Table 4.1.4: Sink AXI Write Transaction Characteristics

Parameter	Value	Description
Data Width	512 bits	Configurable
Burst Length	1-256 beats	Based on transfer size
Burst Type	INCR	Incrementing burst
Max Outstanding AW	8	Configurable
W FIFO Depth	64	Configurable
B FIFO Depth	16	Configurable

30.3.4 Timing Diagram

30.3.5 Figure 4.1.2: Sink AXI Write Burst Timing



TODO: Replace with simulation-generated waveform

30.4 Source AXI Read Master Interface

30.4.1 Purpose

Reads source data from system memory into SRAM. Supports burst transactions for efficient memory bandwidth.

30.4.2 Signal Table

Table 4.1.5: Source AXI Read Master Interface Signals

Signal	Direction	Width	Description
`m_axi_arvalid`	output	1	AR channel valid
`m_axi_arready`	input	1	AR channel ready
`m_axi_araddr`	output	AW	Read address
`m_axi_arid`	output	ID_W	Transaction ID
`m_axi_arlen`	output	8	Burst length (0-255)
`m_axi_arsize`	output	3	Burst size
`m_axi_arburst`	output	2	Burst type
`m_axi_rvalid`	input	1	R channel valid
`m_axi_rready`	output	1	R channel ready
`m_axi_rdata`	input	DW	Read data
`m_axi_rid`	input	ID_W	Response ID
`m_axi_rresp`	input	2	Read response
`m_axi_rlast`	input	1	Last beat

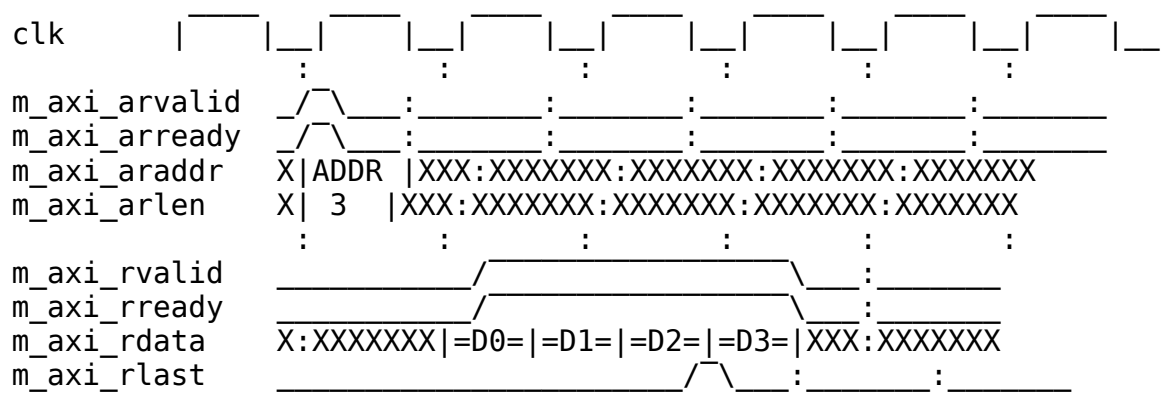
30.4.3 Transaction Characteristics

Table 4.1.6: Source AXI Read Transaction Characteristics

Parameter	Value	Description
Data Width	512 bits	Configurable
Burst Length	1-256 beats	Based on transfer size
Burst Type	INCR	Incrementing burst
Max Outstanding AR	8	Configurable
R FIFO Depth	64	Configurable

30.4.4 Timing Diagram

30.4.5 Figure 4.1.3: Source AXI Read Burst Timing



TODO: Replace with simulation-generated waveform

30.5 AXI Protocol Compliance

30.5.1 Supported Features

Table 4.1.7: AXI Feature Support

Feature	Descriptor	Sink Write	Source Read
Burst Type INCR	Yes	Yes	Yes
Burst Type FIXED	No	No	No

Feature	Descriptor	Sink Write	Source Read
Burst Type WRAP	No	No	No
Exclusive Access	No	No	No
Locked Access	No	No	No
Unaligned Access	No	Yes	Yes
Narrow Transfers	No	Yes	Yes

30.5.2 Error Handling

- **SLVERR:** Transaction aborted, error reported to scheduler
- **DECERR:** Transaction aborted, error reported to scheduler
- **Timeout:** Configurable watchdog, error reported to scheduler

---

Last Updated: 2025-01-10

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31 AXI-Stream Interface Specification

Last Updated: 2025-01-10

31.1 Overview

The RAPIDS Beats architecture provides optional AXI-Stream (AXIS) wrappers for network interfaces: 1. **AXIS Sink Slave:** Receives network data for memory writes 2. **AXIS Source Master:** Transmits network data from memory reads

These wrappers are available when `ENABLE_AXIS_WRAPPERS = 1`.

31.2 AXIS Sink Slave Interface

31.2.1 Purpose

Receives streaming network data and routes to per-channel SRAM buffers based on TID.

31.2.2 Signal Table

Table 4.2.1: AXIS Sink Slave Interface Signals

Signal	Direction	Width	Description
`s_axis_tvalid`	input	1	Data valid
`s_axis_tready`	output	1	Ready to accept data
`s_axis_tdata`	input	DATA_WIDTH	Data payload
`s_axis_tkeep`	input	DATA_WIDTH/8	Byte enables
`s_axis_tlast`	input	1	Last beat of packet
`s_axis_tid`	input	TID_WIDTH	Stream ID (channel select)
`s_axis_tdest`	input	TDEST_WIDTH	Destination routing
`s_axis_tuser`	input	TUSER_WIDTH	User sideband

31.2.3 Signal Details

31.2.3.1 TID (Transaction ID)

TID is used for per-channel routing:

Table 4.2.2: TID Channel Mapping

TID Value	Destination
0	Channel 0 SRAM
1	Channel 1 SRAM
2	Channel 2 SRAM
3	Channel 3 SRAM
4	Channel 4 SRAM

TID Value	Destination
5	Channel 5 SRAM
6	Channel 6 SRAM
7	Channel 7 SRAM

31.2.3.2 TKEEP (Byte Enables)

TKEEP indicates valid bytes within TDATA:

Table 4.2.3: TKEEP Configuration

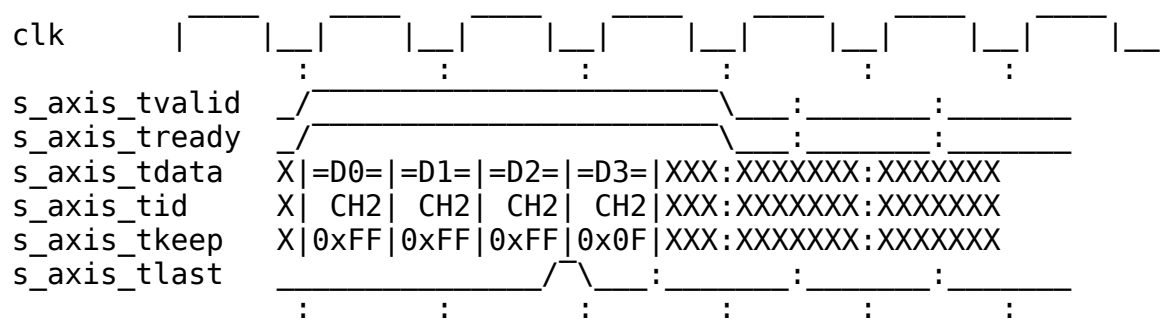
DATA_WIDTH	TKEEP Width	Usage
512 bits	64 bits	Per-byte valid
256 bits	32 bits	Per-byte valid

31.2.3.3 TLAST (Packet Boundary)

TLAST marks the end of an AXIS packet. This is mapped to the internal `fill_last` signal for packet boundary tracking.

31.2.4 Timing Diagram

31.2.5 Figure 4.2.1: AXIS Sink Slave Timing



TODO: Replace with simulation-generated waveform

31.2.6 Backpressure

The sink interface asserts backpressure (`s_axis_tready = 0`) when: - Target channel SRAM is full - No space allocated for incoming data - Internal pipeline stalls

---

31.3 AXIS Source Master Interface

31.3.1 Purpose

Transmits streaming network data from per-channel SRAM buffers with TID indicating source channel.

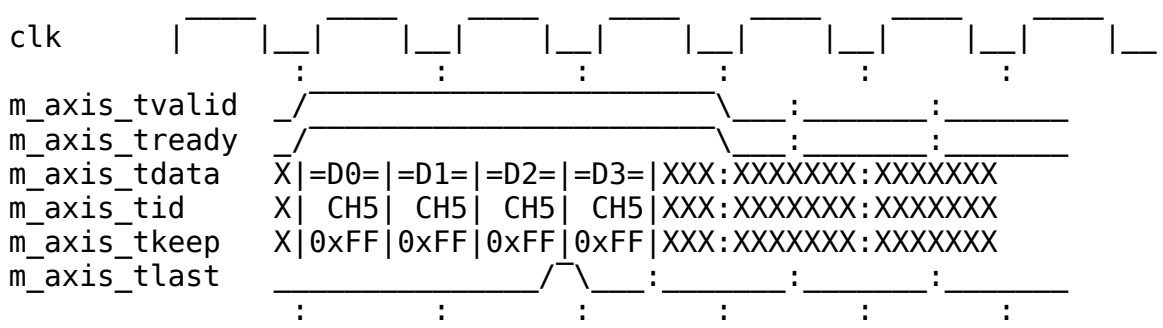
31.3.2 Signal Table

Table 4.2.4: AXIS Source Master Interface Signals

Signal	Direction	Width	Description
m_axis_tvalid	output	1	Data valid
m_axis_tready	input	1	Ready to accept data
m_axis_tdata	output	DATA_WIDTH	Data payload
m_axis_tkeep	output	DATA_WIDTH/8	Byte enables
m_axis_tlast	output	1	Last beat of packet
m_axis_tid	output	TID_WIDTH	Stream ID (source channel)
m_axis_tdest	output	TDEST_WIDTH	Destination routing
m_axis_tuser	output	TUSER_WIDTH	User sideband

31.3.3 Timing Diagram

31.3.4 Figure 4.2.2: AXIS Source Master Timing



TODO: Replace with simulation-generated waveform

31.3.5 Flow Control

The source interface respects network backpressure: - Data held when `m_axis_tready = 0` - No data loss on backpressure - SRAM drain pauses until ready

---

31.4 AXIS vs Custom Fill/Drain Interfaces

31.4.1 Comparison

Table 4.2.5: Interface Comparison

Aspect	Custom Fill/Drain	AXIS
Standard	Proprietary	AXI-Stream
Channel ID	fill_id / drain_id	TID
Packet Boundary	fill_last / drain_last	TLAST
Byte Enables	fill_strb / drain_strb	TKEEP
Interoperability	RAPIDS only	Industry standard

31.4.2 Selection Guide

Table 4.2.6: Interface Selection Guide

Use Case	Recommended Interface
Internal RAPIDS testing	Custom Fill/Drain
Network IP integration	AXIS
FPGA board-level design	AXIS
Maximum performance	Custom Fill/Drain

31.5 Configuration Parameters

```
// AXIS Interface Parameters
parameter int DATA_WIDTH = 512;       // Data payload width
parameter int TID_WIDTH = 3;          // Stream ID width (log2
channels)
parameter int TDEST_WIDTH = 1;        // Destination width
parameter int TUSER_WIDTH = 1;        // User sideband width
```

Last Updated: 2025-01-10

32 MonBus Interface Specification

Last Updated: 2025-01-10

32.1 Overview

The MonBus (Monitor Bus) is a 64-bit packet-based interface for reporting internal events, status, and performance data. All RAPIDS subsystems generate MonBus packets that are aggregated and output through a unified interface.

32.2 MonBus Signal Interface

32.2.1 Signal Table

Table 4.3.1: MonBus Interface Signals

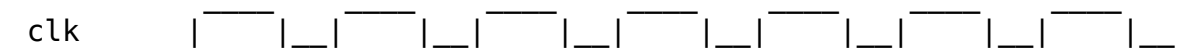
Signal	Direction	Width	Description
`monbus_pkt_valid`	output	1	Packet valid
`monbus_pkt_ready`	input	1	Consumer ready
`monbus_pkt_data`	output	64	Packet data

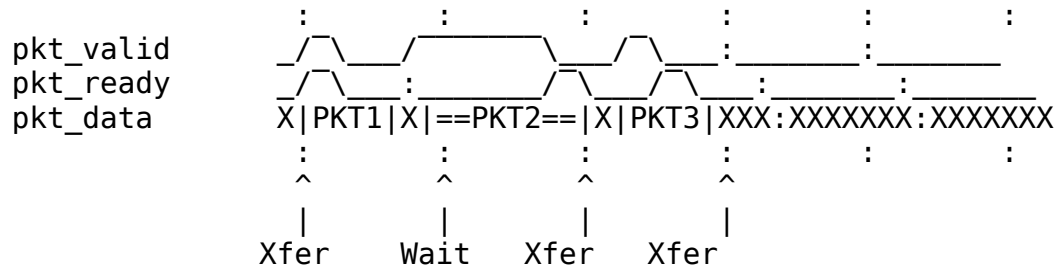
32.2.2 Handshaking Protocol

Standard valid/ready handshaking: - Data transfers when `valid & ready` both high - Producer holds `valid` until `ready` seen - Consumer asserts `ready` when able to accept

32.2.3 Timing Diagram

32.2.4 Figure 4.3.1: MonBus Packet Transfer Timing





TODO: Replace with simulation-generated waveform

32.3 MonBus Packet Format

32.3.1 64-bit Packet Structure

```
63      56 55      48 47      40 39      32 31              0
+--------+---------+---------+---------+------------------+
| Type   | Unit ID | Agent ID| Channel | Payload (32 bits) |
+--------+---------+---------+---------+------------------+
```

32.3.2 Figure 4.3.2: MonBus Packet Format

Table 4.3.2: MonBus Packet Fields

Field	Bits	Description
Type	[63:56]	Packet type code
Unit ID	[55:48]	Reporting unit identifier
Agent ID	[47:40]	Source agent within unit
Channel	[39:32]	Channel number (0-7)
Payload	[31:0]	Type-specific payload

32.4 Packet Types

32.4.1 Type Code Definitions

Table 4.3.3: MonBus Packet Types

Type Code	Name	Description
0x01	DESC_FETCH	Descriptor fetch started
0x02	DESC_COMPLETE	Descriptor fetch completed
0x03	SCHED_START	Scheduler started transfer
0x04	SCHED_DONE	Scheduler transfer complete
0x10	AXI_RD_START	AXI read burst started
0x11	AXI_RD_DONE	AXI read burst completed
0x12	AXI_WR_START	AXI write burst started
0x13	AXI_WR_DONE	AXI write burst completed
0x20	SRAM_FILL	SRAM fill event
0x21	SRAM_DRAIN	SRAM drain event
0x30	ERROR	Error event
0x31	TIMEOUT	Timeout event
0xFF	HEARTBEAT	Periodic heartbeat

32.4.2 Payload Formats by Type

32.4.2.1 DESC_FETCH / DESC_COMPLETE (0x01, 0x02)

Table 4.3.4: Descriptor Packet Payload

Bits	Description
[31:16]	Descriptor address [47:32]
[15:0]	Descriptor address [31:16]

32.4.2.2 SCHED_START / SCHED_DONE (0x03, 0x04)

Table 4.3.5: Scheduler Packet Payload

Bits	Description
[31:16]	Transfer ID
[15:0]	Beat count

32.4.2.3 AXI_RD/WR_START/DONE (0x10-0x13)

Table 4.3.6: AXI Event Packet Payload

Bits	Description
[31:24]	AXI ID
[23:16]	Burst length
[15:0]	Reserved

32.4.2.4 ERROR (0x30)

Table 4.3.7: Error Packet Payload

Bits	Description
[31:24]	Error code
[23:16]	Error source
[15:0]	Error details

32.5 MonBus Source Aggregation

32.5.1 RAPIDS Core MonBus Sources

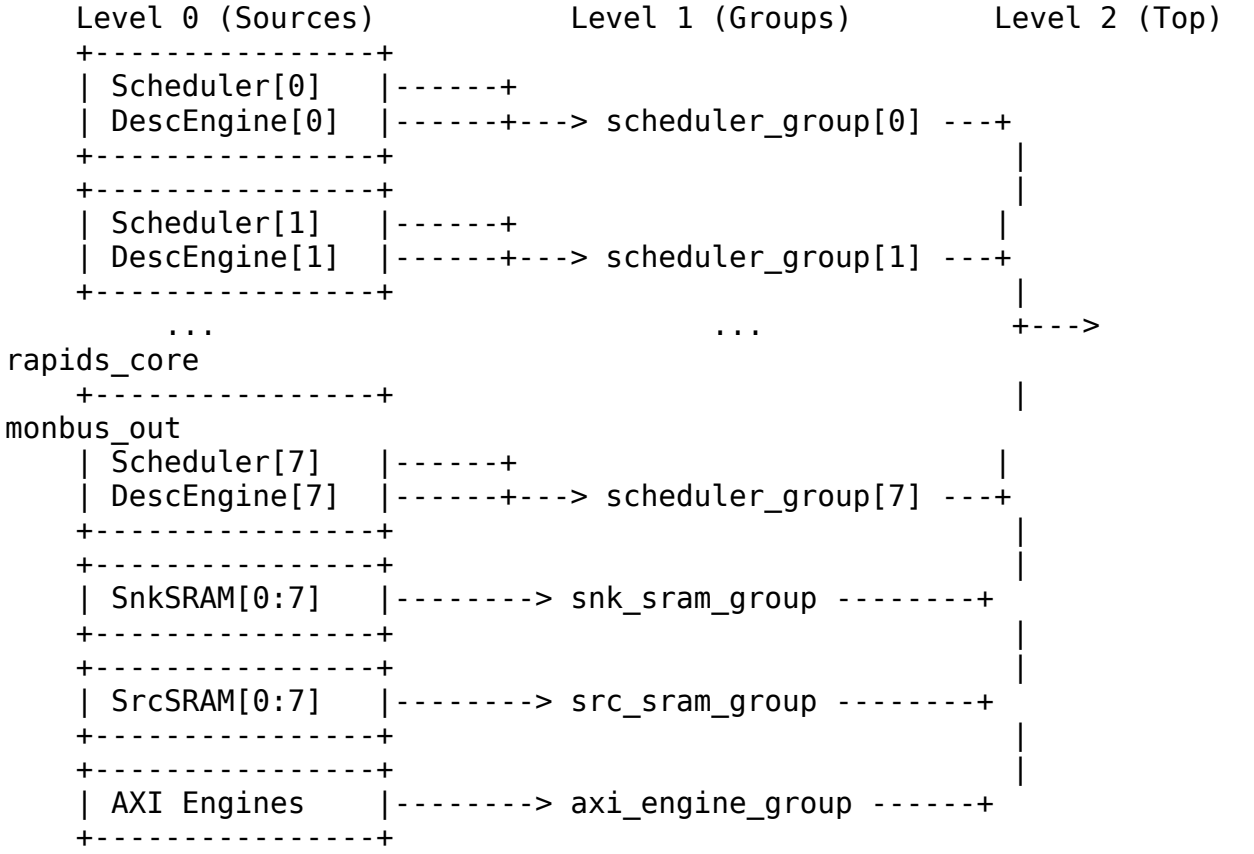
Table 4.3.8: MonBus Source Assignment

Source ID	Agent	Description
0-7	Scheduler[0:7]	Per-channel scheduler events
8-15	DescEngine[0:7]	Per-channel descriptor events
16-23	SnkSRAM[0:7]	Sink SRAM controller events
24-31	SrcSRAM[0:7]	Source SRAM controller events

Source ID	Agent	Description
32	AXI Write Engine	Write transaction events
33	AXI Read Engine	Read transaction events
34	Error Handler	Error events

32.5.2 Aggregation Hierarchy

32.5.3 Figure 4.3.3: MonBus Aggregation Tree



32.6 Integration Guidelines

32.6.1 Consumer Requirements

MonBus consumers must: 1. Provide sufficient ready cycles to prevent packet loss 2. Handle bursty traffic (multiple sources may have packets simultaneously) 3. Process or buffer packets at wire rate

32.6.2 Recommended Consumer Patterns

```
// Pattern 1: Direct FIFO buffering
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_monbus_fifo (
    .i_clk      (clk),
    .i_rst_n    (rst_n),
    .i_valid    (monbus_pkt_valid),
    .i_data     (monbus_pkt_data),
    .o_ready    (monbus_pkt_ready),
    .o_valid    (fifo_valid),
    .o_data     (fifo_data),
    .i_ready    (consumer_ready)
);

// Pattern 2: Software register interface
always_ff @(posedge clk) begin
    if (monbus_pkt_valid && monbus_pkt_ready) begin
        sw_monbus_data <= monbus_pkt_data;
        sw_monbus_valid <= 1'b1;
    end
end
assign monbus_pkt_ready = !sw_monbus_valid || sw_read;
```

32.7 Timing Requirements

Table 4.3.9: MonBus Timing Parameters

Parameter	Value	Description
Min Ready	1 cycle	Must assert ready within N cycles
Max Latency	Configurable	Before packet dropped
Packet Rate	Variable	Depends on activity

Last Updated: 2025-01-10

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33 Product Requirements Document (PRD)

33.1 RAPIDS - Rapid AXI Programmable In-band Descriptor System

Version: 1.0 **Date:** 2025-09-30 **Status:** Active Development - Validation in Progress **Owner:** RTL Design Sherpa Project **Parent Document:** `/PRD.md`

33.2 1. Executive Summary

The Rapid AXI Programmable In-band Descriptor System (RAPIDS) is a custom hardware accelerator designed for efficient memory-to-memory data movement with network interface integration. It demonstrates complex FSM coordination, descriptor-based DMA operations, and comprehensive monitoring capabilities.

33.2.1 1.1 Quick Stats

- **Modules:** 17 SystemVerilog files
- **Architecture:** 3 major blocks (Scheduler, Sink, Source)
- **Interfaces:** AXI4 (memory), Network (network), AXIL4 (control), MonBus (monitoring)
- **Test Coverage:** ~80% functional (basic scenarios validated)
- **Status:** Active validation, known issues documented

33.2.2 1.2 Subsystem Goals

- **Primary:** Demonstrate complex accelerator design patterns
- **Secondary:** Provide DMA-style memory transfer capability
- **Tertiary:** Educational reference for descriptor-based engines

33.3 2. Documentation Structure

This PRD provides a high-level overview. **Detailed specifications are maintained separately:**

33.3.1 📚 Complete RAPIDS Specification

Location: `projects/components/rapids/docs/rapids_spec/`

- **Index** - Complete specification structure

33.3.1.1 Chapter 1: Overview

- Overview
- Architecture Requirements
- Clocking and Reset
- Acronyms
- References

33.3.1.2 Chapter 2: Block Specifications

- Blocks Overview

Scheduler Group: - Scheduler Group - Scheduler - Task management FSM - Descriptor Engine - Descriptor parsing - Program Engine - Program sequencing

Sink Data Path (Network → Memory): - Sink Data Path - Network Slave - Network ingress - Sink SRAM Control - Buffer management - Sink AXI Write Engine - Memory writes

Source Data Path (Memory → Network): - Source Data Path - Network Master - Network egress - Source SRAM Control - Buffer management - Source AXI Read Engine - Memory reads

Monitoring: - MonBus AXIL Group - Control/status - FSM Summary - State machine overview

33.3.1.3 Chapter 3: Interfaces

- Top-Level Ports
- AXIL4 Interface
- AXI4 Interface
- Network Interface
- MonBus Interface

33.3.1.4 Chapter 4 & 5: Programming

- Programming Model
- Register Definitions

33.3.2 🐛 Known Issues

Location: `projects/components/rapids/known_issues/`

- **Scheduler** - Credit counter initialization bug
- **Sink Data Path** - Minor issues
- **Sink SRAM Control** - Edge cases

33.3.3 📖 Other Documentation

- **README** - Quick start and integration guide
- **CLAUDE** - AI assistance guide for this subsystem
- **TASKS** - Current work items (to be created)
- **Validation Report** - Test results

33.4 2.4 Organizational Standards - RAPIDS Code Location

⚠️ MANDATORY: All RAPIDS-specific code must be in the project area ⚠️

33.4.1 Code Organization Principle

"All RAPIDS-specific verification code MUST reside in `projects/components/rapids/dv/` for easy discovery."

This subsystem follows the repository-wide organizational standard (see `/PRD.md` Section 2.3) requiring all project-specific code to be located in the project area, NOT the framework area.

33.4.2 RAPIDS Directory Structure

```
projects/components/rapids/
├── rtl/                              # RTL source code
│   ├── includes/                     # RAPIDS packages (rapids_pkg.sv)
│   ├── rapids_fub/                   # Functional unit blocks
│   └── rapids_macro/                 # Top-level integration
│
└── dv/                               # Design verification (all RAPIDS-
specific)
    ├── tbclasses/                    # ★ RAPIDS TB classes HERE (not
framework!)
    │   ├── scheduler_tb.py           # Scheduler testbench class
    │   ├── descriptor_engine_tb.py
    │   ├── program_engine_tb.py
    │   └── rapids_integration_tb.py
    │
    ├── components/                   # ★ RAPIDS-specific BFMs
    │   └── (project-specific components if needed)
    │
    ├── scoreboards/                  # ★ RAPIDS-specific scoreboards
    │   └── (verification scoreboards)
    │
    └── tests/                        # Test runners (import TB classes)
        ├── fub_tests/                # Functional unit block tests
        │   ├── scheduler/
        │   │   └── test_scheduler.py
```

```
│   ├── descriptor_engine/
│   └── program_engine/
├── integration_tests/    # Multi-block scenarios
└── system_tests/         # Full RAPIDS operation
```

33.4.3 What Goes Where?

Code Type	✓ CORRECT Location	✗ WRONG Location
RAPIDS TB Classes	`projects/components/rapids/dv/tbclasses/`	`bin/TBClasses/rapids/`
RAPIDS-Specific BFMs	`projects/components/rapids/dv/components/`	`bin/TBClasses/components/rapids/`
RAPIDS Scoreboards	`projects/components/rapids/dv/scoreboards/`	`bin/TBClasses/scoreboards/rapids/`
Test Runners	`projects/components/rapids/dv/tests/`	Anywhere else
Shared AXI4/APB BFMs	`bin/TBClasses/components/{protocol}/`	Project area

33.4.4 Import Pattern for RAPIDS Tests

✓ CORRECT - Import from Project Area:

```python
# Import framework utilities (PYTHONPATH includes bin/)
import os, sys
from TBClasses.shared.utilities import get_repo_root
from TBClasses.shared.tbbase import TBBase

# Add repo root to Python path using robust git-based method
repo_root = get_repo_root()
sys.path.insert(0, repo_root)

# Import RAPIDS TB classes from PROJECT AREA
from projects.components.rapids.dv.tbclasses.scheduler_tb import
SchedulerTB
from projects.components.rapids.dv.tbclasses.descriptor_engine_tb
import DescriptorEngineTB

# Shared framework components
from CocoTBFramework.components.axi4.axi4_master import AXI4Master
```

✗ WRONG - Don't Import from Framework:

```
# DON'T DO THIS!
from TBClasses.rapids.scheduler_tb import SchedulerTB  # ✗ WRONG!
```

33.4.5 Benefits of This Organization

1. **Easy Discovery** - All RAPIDS code in ONE place: `projects/components/rapids/`
2. **Clear Ownership** - RAPIDS team owns their `dv/` area completely
3. **No Confusion** - Never wonder "where does this TB class live?"
4. **Maintainability** - Changes isolated to RAPIDS area don't affect other projects
5. **Framework Stays Clean** - Only truly shared cross-project code in framework

33.4.6 Compliance Status

✓ **RAPIDS is now compliant** - All TB classes moved to project area as of 2025-10-18

Migration History: - **Before:** TB classes incorrectly in `bin/TBClasses/rapids/` - **After:** TB classes correctly in `projects/components/rapids/dv/tbclasses/` - **Test Imports:** Updated to import from project area

📖 **Complete Documentation:** See `/PRD.md` Section 2.3 for repository-wide organizational standards.

33.5 3. Architecture Overview

33.5.1 3.1 Top-Level Block Diagram

```
RAPIDS (Rapid AXI Programmable In-band Descriptor System)
├── Scheduler Group
│   ├── Scheduler          (Task FSM, credit management)
│   ├── Descriptor Engine  (Descriptor FIFO, parsing)
│   └── Program Engine     (Program sequencing, alignment)
│
├── Sink Data Path (Network → SRAM → System Memory)
│   ├── Network Slave        (Network ingress interface)
│   ├── Sink SRAM Control (Buffering, flow control)
│   └── Sink AXI Writer   (AXI4 write to system memory)
│
├── Source Data Path (System Memory → SRAM → Network)
│   ├── Source AXI Reader (AXI4 read from system memory)
│   ├── Source SRAM Control (Buffering, flow control)
│   └── Network Master       (Network egress interface)
│
```

```
└── MonBus AXIL Group
    ├── AXIL4 Slave       (Control/status registers)
    └── MonBus Reporter   (Monitor packet generation)
```

📖 **See:** `rapids_spec/ch02_blocks/00_overview.md` for detailed architecture

33.5.2 3.2 Data Flow

Sink Path (Receive): 1. Network packets arrive via Network Slave 2. Buffered in Sink SRAM 3. DMA'd to system memory via AXI4 Write Engine 4. Completion reported via MonBus

Source Path (Transmit): 1. Descriptor specifies data location in system memory 2. Source AXI Reader fetches data to Source SRAM 3. Network Master transmits to network 4. Completion reported via MonBus

Scheduler Coordination: - Parses descriptors from Descriptor Engine - Manages credit-based flow control - Sequences program engine operations - Coordinates sink/source data paths

---

33.6 4. Key Features

33.6.1 4.1 Descriptor-Based Operation

Feature	Status	Description
Descriptor FIFO	√	Queued descriptor processing
Multi-field parsing	√	Address, length, control fields
Chained descriptors	⏳	Future enhancement
Completion reporting	√	Via MonBus packets

33.6.2 4.2 Data Path Features

Feature	Status	Description
SRAM buffering	√	Decouple network from memory
AXI4 burst support	√	Efficient memory transfers
Backpressure handling	√	Flow control on all interfaces
Data alignment	√	Handle unaligned

Feature	Status	Description
		transfers

33.6.3 4.3 Scheduler Features

Feature	Status	Description
Task FSM	√	Multi-state coordination
Credit management	√	Exponential encoding (0→1, 1→2, 2→4, ..., 15→∞)
Program sequencing	√	Coordinated operations
Error detection	√	Timeout, overflow detection

Credit Management Details: - Uses **exponential credit encoding** for compact configuration - 4-bit `cfg_initial_credit` decodes to actual credit counts: - `0` → 1 credit (2^0), `4` → 16 credits (2^4), `8` → 256 credits (2^8) - `15` → ∞ (unlimited credits, 0xFFFFFFFF) - Encoding applied at initialization; runtime operations are linear (increment/decrement by 1) - Provides wide range (1 to 16384) with minimal configuration overhead

📖 **See:** `rapids_spec/ch02_blocks/01_01_scheduler.md` for complete encoding table

33.6.4 4.4 Monitoring Integration

Feature	Status	Description
MonBus packets	√	StandardAMBA 64-bit format
Descriptor events	√	Start/complete reporting
Error events	√	Timeout, overflow, underflow
Performance metrics	⏳	Future enhancement

33.7 5. Interfaces

33.7.1 5.1 External Interfaces

Interface	Type	Width	Purpose
AXIL4	Slave	32-bit	Control/status registers
AXI4 (Sink)	Master	Configurable	Write to

Interface	Type	Width	Purpose
			system memory
AXI4 (Source)	Master	Configurable	Read from system memory
Network (Sink)	Slave	Configurable	Network ingress
Network (Source)	Master	Configurable	Network egress
MonBus	Master	64-bit	Monitor packet output

📖 **See:** `rapids_spec/ch03_interfaces/` for complete interface specs

33.7.2 5.2 Configuration Parameters

```
// Example RAPIDS instantiation parameters
rapids_top #(
    .AXI_ADDR_WIDTH(32),
    .AXI_DATA_WIDTH(64),
    .Network_DATA_WIDTH(64),
    .SRAM_DEPTH(1024),
    .MAX_DESCRIPTORS(16)
) u_rapids (
    .aclk               (clk),
    .aresetn            (rst_n),
    // AXIL4 control interface
    .s_axil_*           (...),
    // AXI4 memory interfaces
    .m_axi_sink_*       (...),
    .m_axi_source_*     (...),
    // Network network interfaces
    .s_network_*           (...),
    .m_network_*           (...),
    // MonBus output
    .monbus_pkt_*       (...)
);
```

33.8 6. Use Cases

33.8.1 6.1 DMA-Style Transfers

Scenario: Move data from network to system memory

Flow: 1. Software writes descriptor to Descriptor Engine 2. Scheduler parses descriptor, activates Sink path 3. Network packets arrive via Network Slave 4. Data buffered in Sink SRAM 5. AXI4 Write Engine DMAs to system memory 6. Completion packet on MonBus

33.8.2 6.2 Network Packet Processing

Scenario: Read data from memory, transmit to network

Flow: 1. Descriptor specifies source address, length 2. Source AXI Reader fetches data to SRAM 3. Network Master transmits to network 4. Completion/error reporting via MonBus

33.8.3 6.3 Custom Data Path Acceleration

Educational value: Shows how to build custom accelerators - Descriptor-based control - Multi-block FSM coordination - Buffering strategies - Error handling - Performance monitoring

---

33.9 7. Test Coverage

33.9.1 7.1 Current Status

Overall: ~85% functional coverage (basic scenarios validated, descriptor engine complete)

Component	Test Coverage	Status
Scheduler	~95%	Credit encoding fixed and verified (43/43 tests passing)
Descriptor Engine	√ 100%	**All tests passing** (14/14 tests, 100% success rate)
Program Engine	~85%	Alignment tested
Sink Data Path	~75%	Basic flows working
Source Data Path	~70%	Basic flows working

Component	Test Coverage	Status
SRAM Controllers	~80%	Buffer management tested
Integration	~60%	More stress testing needed

Test Location: `projects/components/rapids/dv/tests/fub_tests/` and `projects/components/rapids/dv/tests/integration_tests/`

Recent Achievements: - √ **Descriptor Engine (2025-10-13):** Achieved 100% test pass rate using continuous background monitoring pattern - 14/14 tests passing across all test levels (basic, medium, full) - All test classes passing (APB_ONLY, MIXED) - All delay profiles passing (fast_producer, fast_consumer, fixed_delay, minimal_delay) - Applied continuous monitoring methodology for asynchronous output capture

📖 **See:** `docs/RAPIDS_Validation_Status_Report.md` for detailed results

33.9.2 7.2 Test Strategy

FUB (Functional Unit Block) Tests: - Individual block testing - Located in `projects/components/rapids/dv/tests/fub_tests/` - Focus on module-level functionality

Integration Tests: - Multi-block scenarios - Located in `projects/components/rapids/dv/tests/integration_tests/` - End-to-end data flow validation

System Tests: - Full RAPIDS operation - Located in `projects/components/rapids/dv/tests/system_tests/` - Realistic traffic patterns

---

33.10 8. Known Issues Summary

33.10.1 8.1 Critical Issues

√ **Scheduler Credit Counter Initialization - FIXED (2025-10-11)** - **File:** `projects/components/rapids/rtl/rapids_fub/scheduler.sv:567-570` - **Issue:** Credit counter was initializing to 0 instead of using exponential encoding - **Fix Applied:** Implemented exponential credit encoding - **Status:** Fixed, awaiting test verification - **Documentation:** `known_issues/scheduler.md`

Fix Details:

```
// Previous (wrong):
r_descriptor_credit_counter <= 32'h0;
```

```
// Fixed - Exponential encoding:
// 0→1, 1→2, 2→4, 3→8, ..., 14→16384, 15→∞
r_descriptor_credit_counter <= (cfg_initial_credit == 4'hF) ?
32'hFFFFFFFF :
                               (cfg_initial_credit == 4'h0) ?
32'h00000001 :
                               (32'h1 << cfg_initial_credit);
```

Encoding Rationale: - Compact 4-bit configuration covers 1 to 16384 credits - Fine-grained control for low traffic (1, 2, 4, 8) - High-throughput support (256, 1024, 16384) - Special unlimited mode (15 → ∞) - Exponential encoding applied at initialization only; runtime operations are linear

33.10.2 8.2 Medium Priority Issues

Descriptor Engine Edge Cases: - Some stress test failures under high load - Edge case handling needs improvement - **Priority:** P2

SRAM Control Timing: - Rare timing issues in back-to-back operations - **Priority:** P2

📖 **See:** `known_issues/` directory for complete issue tracking

33.11 9. Integration Guidelines

33.11.1 9.1 Quick Start

```
rapids_top #(
    .AXI_ADDR_WIDTH(32),
    .AXI_DATA_WIDTH(64),
    .Network_DATA_WIDTH(64)
) u_rapids (
    // Clocking & Reset
    .aclk               (system_clk),
    .aresetn            (system_rst_n),

    // AXIL4 Control (connect to control fabric)
    .s_axil_awaddr      (ctrl_awaddr),
    .s_axil_awvalid     (ctrl_awvalid),
    .s_axil_awready     (ctrl_awready),
    // ... other AXIL signals

    // AXI4 Memory (connect to memory controller)
    .m_axi_sink_awaddr  (mem_awaddr),
    .m_axi_sink_awvalid (mem_awvalid),
    // ... AXI write channel for sink
    // ... AXI read channel for source
```

```
    // Network Network (connect to network fabric)
    .s_network_tdata       (net_rx_data),
    .s_network_tvalid      (net_rx_valid),
    // ... Network slave (receive)
    // ... Network master (transmit)

    // MonBus Output (connect to monitor aggregator)
    .monbus_pkt_valid   (rapids_mon_valid),
    .monbus_pkt_ready   (rapids_mon_ready),
    .monbus_pkt_data    (rapids_mon_data)
);
```

33.11.2 9.2 Configuration Steps

1. **Initialize via AXIL4:**
   - Configure SRAM depths
   - Set timeout thresholds
   - Enable credit management (or disable if using workaround)
2. **Load Descriptors:**
   - Write descriptors to Descriptor Engine FIFO
   - Each descriptor specifies: address, length, control bits
3. **Enable Operation:**
   - Set enable bits via AXIL4 registers
   - Monitor MonBus for completion/error packets

📖 **See:** `rapids_spec/ch04_programming_models/01_programming.md` for register details

33.12 10. Development Status

33.12.1 10.1 Current Phase

Phase: Validation and Bug Fixing (In Progress)

- √ Core architecture implemented
- √ Basic functionality verified
- √ Scheduler credit counter bug fixed (exponential encoding implemented)
- ⏳ Credit management tests need verification (remove workarounds)
- ⏳ Stress testing ongoing
- ⏳ Edge case refinement

📖 **See:** `TASKS.md` for detailed work items

33.12.2 10.2 Roadmap

Near-Term (Q4 2025): - ✓ Fix scheduler credit counter bug (completed 2025-10-11) - ⏳ Verify credit management tests (remove workarounds) - ⏳ Complete descriptor engine stress testing - ⏳ Integration test suite expansion - ⏳ Performance benchmarking

Long-Term (2026+): - Chained descriptor support - Advanced error recovery - Performance optimizations - Multi-channel support

---

33.13 11. Performance Characteristics

33.13.1 11.1 Throughput

Target: Match network/memory interface bandwidth

Bottlenecks: - SRAM buffer size - AXI4 burst efficiency - Scheduler overhead

Optimization: - Increase SRAM depth for larger packets - Tune AXI4 burst parameters - Pipeline scheduler operations

33.13.2 11.2 Latency

Components: - Descriptor parsing: ~10 cycles - SRAM buffering: Configurable depth - AXI4 memory access: System dependent - End-to-end: Typically <100 cycles for small packets

33.13.3 11.3 Resource Utilization

Area: - Scheduler: ~2K LUTs - Each data path: ~3K LUTs - SRAM buffers: Configurable (dominant area) - Total: ~10K LUTs + SRAM

Power: - Clock gating opportunities in idle blocks - SRAM power depends on depth/width

---

33.14 12. Verification Infrastructure

33.14.1 12.1 Test Organization

Location: `projects/components/rapids/dv/tests/`

Structure:

```
projects/components/rapids/dv/tests/
├── fub_tests/              # Functional Unit Block tests
│   ├── scheduler/
│   ├── descriptor_engine/
│   ├── program_engine/
```

```
│   ├── network_interfaces/
│   └── simple_sram/
├── integration_tests/      # Multi-block scenarios
└── system_tests/           # Full RAPIDS operation
```

33.14.2 12.2 CocoTB Framework

Location: `bin/TBClasses/rapids/`

Components: - RAPIDS-specific drivers - Descriptor generators - Traffic patterns - Monitor checkers

📖 **See:** `docs/markdown/TBClasses/` (once created)

33.14.3 12.2.1 MANDATORY: BFM Usage for FUB Tests

⚠️ CRITICAL DESIGN REQUIREMENT ⚠️

All RAPIDS FUB (Functional Unit Block) level tests MUST use CocoTB Framework BFMs. Manual handshake driving is NOT allowed.

Required BFM Components:

Interface Type	Framework Location	BFM Component
Custom valid/ready	`bin/TBClasses/components/gaxi/`	GAXI Master/Slave
AXI4	`bin/TBClasses/components/axi4/`	AXI4 Master/Slave
AXI4-Lite (AXIL)	`bin/TBClasses/components/axil4/`	AXIL Master/Slave
APB	`bin/TBClasses/components/apb/`	APB Master/Slave
AXI-Stream (AXIS)	`bin/TBClasses/components/axis4/`	AXIS Master/Slave
Network	`bin/TBClasses/components/network/`	Network Master/Slave
MonBus	`bin/TBClasses/components/monbus/`	MonBus drivers

Rationale: 1. **Consistency**: All tests use standardized handshake protocols 2. **Correctness**: BFMs handle complex timing scenarios (backpressure, randomization) 3. **Reusability**: Same BFM across all RAPIDS tests 4. **Maintainability**: Fix once in BFM, all tests benefit 5. **Coverage**: BFMs include comprehensive timing profiles

Example - Program Engine:

```
# ✗ WRONG: Manual handshake (violates design requirement)
async def send_request(self, addr, data):
```

```
    self.dut.program_valid.value = 1
    self.dut.program_pkt_addr.value = addr
    # ... manual handshaking logic ...

# ✓ CORRECT: Use GAXI Master BFM
from CocoTBFramework.components.gaxi.gaxi_master import GAXIMaster

class ProgramEngineTB(TBBase):
    def __init__(self, dut):
        super().__init__(dut)
        self.program_master = GAXIMaster(
            dut=dut,
            clock=dut.clk,
            valid_signal='program_valid',
            ready_signal='program_ready',
            data_signals=['program_pkt_addr', 'program_pkt_data'],
            data_widths=[64, 32]
        )

    async def send_request(self, addr, data):
        await self.program_master.write({'program_pkt_addr': addr,
'program_pkt_data': data})
```

📖 **See:** - `projects/components/rapids/CLAUDE.md` - Rule #1 for complete BFM usage guidelines - `bin/TBClasses/components/gaxi/README.md` - GAXI BFM documentation - `bin/TBClasses/components/axi4/README.md` - AXI4 BFM documentation

33.14.4 12.3 Test File Structure (Standard Pattern)

⚠️ **MANDATORY: All RAPIDS tests must follow this structure** ⚠️

RAPIDS tests follow the same pattern as AMBA tests for consistency across the repository:

```
# Example:
projects/components/rapids/dv/tests/fub_tests/scheduler/test_scheduler
.py

import os
import pytest
import cocotb
from cocotb_test.simulator import run

# Import REUSABLE testbench class (NOT defined in this file!)
from TBClasses.rapids.scheduler_tb import SchedulerTB
from TBClasses.shared.utilities import get_paths, create_view_cmd
from TBClasses.shared.tbbase import TBBase
```

```
#
================================================================================
=====
# COCOTB TEST FUNCTIONS - prefix with "cocotb_" to prevent pytest
collection
#
================================================================================
=====

@cocotb.test(timeout_time=100, timeout_unit="ms")
async def cocotb_test_basic_flow(dut):
    """Test basic descriptor flow."""
    tb = SchedulerTB(dut)
    await tb.setup_clocks_and_reset()  # Mandatory init method
    await tb.initialize_test()
    result = await tb.test_basic_descriptor_flow()
    assert result, "Basic descriptor flow test failed"

@cocotb.test(timeout_time=100, timeout_unit="ms")
async def cocotb_test_credit_encoding(dut):
    """Test exponential credit encoding."""
    tb = SchedulerTB(dut)
    await tb.setup_clocks_and_reset()  # Mandatory init method
    await tb.initialize_test()
    result = await tb.test_exponential_encoding_all_values()
    assert result, "Credit encoding test failed"

#
================================================================================
=====
# PARAMETER GENERATION - at bottom of file
#
================================================================================
=====

def generate_scheduler_test_params():
    """Generate test parameters for scheduler tests."""
    return [
        # (channel_id, num_channels, data_width, credit_width)
        (0, 8, 512, 8),  # Standard configuration
        # Add more parameter sets as needed
    ]

scheduler_params = generate_scheduler_test_params()

#
================================================================================
```

```
=====
# PYTEST WRAPPER FUNCTIONS - at bottom of file
#
================================================================================
=====

@pytest.mark.parametrize("channel_id, num_channels, data_width,
credit_width", scheduler_params)
def test_basic_flow(request, channel_id, num_channels, data_width,
credit_width):
    """
    Scheduler basic flow test.

    Run with: pytest
projects/components/rapids/dv/tests/fub_tests/scheduler/test_scheduler
.py::test_basic_flow -v
    """
    module, repo_root, tests_dir, log_dir, rtl_dict = get_paths({
        'rtl_rapids_fub': '../../rtl/rapids_fub'
    })

    dut_name = "scheduler"
    toplevel = dut_name

    verilog_sources = [
        os.path.join(repo_root, 'rtl', 'amba', 'includes',
'monitor_pkg.sv'),
        os.path.join(repo_root, 'rtl', 'rapids', 'includes',
'rapids_pkg.sv'),
        os.path.join(rtl_dict['rtl_rapids_fub'], f'{dut_name}.sv'),
    ]

    # Format parameters for unique test name
    cid_str = TBBase.format_dec(channel_id, 2)
    nc_str = TBBase.format_dec(num_channels, 2)
    dw_str = TBBase.format_dec(data_width, 4)
    cw_str = TBBase.format_dec(credit_width, 2)
    test_name_plus_params =
f"test_{dut_name}_cid{cid_str}_nc{nc_str}_dw{dw_str}_cw{cw_str}"

    # Add worker ID for pytest-xdist parallel execution
    worker_id = os.environ.get('PYTEST_XDIST_WORKER', '')
    if worker_id:
        test_name_plus_params = f"{test_name_plus_params}_{worker_id}"

    log_path = os.path.join(log_dir, f'{test_name_plus_params}.log')
```

```
    sim_build = os.path.join(tests_dir, 'local_sim_build',
test_name_plus_params)
    os.makedirs(sim_build, exist_ok=True)
    os.makedirs(log_dir, exist_ok=True)

    rtl_parameters = {
        'CHANNEL_ID': channel_id,
        'NUM_CHANNELS': num_channels,
        'DATA_WIDTH': data_width,
        'CREDIT_WIDTH': credit_width,
        # Add other RTL parameters as needed
    }

    extra_env = {
        'LOG_PATH': log_path,
        'TEST_CHANNEL_ID': str(channel_id),
        'TEST_NUM_CHANNELS': str(num_channels),
        'TEST_DATA_WIDTH': str(data_width),
    }

    compile_args = ["-Wno-TIMESCALEMOD"]
    sim_args = []
    plusargs = []

    cmd_filename = create_view_cmd(log_dir, log_path, sim_build,
module, test_name_plus_params)

    try:
        run(
            python_search=[tests_dir],
            verilog_sources=verilog_sources,
            includes=[
                os.path.join(repo_root, 'rtl', 'rapids', 'includes'),
                os.path.join(repo_root, 'rtl', 'amba', 'includes'),
            ],
            toplevel=toplevel,
            module=module,
            testcase="cocotb_test_basic_flow",  # ← cocotb test
function name
            parameters=rtl_parameters,
            sim_build=sim_build,
            extra_env=extra_env,
            waves=False,
            keep_files=True,
            compile_args=compile_args,
            sim_args=sim_args,
            plusargs=plusargs,
```

```
        )

        print(f"✓ Scheduler basic flow test completed!")
        print(f"Logs: {log_path}")

    except Exception as e:
        print(f"✗ Scheduler basic flow test failed: {str(e)}")
        print(f"Logs preserved at: {log_path}")
        raise
```

Key Structure Requirements:

1. **Testbench Class Location:**
   - ALWAYS in `bin/TBClasses/rapids/`
   - NEVER inline in test file
   - Reusable across multiple test files
2. **CocoTB Test Functions:**
   - Prefix with `cocotb_` to prevent pytest collection
   - Located at top of test file
   - Use `@cocotb.test()` decorator
   - Call testbench methods
3. **Parameter Generation:**
   - Function returns list of parameter tuples
   - Located near bottom of file (before pytest wrappers)
   - Stored in variable (e.g., `scheduler_params`)
4. **Pytest Wrapper Functions:**
   - Located at bottom of file
   - Use `@pytest.mark.parametrize()` with parameter variable
   - Build unique test names with `TBBase.format_dec()`
   - Call `run()` with `testcase="cocotb_test_function_name"`
   - Handle parallel execution (`PYTEST_XDIST_WORKER`)
5. **Mandatory TB Methods:**
   - `async def setup_clocks_and_reset(self)` - Complete initialization
   - `async def assert_reset(self)` - Assert reset signal(s)
   - `async def deassert_reset(self)` - Deassert reset signal(s)

📖 **See:** - `val/amba/test_apb_slave.py` - Reference example - `projects/components/rapids/CLAUDE.md` - Detailed TB requirements

33.15 13. Quick Reference

33.15.1 13.1 Key Files

File	Purpose
`projects/components/rapids/PRD.md`	This document (overview)
`projects/components/rapids/README.md`	Quick start guide
`projects/components/rapids/CLAUDE.md`	AI assistance guide
`projects/components/rapids/TASKS.md`	Work items (to be created)
`projects/components/rapids/docs/rapids_spec/`	**Complete specification**
`projects/components/rapids/known_issues/`	Bug tracking
`docs/RAPIDS_Validation_Status_Report.md`	Test results

33.15.2 13.2 Commands

```
# Run RAPIDS tests
pytest projects/components/rapids/dv/tests/fub_tests/ -v          #
Individual blocks
pytest projects/components/rapids/dv/tests/integration_tests/ -v   #
Multi-block
pytest projects/components/rapids/dv/tests/system_tests/ -v        #
Full system

# Run specific FUB test
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ -v

# Lint RAPIDS RTL
verilator --lint-only
projects/components/rapids/rtl/rapids_fub/scheduler.sv

# View specifications
cat projects/components/rapids/docs/rapids_spec/rapids_index.md
cat
projects/components/rapids/docs/rapids_spec/ch02_blocks/01_01_schedule
r.md
```

33.16 14. Success Criteria

33.16.1 14.1 Functional

- ✓ All major blocks implemented
- ✓ Basic data flows working
- ✓ Scheduler credit bug fixed (exponential encoding implemented)
- ⏳ Credit management tests verified (remove workarounds, run full suite)
- ⏳ 100% descriptor test pass rate (currently ~80%)
- ⏳ Stress tests passing

33.16.2 14.2 Quality

- ⏳ >90% functional coverage (currently ~80%)
- ⏳ >85% code coverage
- ✓ All FSMs documented
- ⏳ Integration guide complete

33.16.3 14.3 Documentation

- ✓ Complete specification in rapids_spec/
- ✓ Known issues documented
- ⏳ Integration examples
- ⏳ Performance characterization

33.17 15. Educational Value

RAPIDS demonstrates: - ✓ Complex FSM coordination (scheduler ↔ data paths) - ✓ Descriptor-based DMA design patterns - ✓ Buffer management strategies - ✓ Credit-based flow control with exponential encoding - ✓ Multi-interface integration - ✓ Comprehensive monitoring - ✓ Error detection and reporting - ✓ Compact configuration encoding strategies

Target Audience: - Advanced RTL designers - Accelerator architects - DMA engine developers - System integration engineers

33.18 15. Attribution and Contribution Guidelines

33.18.1 15.1 Git Commit Attribution

When creating git commits for RAPIDS documentation or implementation:

Use:

```
Documentation and implementation support by Claude.
```

Do NOT use:

```
Co-Authored-By: Claude <noreply@anthropic.com>
```

Rationale: RAPIDS documentation and organization receives AI assistance for structure and clarity, while design concepts and architectural decisions remain human-authored.

33.19 16. Documentation Generation

33.19.1 16.1 Generating PDF/DOCX from Specification

Tool: `/mnt/data/github/rtldesignsherpa/bin/md_to_docx.py`

Use this tool to convert the linked specification index into a single all-inclusive PDF or DOCX file.

Basic Usage:

```
# Generate DOCX from rapids_spec index
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    --toc \
    --title-page

# Generate both DOCX and PDF
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    --toc \
    --title-page \
    --pdf

# With custom template (optional)
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
```

```
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    -t path/to/template.dotx \
    --toc \
    --title-page \
    --pdf
```

Key Features: - **Recursive Collection:** Follows all markdown links in the index file - **Heading Demotion:** Automatically adjusts heading levels for included files - **Table of Contents:** `--toc` flag generates automatic ToC - **Title Page:** `--title-page` flag creates title page from first heading - **PDF Export:** `--pdf` flag generates both DOCX and PDF - **Image Support:** Resolves images relative to source directory - **Template Support:** Optional custom DOCX/DOTX template via `-t` flag

Common Workflow:

```
# 1. Update version number in index file (rapids_index.md)
# 2. Generate documentation
cd /mnt/data/github/rtldesignsherpa
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    --toc --title-page --pdf

# 3. Output files created:
#    - RAPIDS_Specification_v0.25.docx
#    - RAPIDS_Specification_v0.25.pdf (if --pdf used)
```

Debug Mode:

```
# Generate debug markdown to see combined output
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o output.docx \
    --debug-md

# This creates debug.md showing the complete merged content
```

Tool Requirements: - Python 3.6+ - Pandoc installed and in PATH - For PDF generation: LaTeX (e.g., texlive) or use Pandoc's built-in PDF writer

📖 **See:** `/mnt/data/github/rtldesignsherpa/bin/md_to_docx.py` for complete implementation details

33.20 16.2 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

`/mnt/data/github/rtldesignsherpa/projects/components/rapids/docs/`

Quick Command: Use the provided shell script:

```
cd /mnt/data/github/rtldesignsherpa/projects/components/rapids/docs
./generate_pdf.sh
```

The shell script will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the rapids_spec index file 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

📖 **See:** `bin/md_to_docx.py` for complete implementation details

33.21 17. References

33.21.1 16.1 Internal Documentation

- **Complete Spec:** `rapids_spec/` ← **Primary technical reference**
- **Validation:** `docs/RAPIDS_Validation_Status_Report.md`
- **Master PRD:** `/PRD.md`
- **Repository Guide:** `/CLAUDE.md`

33.21.2 16.2 Related Subsystems

- **AMBA:** `rtl/amba/` - Monitor infrastructure used in RAPIDS
- **Common:** `rtl/common/` - Building blocks (counters, FIFOs, etc.)
- **CocoTB Framework:** `bin/TBClasses/rapids/`

33.21.3 16.3 External References

- AXI4 Specification: ARM IHI0022E
- AXIL4 Specification: ARM IHI0022E (subset)
- Network interface specs (custom Network protocol)

Document Version: 1.0 **Last Updated:** 2025-09-30 **Review Cycle:** Monthly during active development **Next Review:** 2025-10-30 **Owner:** RTL Design Sherpa Project

---

33.22 Navigation

- **← Back to Root:** `/PRD.md`
- **Complete Specification:** `rapids_spec/rapids_index.md`

- **Quick Start:** `README.md`
- **AI Guidance:** `CLAUDE.md`
- **Tasks:** `TASKS.md` (to be created)
- **Issues:** `known_issues/`

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34 Claude Code Guide: RAPIDS Subsystem

Version: 1.0 **Last Updated:** 2025-10-11 **Purpose:** AI-specific guidance for working with RAPIDS subsystem

---

34.1 Quick Context

What: Rapid AXI Programmable In-band Descriptor System - Custom DMA-style accelerator with network interfaces **Status:** 🟡 Active development - ~80% test coverage, validation in progress **Your Role:** Help users understand architecture, fix bugs, extend functionality

📖 **Complete Specification:** `projects/components/rapids/docs/rapids_spec/` ← **Always reference this for technical details**

34.2 📖 Global Requirements Reference

IMPORTANT: Check `/GLOBAL_REQUIREMENTS.md` before starting RAPIDS work

All mandatory requirements are consolidated in the global requirements document: - **See:** `/GLOBAL_REQUIREMENTS.md` - Repository-wide mandatory requirements - **RAPIDS-Specific:** Attribution format, BFM usage requirements - **Universal:** TB location, three methods, TBBase inheritance, 100% success

This CLAUDE.md provides RAPIDS-specific guidance. Also review: - Root `/CLAUDE.md` - Repository-wide patterns - `projects/components/CLAUDE.md` - Project area standards (reset macros, FPGA attributes) - `bin/TBClasses/CLAUDE.md` - Framework usage patterns

---

34.3 Critical Rules for This Subsystem

34.3.1 Rule #0: Attribution Format for Git Commits

IMPORTANT: When creating git commit messages for RAPIDS documentation or code:

Use:

```
Documentation and implementation support by Claude.
```

Do NOT use:

```
Co-Authored-By: Claude <noreply@anthropic.com>
```

Rationale: RAPIDS receives AI assistance for structure and clarity, while design concepts and architectural decisions remain human-authored.

34.3.2 Rule #0.1: Testbench Location and Test Structure (MANDATORY)

📖 **See:** `/GLOBAL_REQUIREMENTS.md` Section 2.1 for complete requirement

RAPIDS-Specific Directory Structure:

```
projects/components/rapids/dv/
├── tbclasses/                    # ★ RAPIDS TB classes (project
area!)
│   ├── scheduler_tb.py           # Scheduler testbench
│   ├── descriptor_engine_tb.py   # Descriptor engine testbench
│   └── rapids_integration_tb.py  # Integration testbench
└── tests/                        # Test runners
    ├── fub_tests/scheduler/test_scheduler.py
    ├── fub_tests/descriptor_engine/test_desc_engine.py
    └── integration_tests/test_rapids_integration.py
```

RAPIDS Import Pattern:

```python
# Import framework utilities (PYTHONPATH includes bin/)
import os, sys
from TBClasses.shared.utilities import get_repo_root
from TBClasses.shared.tbbase import TBBase

# Add repo root to Python path using robust git-based method
repo_root = get_repo_root()
sys.path.insert(0, repo_root)

# Import RAPIDS TB from project area
from projects.components.rapids.dv.tbclasses.scheduler_tb import
SchedulerTB
```

RAPIDS Test File Organization:

```python
# 1. CocoTB functions at top (prefix "cocotb_test_*")
@cocotb.test(timeout_time=100, timeout_unit="ms")
async def cocotb_test_basic_flow(dut):
    tb = SchedulerTB(dut)
    await tb.setup_clocks_and_reset()
    # ... test logic

# 2. Parameter generation near bottom
def generate_scheduler_test_params():
    return [(0, 8, 512, 8)]  # (channel_id, num_channels, data_width,
credit_width)

scheduler_params = generate_scheduler_test_params()

# 3. Pytest wrappers at bottom
@pytest.mark.parametrize("channel_id, ...", scheduler_params)
def test_basic_flow(request, channel_id, ...):
    run(..., testcase="cocotb_test_basic_flow", ...)
```

Why RAPIDS Tests Use Project Area: 1. **Reusability:** Same TB in FUB tests, integration tests, system tests 2. **Composition:** Scheduler TB + Descriptor TB → Integration TB 3. **Discovery:** All RAPIDS code under `projects/components/rapids/`

📖 **Complete Pattern:** `val/amba/test_apb_slave.py` lines 1251-1346 (reference example)

📖 **Queue-Based Verification Pattern:** See `/GLOBAL_REQUIREMENTS.md` Section 2.4

RAPIDS-Specific Usage:

RAPIDS uses queue-based verification for in-order verification of program engine, descriptor engine, and AXI transactions.

Example - Program Engine Verification:

```python
# Direct queue access for in-order RAPIDS operations
class ProgramEngineTB(TBBase):
    async def verify_write_operation(self, expected_addr,
expected_data):
        """RAPIDS program engine: In-order writes, simple queue
verification"""
        # Wait for AXI transaction
        await self.wait_for_transaction(self.aw_monitor)

        # Get from queue - program engine writes are always in-order
        aw_pkt = self.aw_monitor._recvQ.popleft()
        w_pkt = self.w_monitor._recvQ.popleft()

        # Verify
```

```
        assert aw_pkt.addr == expected_addr
        assert w_pkt.data == expected_data
```

When RAPIDS Uses Memory Models: - ✗ Descriptor engine (in-order) - Use queue access - ✗ Program engine (in-order) - Use queue access - √ Integration tests with multiple masters - Memory model tracks state

📖 **Complete Patterns:** `docs/VERIFICATION_ARCHITECTURE_GUIDE.md`

34.3.3 Rule #0.5: Three Mandatory TB Methods (MANDATORY)

📖 **See:** `/GLOBAL_REQUIREMENTS.md` Section 2.2 for complete requirement

RAPIDS-Specific Context:

Many RAPIDS modules (especially scheduler and credit management) require configuration signals set BEFORE reset is released. This is because some counters/state are initialized during reset based on these config values.

Critical Example - Exponential Credit Encoding:

```
async def setup_clocks_and_reset(self):
    """RAPIDS-specific: Config signals MUST be set before reset"""
    # Start clock
    await self.start_clock('clk', freq=10, units='ns')

    # ⚠️ CRITICAL: Set exponential credit config BEFORE reset
    # Scheduler's credit counter initializes during reset based on
this value
    self.dut.cfg_initial_credit.value = 4  # 4 = 16 credits (2^4)
    self.dut.cfg_use_credit.value = 1

    # Now perform reset sequence
    await self.assert_reset()
    await self.wait_clocks('clk', 10)
    await self.deassert_reset()
    await self.wait_clocks('clk', 5)

async def assert_reset(self):
    """Assert active-low reset"""
    self.dut.rst_n.value = 0

async def deassert_reset(self):
    """Release active-low reset"""
    self.dut.rst_n.value = 1
```

Why Config-Before-Reset Matters for RAPIDS: - Scheduler credit counter: Initialized to `(1 << cfg_initial_credit)` during reset - Descriptor engine depth: Configuration read during reset sequence - SRAM parameters: Must be stable before memory controllers come out of reset

Common RAPIDS Configs to Set Before Reset: - `cfg_initial_credit` - Exponential credit encoding (0→1, 1→2, 2→4, etc.) - `cfg_use_credit` - Enable/disable credit-based flow control - `cfg_timeout_threshold` - Watchdog timer configuration - `cfg_sram_depth` - SRAM buffer size parameters

---

34.3.4 Rule #0.75: Audit Signal Naming BEFORE Writing Testbenches

⚠️ CRITICAL: Check for Signal Naming Conflicts BEFORE Factory Usage ⚠️

Before writing testbench code that uses AXI factory functions, audit your RTL for signal naming conflicts.

The Problem: AXI factory pattern matching searches for signals using prefix + channel patterns (e.g., `{prefix}ar_valid`, `{prefix}r_ready`). If you have: - Internal signals: `desc_valid`, `desc_ready` (simple handshake) - External AXI ports: `desc_ar_valid`, `desc_ar_ready` (AXI AR channel)

Both match `desc_*valid` → Factory finds BOTH signals → Initialization FAILS!

Solution - Signal Naming Audit Tool:

```
# Audit single file before writing testbench
../../bin/audit_signal_naming_conflicts.py
projects/components/rapids/rtl/rapids_macro/scheduler_group.sv

# Audit entire RAPIDS subsystem
../../bin/audit_signal_naming_conflicts.py
projects/components/rapids/rtl/

# Generate markdown report
../../bin/audit_signal_naming_conflicts.py
projects/components/rapids/rtl/ --markdown signal_conflicts.md
```

Known RAPIDS Conflicts: See `projects/components/rapids/known_issues/scheduler_group_signal_naming_conflicts.md` for documented conflicts in scheduler_group.sv: - `desc` prefix: 4 internal + 4 external (AR/R channels) - `prog` prefix: 4 internal + 6 external (AW/W/B channels)

Recommended Workflow: 1. √ Write RTL module with signals 2. √ **Run audit script to detect conflicts** 3. √ Fix naming conflicts (rename internal signals with `_to_sched` suffix) 4. √ Write testbench using factory pattern matching

Three Solutions When Conflicts Found: 1. **Rename internal signals** (recommended): `desc_valid` → `desc_to_sched_valid` 2. **Use explicit signal_map**: Bypass pattern matching with manual signal mapping 3. **Test at higher level**: Where internal signals aren't visible (e.g., rapids_top)

📖 **Complete Guide:** `bin/SIGNAL_NAMING_AUDIT.md`

34.3.5 Rule #1: MANDATORY BFM Usage for FUB-Level Tests

⚠️ CRITICAL DESIGN INTENTION - USE FRAMEWORK BFMs ⚠️

For all RAPIDS FUB (Functional Unit Block) level tests, you MUST use appropriate BFMs from the CocoTB Framework:

Interface Type → Required BFM Component:

Interface Type	Framework Component Location	Usage
Custom valid/ready	`bin/TBClasses/components/gaxi/`	**GAXI Master/Slave BFMs**
AXI4	`bin/TBClasses/components/axi4/`	AXI4 Master/Slave drivers
AXI4-Lite (AXIL)	`bin/TBClasses/components/axil4/`	AXIL Master/Slave drivers
APB	`bin/TBClasses/components/apb/`	APB Master/Slave drivers
AXI-Stream (AXIS)	`bin/TBClasses/components/axis4/`	AXIS Master/Slave drivers
Network	`bin/TBClasses/components/network/`	Network Master/Sl

Interface Type	Framework Component Location	Usage
		ave drivers
MonBus	`bin/TBClasses/components/monbus/`	MonBus drivers

Critical Rules:

1. **NEVER manually drive valid/ready handshakes** - Use GAXI BFM components
2. **NEVER create custom protocol drivers** - Framework components already exist
3. **NEVER embed simple handshake logic** - Extract to GAXI Master/Slave

Example - Program Engine Interface:

```
# ✗ WRONG: Manual handshake driving
async def send_program_request(self, addr, data):
    # Manually driving program_valid/program_ready - DON'T DO THIS!
    self.dut.program_valid.value = 1
    self.dut.program_pkt_addr.value = addr
    self.dut.program_pkt_data.value = data
    while int(self.dut.program_ready.value) == 0:
        await self.wait_clocks(self.clk_name, 1)
    await self.wait_clocks(self.clk_name, 1)
    self.dut.program_valid.value = 0

# ✓ CORRECT: Use GAXI Master BFM
from CocoTBFramework.components.gaxi.gaxi_master import GAXIMaster

class ProgramEngineTB(TBBase):
    def __init__(self, dut):
        super().__init__(dut)
        # Create GAXI master for program interface
        self.program_master = GAXIMaster(
            dut=dut,
            clock=dut.clk,
            valid_signal='program_valid',
            ready_signal='program_ready',
            data_signals=['program_pkt_addr', 'program_pkt_data'],
            data_widths=[64, 32],
            name='program_interface'
        )

    async def send_program_request(self, addr, data):
```

```
        # Use GAXI master for proper handshaking
        await self.program_master.write({'program_pkt_addr': addr,
'program_pkt_data': data})
```

Why This Rule Exists:

1. **Consistency**: All tests use same handshake protocol
2. **Correctness**: GAXI BFMs handle complex timing scenarios correctly
3. **Reusability**: Same BFM used across all RAPIDS tests
4. **Maintainability**: Fix once in BFM, all tests benefit
5. **Coverage**: GAXI BFMs include timing randomization and backpressure

When Manual Driving is Acceptable:

- ✓ **Quick debug/prototype** - Temporary, will be replaced with BFM
- ✓ **Clock/reset initialization** - Not part of protocol handshaking
- ✗ **Production testbenches** - MUST use BFMs

Search Before Creating:

```
# Find existing GAXI components
find bin/TBClasses/components/gaxi/ -name "*.py"

# Find example usage
grep -r "GAXIMaster\|GAXISlave"
projects/components/rapids/dv/tests/fub_tests/
grep -r "from.*gaxi" bin/TBClasses/
```

BFM Selection Guide:

```
Need to drive interface with valid/ready?
├─ Standard protocol (AXI4, APB, AXIS, Network)?
│  └─ Use protocol-specific BFM (axi4/, apb/, axis4/, network/)
│
├─ Custom RAPIDS interface with valid/ready signals?
│  └─ Use GAXI Master/Slave BFM (gaxi/)
│
└─ MonBus monitor packet interface?
   └─ Use MonBus BFM (monbus/)
```

📖 **See:** - `bin/TBClasses/components/gaxi/README.md` - GAXI BFM documentation - `bin/TBClasses/components/axi4/README.md` - AXI4 BFM documentation - `val/amba/test_*.py` - Reference examples using framework BFMs

---

34.3.6 Rule #2: Always Reference Detailed Specification

This subsystem has extensive documentation in `projects/components/rapids/docs/rapids_spec/`

Before answering technical questions:

```
# Check complete specification
ls projects/components/rapids/docs/rapids_spec/
cat projects/components/rapids/docs/rapids_spec/rapids_index.md
cat
projects/components/rapids/docs/rapids_spec/ch02_blocks/01_01_schedule
r.md
```

Your answer should: 1. Provide direct answer/code 2. **Then link to detailed spec:** "See `projects/components/rapids/docs/rapids_spec/{file}.md` for complete specification"

34.3.7 Rule #3: Know the Known Issues

Fixed Critical Issue: - √ **Scheduler Credit Counter - FIXED** (projects/components/rapids/rtl/rapids_fub/scheduler.sv:567-570) - Was: Credit counter hardcoded to 0 - Now: Implements exponential encoding (0→1, 1→2, 2→4, ..., 15→∞) - Impact: Credit-based flow control now functional with proper encoding - Tests: Verify exponential decoding works correctly - Priority: P0 (Critical - COMPLETED)

Always check: `projects/components/rapids/known_issues/` before diagnosing bugs

```
ls projects/components/rapids/known_issues/
cat projects/components/rapids/known_issues/scheduler.md
```

34.3.8 Rule #4: RAPIDS is Complex - Understand Block Interactions

Key Interaction Patterns:

1. **Descriptor Flow:**

   ```
   Software → AXIL4 → Descriptor Engine → Scheduler → Data Paths
   ```

2. **Sink Data Path (Network → Memory):**

   ```
   Network Slave → Sink SRAM Control → Sink AXI Write Engine →
   System Memory
   ```

3. **Source Data Path (Memory → Network):**

   ```
   System Memory → Source AXI Read Engine → Source SRAM Control →
   Network Master
   ```

4. **Monitoring:**

```
All Blocks → MonBus Reporter → MonBus Output
```

Never work on one block in isolation without understanding its upstream/downstream dependencies!

34.3.9 Rule #5: Test Strategy is Multi-Layered

RAPIDS testing follows this hierarchy:

1. **FUB (Functional Unit Block) Tests:** `projects/components/rapids/dv/tests/fub_tests/`
   - Individual block testing
   - Focus: Module-level functionality
   - Example: Scheduler FSM, Descriptor Engine FIFO
2. **Integration Tests:** `projects/components/rapids/dv/tests/integration_tests/`
   - Multi-block scenarios
   - Focus: Block-to-block interfaces
   - Example: Scheduler + Descriptor Engine, Sink data path end-to-end
3. **System Tests:** `projects/components/rapids/dv/tests/system_tests/`
   - Full RAPIDS operation
   - Focus: Realistic traffic patterns
   - Example: Complete DMA transfer with monitoring

When creating/modifying tests, ensure appropriate test layer is used!

34.4 Architecture Quick Reference

34.4.1 Block Organization

```
RAPIDS Architecture (17 SystemVerilog modules)
├── Scheduler Group
│   ├── scheduler.sv              (Task FSM, credit management)
│   ├── descriptor_engine.sv      (Descriptor FIFO, parsing)
│   └── program_engine.sv         (Program sequencing, alignment)
│
├── Sink Data Path (Network → Memory)
│   ├── network_slave.sv             (Network ingress)
│   ├── sink_sram_control.sv      (Buffer management)
│   └── sink_axi_write_engine.sv  (AXI4 write to system memory)
│
├── Source Data Path (Memory → Network)
```

```
│   ├── source_axi_read_engine.sv (AXI4 read from system memory)
│   ├── source_sram_control.sv    (Buffer management)
│   └── network_master.sv            (Network egress)
│
└── MonBus AXIL Group
    ├── axil4_slave.sv            (Control/status registers)
    └── monbus_reporter.sv        (Monitor packet generation)
```

📖 **See:** `projects/components/rapids/docs/rapids_spec/ch02_blocks/00_overview.md` for detailed block descriptions

34.4.2 Module Quick Reference

Module	Location	Purpose	Documentation
scheduler.sv	`rapids_fub/`	Task FSM, credit management	`rapids_spec/ch02_blocks/01_01_scheduler.md`
descriptor_engine.sv	`rapids_fub/`	Descriptor FIFO, parsing	`rapids_spec/ch02_blocks/01_02_descriptor_engine.md`
program_engine.sv	`rapids_fub/`	Program sequencing	`rapids_spec/ch02_blocks/01_03_program_engine.md`
network_slave.sv	`rapids_fub/`	Network ingress	`rapids_spec/ch02_blocks/02_01_network_slave.md`
sink_sram_control.sv	`rapids_fub/`	Sink buffer management	`rapids_spec/ch02_blocks/02_02_sink_sram_control.md`
sink_axi_write_engine.sv	`rapids_fub/`	Memory writes	`rapids_spec/ch02_blocks/02_03_sink_axi_write_engine.md`
source_axi_read_engine.sv	`rapids_fub/`	Memory reads	`rapids_spec/ch02_blocks/03_03_source_axi_read_engine.md`
source_sram_control.sv	`rapids_fub/`	Source buffer management	`rapids_spec/ch02_blocks/03_02_source_sram_control.md`
network_master.sv	`rapids_fub/`	Network egress	`rapids_spec/ch02_blocks/`

Module	Location	Purpose	Documentation
			03_01_network_master.md
rapids_top.sv	rapids_macro/	Top-level integration	rapids_spec/ ch03_interfaces/ 01_top_level.md

34.4.3 Interface Summary

Interface	Type	Width	Purpose	Specification
AXIL4	Slave	32-bit	Control/ status registers	rapids_spec/ ch03_interfaces/ 02_axil_interface_spec.md
AXI4 (Sink)	Master	Configurable	Write to system memory	rapids_spec/ ch03_interfaces/ 03_axi4_interface_spec.md
AXI4 (Source)	Master	Configurable	Read from system memory	rapids_spec/ ch03_interfaces/ 03_axi4_interface_spec.md
Network (Sink)	Slave	Configurable	Network ingress	rapids_spec/ ch03_interfaces/ 04_network_interface_spec.md
Network (Source)	Master	Configurable	Network egress	rapids_spec/ ch03_interfaces/ 04_network_interface_spec.md
MonBus	Master	64-bit	Monitor packet output	rapids_spec/ ch03_interfaces/ 05_monbus_interface_spec.md

34.5 Common User Questions and Responses

34.5.1 Q: "How does the scheduler work?"

A: Direct answer:

The scheduler is a complex FSM that coordinates RAPIDS operations:

1. **Idle State:** Waits for descriptors from Descriptor Engine
2. **Parse State:** Extracts descriptor fields (address, length, control)

3. **Credit Check:** Verifies credit availability (if credit mode enabled)
4. **Execute State:** Activates appropriate data path (sink or source)
5. **Monitor State:** Tracks operation progress
6. **Complete State:** Generates completion packets, updates credits

Key FSM States: - `IDLE` → `PARSE` → `CREDIT_CHECK` → `EXECUTE` → `MONITOR` → `COMPLETE` → `IDLE`

Credit Management: - √ **FIXED:** Exponential credit encoding now implemented (lines 567-570) - **Encoding:** 0→1, 1→2, 2→4, 3→8, ..., 14→16384, 15→∞ (unlimited) - **See Q&A below** for complete encoding table and details

📖 **See:** - `projects/components/rapids/docs/rapids_spec/ch02_blocks/01_01_scheduler.md` - Complete FSM specification - `projects/components/rapids/known_issues/scheduler.md` - Known bugs and workarounds

34.5.2 Q: "How do I configure RAPIDS?"

A: Configuration via AXIL4 interface:

```
// 1. Initialize RAPIDS via AXIL4 writes
// Configure SRAM depths
write_axil(ADDR_SINK_SRAM_DEPTH, 1024);
write_axil(ADDR_SOURCE_SRAM_DEPTH, 1024);

// Set timeout thresholds
write_axil(ADDR_TIMEOUT_THRESHOLD, 1000);

// Configure initial credits (exponential encoding)
// 0→1, 1→2, 2→4, 3→8, 4→16, etc.
write_axil(ADDR_INITIAL_CREDIT, 4);  // 4 = 16 credits (2^4)
write_axil(ADDR_CREDIT_ENABLE, 1);   // ✓ Enable credit mode (now
fixed!)

// 2. Load descriptors
write_descriptor(addr, length, control_bits);

// 3. Enable operation
write_axil(ADDR_ENABLE, 1);
```

📖 **See:** - `projects/components/rapids/docs/rapids_spec/ch04_programming_models/01_programming.md` - Programming model - `projects/components/rapids/docs/rapids_spec/ch05_registers/01_registers.md` - Register definitions

34.5.3 Q: "What's the data flow for network to memory transfer?"

A: Sink path data flow:

```
1. Network Packet Arrives
   ↓
2. Network Slave receives packet
   - Validates packet format
   - Handshakes with Network interface
   ↓
3. Sink SRAM Control buffers data
   - Writes to SRAM
   - Manages write pointers
   - Handles backpressure
   ↓
4. Sink AXI Write Engine
   - Reads from SRAM
   - Generates AXI4 write transactions
   - Bursts data to system memory
   ↓
5. Completion Reporting
   - Generates MonBus completion packet
   - Updates scheduler state
```

Key Considerations: - SRAM acts as buffer to decouple network from memory timing - AXI4 bursts used for efficient memory access - Backpressure propagates from memory to network

📖 **See:** `projects/components/rapids/docs/rapids_spec/ch02_blocks/02_00_sink_data_path.md`

34.5.4 Q: "What's the credit counter bug and how does exponential encoding work?"

A: Scheduler credit counter exponential encoding:

The scheduler uses **exponential credit encoding** to provide a wide range of credit values (1 to 16384) with a compact 4-bit configuration:

Encoding Table: | `cfg_initial_credit` | Actual Credits | Use Case | |———————|———————|———-| | 0 | 1 | Minimum (2^0) | | 1 | 2 | Very low traffic (2^1) | | 2 | 4 | Low traffic (2^2) | | 3 | 8 | (2^3) | | 4 | 16 | Typical (2^4) | | 5 | 32 | (2^5) | | 6 | 64 | Medium traffic (2^6) | | 7 | 128 | (2^7) | | 8 | 256 | High traffic (2^8) | | 10 | 1024 | Very high traffic (2^10) | | 14 | 16384 | Maximum finite (2^14) | | 15 | ∞ (0xFFFFFFFF) | Unlimited credits |

✓ **FIXED Implementation:**

```
// projects/components/rapids/rtl/rapids_fub/scheduler.sv:567-570
// Exponential credit encoding: 0→1, 1→2, 2→4, ..., 14→16384, 15→∞
r_descriptor_credit_counter <= (cfg_initial_credit == 4'hF) ?
```

```
32'hFFFFFFFF :
                                (cfg_initial_credit == 4'h0) ?
32'h00000001 :
                                (32'h1 << cfg_initial_credit);
```

Was Broken (Before Fix):

```
r_descriptor_credit_counter <= 32'h0;  // ✗ WRONG: Hardcoded to 0
```

Impact (Before Fix): - Credit-based flow control didn't work - All operations blocked if credit mode enabled - Descriptors never processed

Encoding Rationale: - Compact 4-bit config covers wide range: 1 to 16384 credits - Fine-grained control for low traffic (1, 2, 4, 8) - High-throughput support (256, 1024, 16384) - Special unlimited mode (15 → ∞)

Important: Exponential encoding applies **only at initialization**. Once running, the counter operates linearly (increment/decrement by 1).

📖 **See:** - `projects/components/rapids/docs/rapids_spec/ch02_blocks/01_01_scheduler.md` - Complete encoding table - `projects/components/rapids/known_issues/scheduler.md` - Bug details and fix verification

34.5.5 Q: "How do I run RAPIDS tests?"

A: Multi-layered test approach:

```
# 1. FUB (Functional Unit Block) Tests - Individual blocks
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ -v
pytest
projects/components/rapids/dv/tests/fub_tests/descriptor_engine/ -v
pytest projects/components/rapids/dv/tests/fub_tests/ -v  # All FUB
tests

# 2. Integration Tests - Multi-block scenarios
pytest projects/components/rapids/dv/tests/integration_tests/ -v

# 3. System Tests - Full RAPIDS operation
pytest projects/components/rapids/dv/tests/system_tests/ -v

# Run with waveforms for debugging
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ --
vcd=debug.vcd
gtkwave debug.vcd
```

Test Organization: - **FUB tests:** Focus on individual block functionality - **Integration tests:** Verify block-to-block interfaces - **System tests:** Validate complete data flows

Current Status: ~80% functional coverage, basic scenarios validated

📖 **See:** `docs/RAPIDS_Validation_Status_Report.md` for detailed test results

34.6 Integration Patterns

34.6.1 Pattern 1: Basic RAPIDS Instantiation

```
rapids_top #(
    .AXI_ADDR_WIDTH(32),
    .AXI_DATA_WIDTH(64),
    .Network_DATA_WIDTH(64),
    .SRAM_DEPTH(1024),
    .MAX_DESCRIPTORS(16)
) u_rapids (
    // Clock and Reset
    .aclk               (system_clk),
    .aresetn            (system_rst_n),

    // AXIL4 Control Interface
    .s_axil_awaddr      (ctrl_awaddr),
    .s_axil_awvalid     (ctrl_awvalid),
    .s_axil_awready     (ctrl_awready),
    .s_axil_wdata       (ctrl_wdata),
    .s_axil_wstrb       (ctrl_wstrb),
    .s_axil_wvalid      (ctrl_wvalid),
    .s_axil_wready      (ctrl_wready),
    .s_axil_bresp       (ctrl_bresp),
    .s_axil_bvalid      (ctrl_bvalid),
    .s_axil_bready      (ctrl_bready),
    .s_axil_araddr      (ctrl_araddr),
    .s_axil_arvalid     (ctrl_arvalid),
    .s_axil_arready     (ctrl_arready),
    .s_axil_rdata       (ctrl_rdata),
    .s_axil_rresp       (ctrl_rresp),
    .s_axil_rvalid      (ctrl_rvalid),
    .s_axil_rready      (ctrl_rready),

    // AXI4 Memory Interface (Sink - Write)
    .m_axi_sink_awaddr  (mem_sink_awaddr),
    .m_axi_sink_awlen   (mem_sink_awlen),
    .m_axi_sink_awsize  (mem_sink_awsize),
    .m_axi_sink_awburst (mem_sink_awburst),
    .m_axi_sink_awvalid (mem_sink_awvalid),
    .m_axi_sink_awready (mem_sink_awready),
    // ... additional AXI4 sink write channel signals
```

```
    // AXI4 Memory Interface (Source - Read)
    .m_axi_source_araddr  (mem_source_araddr),
    .m_axi_source_arlen   (mem_source_arlen),
    .m_axi_source_arsize  (mem_source_arsize),
    .m_axi_source_arburst (mem_source_arburst),
    .m_axi_source_arvalid (mem_source_arvalid),
    .m_axi_source_arready (mem_source_arready),
    // ... additional AXI4 source read channel signals

    // Network Network Interface (Sink - Receive)
    .s_network_tdata       (net_rx_data),
    .s_network_tvalid      (net_rx_valid),
    .s_network_tready      (net_rx_ready),
    .s_network_tlast       (net_rx_last),
    // ... additional Network sink signals

    // Network Network Interface (Source - Transmit)
    .m_network_tdata       (net_tx_data),
    .m_network_tvalid      (net_tx_valid),
    .m_network_tready      (net_tx_ready),
    .m_network_tlast       (net_tx_last),
    // ... additional Network source signals

    // MonBus Output
    .monbus_pkt_valid   (rapids_mon_valid),
    .monbus_pkt_ready   (rapids_mon_ready),
    .monbus_pkt_data    (rapids_mon_data)
);
```

34.6.2 Pattern 2: Configuration Sequence

```
// Recommended initialization sequence
initial begin
    // 1. Assert reset
    aresetn = 0;
    repeat(10) @(posedge aclk);
    aresetn = 1;

    // 2. Configure RAPIDS via AXIL4
    axil_write(ADDR_SINK_SRAM_DEPTH, 1024);
    axil_write(ADDR_SOURCE_SRAM_DEPTH, 1024);
    axil_write(ADDR_TIMEOUT_THRESHOLD, 1000);
    axil_write(ADDR_INITIAL_CREDIT, 4);  // 4 = 16 credits (2^4,
exponential encoding)
    axil_write(ADDR_CREDIT_ENABLE, 1);  // ✓ Enable credit mode (now
fixed!)
```

```
    // 3. Load descriptors
    for (int i = 0; i < num_descriptors; i++) begin
        axil_write(ADDR_DESC_ADDR, descriptor[i].addr);
        axil_write(ADDR_DESC_LEN, descriptor[i].length);
        axil_write(ADDR_DESC_CTRL, descriptor[i].control);
        axil_write(ADDR_DESC_COMMIT, 1);
    end

    // 4. Enable RAPIDS operation
    axil_write(ADDR_ENABLE, 1);
end
```

34.6.3 Pattern 3: MonBus Integration

```
// Always add downstream FIFO for MonBus
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_rapids_mon_fifo (
    .i_clk      (aclk),
    .i_rst_n    (aresetn),
    .i_data     (monbus_pkt_data),
    .i_valid    (monbus_pkt_valid),
    .o_ready    (monbus_pkt_ready),
    .o_data     (fifo_mon_data),
    .o_valid    (fifo_mon_valid),
    .i_ready    (consumer_ready)
);
```

34.7 Anti-Patterns to Catch

34.7.1 ✗ Anti-Pattern 1: Not Understanding Exponential Credit Encoding

```
✗ WRONG:
.cfg_initial_credit(4'd16),  // Thinks this gives 16 credits - it
doesn't!

✓ CORRECTED:
"Credits use exponential encoding:
- cfg_initial_credit = 4 → 16 credits (2^4)
- cfg_initial_credit = 8 → 256 credits (2^8)
- cfg_initial_credit = 15 → ∞ credits (unlimited)
See
projects/components/rapids/docs/rapids_spec/ch02_blocks/01_01_schedule
r.md for encoding table."
```

34.7.2 ✗ Anti-Pattern 2: Insufficient SRAM Depth

```
✗ WRONG:
.SRAM_DEPTH(16)  // Too small for realistic packets

✓ CORRECTED:
"SRAM depth should match typical packet sizes.
Recommended: 1024-4096 entries depending on data width and packet
sizes."
```

34.7.3 ✗ Anti-Pattern 3: No MonBus Downstream Handling

```
✗ WRONG:
assign monbus_pkt_ready = 1'b1;  // Always ready = potential packet
loss

✓ CORRECTED:
"Connect to FIFO or proper consumer:
gaxi_fifo_sync #(.DATA_WIDTH(64), .DEPTH(256)) u_mon_fifo (
    .i_valid(monbus_pkt_valid),
    .i_data(monbus_pkt_data),
    .o_ready(monbus_pkt_ready),
    ...
);"
```

34.7.4 ✗ Anti-Pattern 4: Testing Individual Blocks in Isolation

```
✗ WRONG:
"Only test scheduler without descriptor engine"

✓ CORRECTED:
"RAPIDS blocks are tightly coupled. Always test:
1. FUB tests for basic block functionality
2. Integration tests for block interactions
3. System tests for complete flows"
```

34.8 Debugging Workflow

34.8.1 Issue: Scheduler Not Processing Descriptors

Check in order: 1. √ Credit configuration correct? (Remember exponential encoding!) 2. √ Descriptors loaded via AXIL4? 3. √ RAPIDS enabled? (`ADDR_ENABLE = 1`) 4. √ Reset properly deasserted? 5. √ Descriptor engine FIFO not empty?

Debug commands:

```
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ -v -s
# Verbose test
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ --
vcd=debug.vcd
gtkwave debug.vcd  # Inspect FSM state transitions
```

34.8.2 Issue: Data Path Stalls

Check in order: 1. √ SRAM depth sufficient? 2. √ Downstream interfaces ready? 3. √ AXI4 backpressure handling? 4. √ Network flow control? 5. √ Buffer overflow/underflow detection?

Waveform Analysis: - Check SRAM read/write pointers - Verify AXI4 handshakes - Inspect Network TREADY signals

34.8.3 Issue: MonBus Packets Not Generated

Check in order: 1. √ Operations completing successfully? 2. √ MonBus ready signal asserted? 3. √ MonBus reporter enabled? 4. √ Downstream FIFO not full?

34.9 Testing Guidance

34.9.1 Test Organization

```
projects/components/rapids/dv/tests/
├── fub_tests/                  # Individual block tests
│   ├── scheduler/
│   │   ├── test_scheduler.py   # FSM, credit management
│   │   └── conftest.py
│   ├── descriptor_engine/
│   │   ├── test_desc_engine.py # FIFO, parsing
│   │   └── conftest.py
│   ├── program_engine/
│   ├── network_interfaces/
│   └── simple_sram/
│
├── integration_tests/          # Multi-block scenarios
│   ├── test_scheduler_desc.py  # Scheduler + Descriptor Engine
│   ├── test_sink_path.py       # Complete sink data path
│   └── test_source_path.py     # Complete source data path
│
└── system_tests/               # Full RAPIDS operation
    ├── test_full_dma.py        # Complete DMA transfer
    └── test_bidirectional.py   # Simultaneous sink/source
```

34.9.2 Running Tests

```
# Single block test
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ -v

# All FUB tests
pytest projects/components/rapids/dv/tests/fub_tests/ -v

# Integration tests
pytest projects/components/rapids/dv/tests/integration_tests/ -v

# System tests
pytest projects/components/rapids/dv/tests/system_tests/ -v

# All RAPIDS tests
pytest projects/components/rapids/dv/tests/ -v

# With waveforms
pytest projects/components/rapids/dv/tests/fub_tests/scheduler/ --
vcd=waves.vcd
```

34.9.3 Test Coverage Status

Current Status: ~80% functional coverage

Component	Coverage	Status
Scheduler	~90%	Exponential credit encoding implemented
Descriptor Engine	~80%	Basic tests passing
Program Engine	~85%	Alignment tested
Sink Data Path	~75%	Basic flows working
Source Data Path	~70%	Basic flows working
Integration	~60%	More stress testing needed

34.10 Key Documentation Links

34.10.1 Always Reference These

Primary Technical Specification: - `projects/components/rapids/docs/rapids_spec/rapids_index.md` - Complete specification index - `projects/components/rapids/docs/rapids_spec/ch01_overview/` - Architecture overview - `projects/components/rapids/docs/rapids_spec/ch02_blocks/` - Block specifications - `projects/components/rapids/docs/rapids_spec/ch03_interfaces/` - Interface specifications - `projects/components/rapids/docs/rapids_spec/ch04_programming_models/` - Programming model - `projects/components/rapids/docs/rapids_spec/ch05_registers/` - Register definitions

This Subsystem: - `projects/components/rapids/PRD.md` - Requirements overview - `projects/components/rapids/README.md` - Quick start guide (if exists) - `projects/components/rapids/TASKS.md` - Current work items - `projects/components/rapids/known_issues/` - Bug tracking

Validation: - `docs/RAPIDS_Validation_Status_Report.md` - Test results and status

Root: - `/PRD.md` - Master requirements - `/CLAUDE.md` - Repository guide

34.11 Quick Commands

```
# View complete specification
cat projects/components/rapids/docs/rapids_spec/rapids_index.md
cat
projects/components/rapids/docs/rapids_spec/ch02_blocks/01_01_schedule
r.md

# Check known issues
ls projects/components/rapids/known_issues/
cat projects/components/rapids/known_issues/scheduler.md

# Run tests
pytest projects/components/rapids/dv/tests/fub_tests/ -v
pytest projects/components/rapids/dv/tests/integration_tests/ -v

# Lint
verilator --lint-only
projects/components/rapids/rtl/rapids_fub/scheduler.sv

# Search for modules
```

```
find projects/components/rapids/rtl/rapids_fub/ -name "*.sv" -exec
grep -H "^module" {} \;
```

34.12 Verification Patterns and Best Practices

34.12.1 Pattern: Continuous Background Monitoring

Problem: When testing components that output data asynchronously (descriptors, packets, etc.), tests that only check outputs at specific points will miss data that arrives during other operations.

Solution: Use continuous background monitoring coroutines that run throughout the test.

Example - Descriptor Engine Testing:

```
class DescriptorEngineTB(TBBase):
    async def run_apb_only_test(self, num_packets: int, profile:
DelayProfile):
        """Test with continuous descriptor monitoring"""
        descriptors_collected = []  # Shared list
        monitor_active = True  # Control flag

        # Background coroutine monitors continuously
        async def descriptor_monitor():
            """Continuously monitor for descriptors throughout the
test"""
            while monitor_active:
                await self.wait_clocks(self.clk_name, 1)
                if int(self.dut.descriptor_valid.value) == 1:
                    desc_data = int(self.dut.descriptor_packet.value)
                    descriptors_collected.append(desc_data)
                    if len(descriptors_collected) % 5 == 1:
                        self.log.info(f"📦 Descriptor
{len(descriptors_collected)}: 0x{desc_data:X}")

        # Start background monitor
        monitor_task = cocotb.start_soon(descriptor_monitor())

        # Send all requests (monitor captures outputs during this
phase)
        for i in range(num_packets):
            # Send APB request
            self.dut.apb_valid.value = 1
            self.dut.apb_addr.value = test_addr
            await self.wait_clocks(self.clk_name, 1)
            # ... wait for ready ...
            self.dut.apb_valid.value = 0
```

```
        # Wait for final descriptors (monitor still running)
        for cycle in range(final_window):
            await self.wait_clocks(self.clk_name, 1)
            if len(descriptors_collected) >= num_packets:
                break

        # Stop background monitor
        monitor_active = False
        await self.wait_clocks(self.clk_name, 2)  # Let monitor finish

        # Verify results
        return len(descriptors_collected) == num_packets
```

Key Points: 1. **Shared State:** Use list/dict accessible from both main test and monitor coroutine 2. **Control Flag:** `monitor_active` allows clean shutdown 3. **Background Task:** `cocotb.start_soon()` runs monitor concurrently 4. **Continuous Capture:** Monitor checks every clock cycle, never missing data 5. **Clean Shutdown:** Set flag to False, wait for monitor to finish

When to Use: - √ Asynchronous output interfaces (descriptors, packets, responses) - √ Tests with rapid-fire input operations - √ Multi-stage pipelines where outputs don't align with inputs - ✗ Simple synchronous request-response patterns

Benefits: - 100% capture rate (no missed transactions) - Decouples input stimulus from output monitoring - Realistic timing coverage (captures outputs at any point)

Results: Descriptor engine tests improved from 42% success (5/12 descriptors) to 100% success (14/14 tests passing) by applying this pattern.

34.12.2 Pattern: Using Existing CocoTBFramework Components

Critical Rule: Always search for existing components before creating new ones.

Framework Structure:

```
bin/TBClasses/
├── components/          # Protocol-specific BFMs and drivers
│   ├── axi4/           # Complete AXI4 infrastructure
│   ├── axil4/          # AXI4-Lite components
│   ├── apb/            # APB drivers and monitors
│   ├── axis4/          # AXI-Stream components
│   └── rapids/           # RAPIDS-specific BFMs
├── tbclasses/          # Reusable testbench classes
│   ├── amba/           # AMBA testbenches
│   ├── rapids/           # RAPIDS testbenches
│   └── shared/         # Common utilities (TBBase, etc.)
└── scoreboards/        # Transaction checkers
```

Decision Tree: Create New BFM vs Use Existing?

```
Need to model protocol behavior?
├─ Is it a standard protocol (AXI4, APB, AXIS)?
│  └─ YES → Use components/axi4/, components/apb/, etc.
│           ✓ DO NOT create new implementation
│
├─ Is it RAPIDS-specific custom behavior?
│  ├─ Is it >100 lines?
│  │  └─ YES → Extract to components/rapids/
│  └─ Is it <50 lines and test-specific?
│     └─ YES → Keep embedded in testbench
│
└─ Is it generic helper (not protocol-specific)?
   └─ Add to tbclasses/shared/utilities.py
```

Example - Descriptor Engine AXI Responder:

```
# ✗ WRONG: Creating new AXI4 read responder BFM
# File: components/rapids/axi_read_responder_bfm.py
class AXIReadResponderBFM:
    """New AXI4 read responder - DON'T DO THIS!"""
    # ... 200 lines of AXI4 protocol implementation ...

# ✓ CORRECT: Use existing AXI4 components
# File: tbclasses/rapids/descriptor_engine_tb.py
async def axi_read_responder(self):
    """Simple test-specific responder - 50 lines, embedded"""
    while True:
        await self.wait_clocks(self.clk_name, 1)
        if int(self.dut.ar_valid.value) == 1 and
int(self.dut.ar_ready.value) == 1:
            # Generate response data
            self.dut.r_valid.value = 1
            self.dut.r_data.value = response_data
            # ... wait for r_ready ...
            self.dut.r_valid.value = 0
```

Why This Matters: - **Existing AXI4 components:** Comprehensive, tested, feature-complete - **Simple embedded responder:** Test-specific, minimal protocol simulation - **No duplication:** Framework already has robust AXI4 infrastructure

BFM Extraction Criteria:

Factor	Extract to BFM	Keep Embedded
Lines of code	>100 lines	<50 lines
Complexity	Complex protocol logic	Simple stimulus/response

Factor	Extract to BFM	Keep Embedded
Reusability	Used across multiple tests	Test-specific
Protocol	Custom RAPIDS-specific	Standard protocol (use framework)
Dependencies	Standalone	Tightly coupled to one test

Examples:

✓ **Extracted to BFM:** `DataMoverBFM` (components/rapids/data_mover_bfm.py) - 150+ lines - Complex RAPIDS data mover protocol - Used across scheduler tests - Reusable for integration tests

✓ **Kept Embedded:** AXI read responder in descriptor engine - 50 lines - Simple test-specific behavior - Framework already has full AXI4 support - No need for extraction

34.12.3 Test Scalability Patterns

Principle: Tests should scale from basic validation to comprehensive stress testing.

Pattern: Delay Profiles for Timing Coverage

```
class DelayProfile(Enum):
    """Delay profiles for comprehensive timing coverage"""
    FAST_PRODUCER = "fast_producer"      # Producer faster than
consumer
    FAST_CONSUMER = "fast_consumer"      # Consumer faster than
producer
    FIXED_DELAY = "fixed_delay"          # Predictable timing
    MINIMAL_DELAY = "minimal_delay"      # Stress test - minimal
delays
    BACKPRESSURE = "backpressure"        # Heavy backpressure
scenarios

# Test method scales with profile
async def run_apb_only_test(self, num_packets: int, profile:
DelayProfile):
    """Scalable test with configurable timing"""
    params = self.delay_params[profile]

    for i in range(num_packets):
        # Apply profile-specific delays
        producer_delay =
self.get_delay_value(params['producer_delay'])
        await self.wait_clocks(self.clk_name, producer_delay)

        # Send request...
```

```
        # Profile-specific backpressure
        if random.random() < params.get('backpressure_freq', 0.1):
            backpressure_cycles = random.randint(5, 25)
            await self.wait_clocks(self.clk_name, backpressure_cycles)
```

Pattern: Hierarchical Test Levels

```
# Test runner with multiple levels
@pytest.mark.parametrize("test_level", ["basic", "medium", "full"])
def test_descriptor_engine(test_level, ...):
    """Hierarchical test levels for different coverage needs"""

    if test_level == "basic":
        # Quick validation: 10 packets, simple timing
        num_packets = 10
        test_class = TestClass.APB_ONLY

    elif test_level == "medium":
        # Moderate coverage: 3 packets × 4 profiles
        num_packets = 3
        test_classes = [TestClass.APB_ONLY, TestClass.MIXED]

    elif test_level == "full":
        # Comprehensive: 5 packets × all profiles × all test classes
        num_packets = 5
        test_classes = [TestClass.APB_ONLY, TestClass.MIXED]
```

Benefits: - **Basic:** Quick smoke tests (CI/CD) - **Medium:** Developer validation - **Full:** Comprehensive regression

Pattern: Parametrized Test Generation

```
def generate_test_params():
    """Generate comprehensive parameter combinations"""
    params = []

    # Basic configurations
    for num_channels in [8, 32, 64]:
        for data_width in [64, 512]:
            for addr_width in [32, 64]:
                params.append((num_channels, data_width, addr_width))

    return params

# Pytest automatically runs all combinations
@pytest.mark.parametrize("num_channels, data_width, addr_width",
generate_test_params())
def test_scheduler(num_channels, data_width, addr_width):
```

```
    """Test scales across all parameter combinations"""
    # ...
```

Test Success Criteria:

⚠️ **CRITICAL:** All tests must achieve 100% success rate.

```
# ✗ WRONG: Accepting partial success
success_rate = (descriptors_received / total_sent) * 100
return success_rate >= 70  # "Mostly good" - NOT ACCEPTABLE

# ✓ CORRECT: Require 100% success
success_rate = (descriptors_received / total_sent) * 100
return success_rate >= 100  # Must receive ALL expected outputs
```

Why 100% Required: - Partial success indicates bugs, not acceptable tolerance - RTL should be deterministic - 100% success is achievable - Lower thresholds mask real issues

---

34.13 Documentation Generation

34.13.1 Generating PDF/DOCX from Specification

Tool: `/mnt/data/github/rtldesignsherpa/bin/md_to_docx.py`

Use this tool to convert the linked specification index into a single all-inclusive PDF or DOCX file.

Basic Usage:

```
# Generate DOCX from rapids_spec index
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    --toc \
    --title-page

# Generate both DOCX and PDF
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    --toc \
    --title-page \
    --pdf

# With custom template (optional)
python bin/md_to_docx.py \
```

```
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    -t path/to/template.dotx \
    --toc \
    --title-page \
    --pdf
```

Key Features: - **Recursive Collection:** Follows all markdown links in the index file - **Heading Demotion:** Automatically adjusts heading levels for included files - **Table of Contents:** `--toc` flag generates automatic ToC - **Title Page:** `--title-page` flag creates title page from first heading - **PDF Export:** `--pdf` flag generates both DOCX and PDF - **Image Support:** Resolves images relative to source directory - **Template Support:** Optional custom DOCX/DOTX template via `-t` flag

Common Workflow:

```
# 1. Update version number in index file (rapids_index.md)
# 2. Generate documentation
cd /mnt/data/github/rtldesignsherpa
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o projects/components/rapids/docs/RAPIDS_Specification_v0.25.docx
\
    --toc --title-page --pdf

# 3. Output files created:
#    - RAPIDS_Specification_v0.25.docx
#    - RAPIDS_Specification_v0.25.pdf (if --pdf used)
```

Debug Mode:

```
# Generate debug markdown to see combined output
python bin/md_to_docx.py \
    projects/components/rapids/docs/rapids_spec/rapids_index.md \
    -o output.docx \
    --debug-md

# This creates debug.md showing the complete merged content
```

Tool Requirements: - Python 3.6+ - Pandoc installed and in PATH - For PDF generation: LaTeX (e.g., texlive) or use Pandoc's built-in PDF writer

📖 **See:** `/mnt/data/github/rtldesignsherpa/bin/md_to_docx.py` for complete implementation details

34.14 PDF Generation Location

IMPORTANT: PDF files should be generated in the docs directory:

`/mnt/data/github/rtldesignsherpa/projects/components/rapids/docs/`

Quick Command: Use the provided shell script:

```
cd /mnt/data/github/rtldesignsherpa/projects/components/rapids/docs
./generate_pdf.sh
```

The shell script will automatically: 1. Use the md_to_docx.py tool from bin/ 2. Process the rapids_spec index file 3. Generate both DOCX and PDF files in the docs/ directory 4. Create table of contents and title page

📖 **See:** `bin/md_to_docx.py` for complete implementation details

34.15 Remember

1. 🎛️ **MANDATORY: Use BFMs** - GAXI Master/Slave for custom valid/ready, protocol-specific BFMs for AXI4/APB/AXIS/Network
2. 📖 **Link to detailed spec** - `projects/components/rapids/docs/rapids_spec/` has complete architecture docs
3. 🔢 **Exponential credit encoding** - Remember: 0→1, 1→2, 2→4, not linear!
4. 🐛 **Check known issues** - Before diagnosing bugs
5. 🔗 **Block interactions** - RAPIDS blocks are tightly coupled
6. 🧪 **Multi-layered testing** - FUB → Integration → System tests
7. 🏗️ **Testbench reuse** - Always create TB classes in `bin/TBClasses/rapids/`
8. 🎯 **Continuous monitoring** - Use background coroutines for asynchronous output capture
9. 🔍 **Search first** - Use existing CocoTBFramework components before creating new ones
10. 📊 **100% success** - All tests must achieve 100% success rate, no exceptions
11. 🏛️ **Three-layer architecture** - TB (infrastructure) + Test (intelligence) + Scoreboard (verification)
12. 🎪 **Queue-based verification** - Use `monitor._recvQ.popleft()` for simple tests, not memory models

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